Portfolio Construction and the Covariance Matrix

Why the optimiser is not the problem, and the matrix is

Abstract

The formula for the optimal portfolio has been known since 1952. The difficulty is in its inputs. The optimiser treats every estimate as exact, and estimates of expected returns and covariances are much noisier than they appear. This chapter explains the damage and the methods that limit it.

Mean-variance optimisation inverts the covariance matrix. Its largest positions are bets between assets it estimates to be nearly identical, which is where the data are weakest. Better optimisation of noisy inputs therefore gives worse portfolios. In one simulation with a correct model and a finite sample, the optimiser reported a Sharpe ratio of 5.3 against an attainable 0.43. The ratio of assets to observations measures the damage. Realised risk exceeds predicted risk by a factor of one over one minus that ratio. Doubling the universe does as much harm as halving the history. Expected returns are the most dangerous input. Their errors cost about 11 times as much as volatility errors and 22 times as much as correlation errors. Without a real return forecast, the best choice is to leave expected returns out. Studies find that equal weighting beats optimisation. This is evidence against sample means, not against optimisation.

Minimum variance, maximum diversification, inverse volatility, risk parity and equal weighting are one formula. They differ in the proxy they use for expected return and in the structure they impose on the covariance matrix. Equal weighting makes the strongest assumptions of all of them. A no-short constraint is equivalent to shrinking the covariance matrix. A turnover penalty is also a form of shrinkage, toward the current portfolio. Shrinkage, no-shorting, position caps and a turnover penalty applied together can reproduce equal weighting without anyone intending it, so measure the distance to it. With proportional costs, rebalance to the edge of a no-trade band whose width grows with the cube root of the cost.

Several small errors have large effects. Stale prices make illiquid assets look like diversifiers, and the optimiser then overweights the positions that are hardest to exit. The standard exponentially weighted estimate has an effective sample size of about 32 observations, which makes it unreliable above about 30 assets. Estimate volatilities with short windows and correlations with long ones. The chapter recommends three practices. First, leave out expected returns unless a real forecast exists. Second, shrink the covariance matrix, and prefer too much shrinkage to too little. Third, track realised over predicted volatility out of sample.

What this is. Portfolio construction is the step between having views and having positions. Investors have lost more real money at this step, to a correct-looking calculation, than anywhere else in quantitative finance. The reason is that the calculation is trivial and the inputs are not trivial. This chapter builds the subject from first principles. It covers four topics. The first is what the optimal portfolio *is*. The second is why feeding it sample estimates makes it fail. The third is what the covariance matrix does in that formula. The fourth is the set of remedies in common use, in enough detail to implement one.

How to read this chapter. Sections 1–2 are the conceptual core. §2 is the one section every reader should read, because it gives the mechanism behind every other section. §3 is history and §4 is the bibliography. §5 is the theory. It covers the closed forms, the regression identity that explains the whole failure, and the random-matrix results that quantify it. §6 and §7 are the two surveys. §6 covers ways to estimate a covariance matrix. §7 covers ways to turn one into weights. §8 shows that most of §6 and §7 is one method with five adjustable choices. Sections 9–11 cover engineering: implementation, costs and evaluation. §12 is the synthesis.

Different readers can start in different places.

- Readers who want only the most important idea should read §2.3, §5.4 and §8.2.
- Readers implementing a system should read §9 and §10, and treat the rest as reference.
- Readers who already know mean-variance theory and want the part that textbooks omit should start at §5.4.

Objectives. After this chapter, you should be able to:

- state the master form $w^\star = \gamma^{-1}\Sigma^{-1}\mu$, and place any named allocation rule in it as a choice of return view, regulariser, constraints and leverage.
- use the eigen-portfolio decomposition and the regression identity to explain why inverting a sample covariance matrix puts the largest bets in the least reliable directions.

- compute the aspect ratio $q = N/T$, or N/T_{eff} for weighted estimators, and predict from it how far realised risk will exceed predicted risk.
- choose and implement a covariance estimator, with linear shrinkage in correlation space as the default.
- recognise shrinkage, constraints and transaction costs as one form of regularisation, and measure their combined strength.
- evaluate a construction method fairly, with realised-over-predicted risk, matched volatility, a sweep over q and paired comparisons.

Relationship to the other chapters. This chapter treats the *cross-sectional* problem. Given many assets, it asks how much of each to hold. A companion chapter, Trend-Following in Financial Markets, covers the *scalar* problem of how large a single position should be. That chapter treats portfolio construction only as far as a trend system needs it. Another chapter, Market Regimes and Hidden Markov Models, argues in its section 12.4 that the covariance matrix is where regime information is most reliably useful. This chapter gives the detail behind that claim. Those chapters ask *what to forecast*. This chapter assumes that the forecasts exist and asks what to do with them. Each chapter stands alone.

A warning about scope. [Practice] Nothing here is investment advice. The numerical illustrations are simulations chosen to isolate a mechanism. They are not backtests of anything tradeable. Where a number comes from a simulation run for this chapter rather than from a published study, the text says so.

Epistemic tags. The chapters in this collection flag claims by status:

- **[Fact]** — replicated across independent datasets or implementations, with broad agreement.
- **[Contested]** — documented, but with live disagreement about magnitude, robustness, or cause.
- **[Hypothesis]** — a proposed mechanism, not decisively tested.
- **[Practice]** — practitioner convention, and the evidence may be private or absent. A [Practice] tag does not say that anyone has refuted the claim.

Tags appear only where the status changes what a reader should do. A tag governs the sentence or clause it opens. Untagged sentences are definitions, derivations or arithmetic: true by construction rather than by evidence.

Notation. The table lists every symbol that recurs in the chapter, with the section that defines or first uses it. Symbols used in only one section, or only in the appendix, are defined where they appear.

Symbol	Meaning	Defined in
N; T	Number of assets; number of observations in the estimation window	§1.4, §2.1
q	**Aspect ratio**, $q = N/T$	§1.4, §2.4
T_{eff}	Effective sample size of a weighted estimator; $(1+\theta)/(1-\theta)$ for an EWMA, and the T that belongs in q	§6.3, §9.4
w; $\mathbf{1}$	Vector of portfolio weights; vector of ones, so $\mathbf{1}'w = 1$ is the fully invested constraint	§1.2, §5.1
$w^\star$	Portfolio that is optimal at the true parameters, $w^\star = \gamma^{-1}\Sigma^{-1}\mu$	§1.2, §1.3
w_{prev}	Portfolio currently held, before a rebalance	§10.2
μ; $\hat{\mu}$	Vector of expected excess returns; an estimate of it, or the return view fed to the optimiser	§1.1, §1.2
Σ	**True** covariance matrix of returns, unobservable	§1.2
S	**Sample** covariance matrix computed from T observations	§2.1
$\hat{\Sigma}$	**Whatever covariance estimator is actually used**: S, a shrunk version of it, a factor model or something else	§1.1
γ	Risk aversion, $\gamma > 0$	§1.3
σ_i; D	Volatility of asset i; $D = \text{diag}(\sigma_1, \dots, \sigma_N)$	§5.3

Symbol	Meaning	Defined in
C; $\bar{\rho}$	Correlation matrix $C = D^{-1}\Sigma D^{-1}$, so $\Sigma = DCD$; average pairwise correlation	§5.3, §6.4
σ_p; $\sigma^\star$	Portfolio volatility $\sqrt{w'\Sigma w}$; volatility target	§7.6, §8.1
λ_i; v_i	Eigenvalues $\lambda_1 \geq ... \geq \lambda_N$ of the covariance matrix; the eigenvectors, read as **eigen-portfolios**	§2.3
V; Λ	Matrix of eigenvectors; $\Lambda = \text{diag}(\lambda_i)$, so $\Sigma = V\Lambda V'$	§6.1
κ	Condition number λ_1/λ_N	§5.5, §5.7
R_i^2; s_i^2	Coefficient of determination from regressing asset i's return on the other $N-1$ assets; the residual variance $\sigma_i^2(1-R_i^2)$	§5.4
α_i; β_{ij}	**Residual alpha**, expected return net of replication; coefficients of the same regression	§5.4
δ; Φ	**Shrinkage intensity**, $\delta \in [0,1]$; the target matrix it shrinks toward	§6.5
ν	**Ridge or turnover penalty** weight, $\nu \geq 0$	§8.2, §10.3
θ	**EWMA decay**, $\theta \in (0,1)$	§6.3
F; Ω; Ψ; K	Factor model $\Sigma = F\Omega F' + \Psi$: $N \times K$ loading matrix; $K \times K$ factor covariance; diagonal matrix of idiosyncratic variances; number of factors	§6.8
Δ; $\mathcal{C}$	Regulariser added to $\hat{\Sigma}$; constraint set	§8.1
Π	Equilibrium excess returns implied by market weights, $\Pi = \gamma\Sigma w_{\text{mkt}}$	§7.8
SR	Sharpe ratio	§5.1

Several symbols carry a qualification:

- Keep three matrices apart: the true Σ, the sample S and the estimator $\hat{\Sigma}$. Hats generally denote estimates.
- γ is always risk aversion, never a significance level.
- C is the correlation matrix, and $\mathcal{C}$ is the constraint set of §8.1. Typography distinguishes these near-identical symbols throughout.
- R_i^2 always denotes the regression of asset i on the other $N-1$ assets, the quantity of §5.4. It never measures goodness of fit for anything else.
- The chapter keeps the three regularisation parameters δ, ν and θ distinct. §10.3 shows that the two uses of ν, as a ridge and as a turnover penalty, are the same thing.
- The Λ_c of §10.2 is a diagonal cost matrix, and it is not the eigenvalue matrix Λ. The GARCH coefficients α_1 and β_1 of A.27 are different quantities from α_i and β_{ij}.

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1. What portfolio construction is

1.1 The wrong intuition

The usual starting intuition is this:

> Portfolio optimisation takes the forecasts and finds the best portfolio.

Each word of that sentence is defensible, but the sentence as a whole is badly wrong. The way it is wrong determines what the entire field studies, so the failure needs a precise statement.

An optimiser cannot tell which parts of its input are forecasts and which are errors. It receives a vector $\hat{\mu}$ and a matrix $\hat{\Sigma}$, and it treats every entry as exact. Suppose the true expected return of asset 17 is 4% and the estimate says 45%. The optimiser does not hedge that possibility, discount it or flag it. It buys asset 17 heavily, because the input says to. Now suppose a *pair* of assets has a true correlation of 0.90 and the sample says 0.97. The optimiser builds a long-short position between them at significant leverage, because at 0.97 that spread looks nearly riskless.

The accurate description therefore contradicts the intuition:

> **Optimisation amplifies whatever the forecasts contain, signal and error alike, in proportion to the confidence the inputs imply. It does not extract value from forecasts.**

The field's history (§3) is therefore a history of better inputs, not of better optimisers. Markowitz solved the optimisation itself in 1952, and computationally it is a first-year exercise. Every advance since then has concerned the inputs. Some advances estimate the inputs better. Some impose structure so that analysts estimate less. Others avoid needing the inputs at all.

1.2 The right starting point

Portfolio construction is a **decision made under uncertainty about the parameters of the decision problem itself.**

That framing has three parts. Keeping them separate prevents most of the confusion in the field.

	The object	Status
Choice variable	w, the weights	Controlled exactly
Parameters	μ, Σ	Never observed; only estimated
Objective	some functional of $w'\mu$ and $w'\Sigma w$	Chosen by the investor. The choice matters less than commonly thought

The classical theory assumes that the parameters are known and studies the choice. The modern theory accepts that they are unknown. It studies the *decision rule*, the map from data to weights, and asks how well that map performs on a finite sample. The two questions have different answers, and the gap between them is the subject of this chapter.

The gap has a precise statement. Write the decision rule as a function $\hat{w}(\text{data})$. Write $w^\star$ for the portfolio that is optimal at the *true* parameters. §1.3 gives its form. The classical question asks whether $w^\star$ is optimal, and the answer is yes by construction. The practical question compares two quantities:

$$\mathbb{E}_{\text{data}}\big[\, U\big(\hat{w}(\text{data});\ \mu, \Sigma\big)\,\big] \quad \text{versus} \quad U(w^\star;\ \mu, \Sigma)$$

The expectation runs over samples, and the utility U is always evaluated at the *true* parameters. The first quantity is what an investor actually experiences. The second is what the optimiser reports. §2 measures the gap between them, and it is much larger than most readers expect.

1.3 The master form

Take the standard mean-variance objective. It maximises expected return, penalised by variance, with risk aversion γ:

$$\max_{w} \; w'\mu - \frac{\gamma}{2} w'\Sigma w$$

Differentiating with respect to w and setting the result to zero gives $\mu - \gamma\Sigma w = 0$. Solving for w gives the **master form** of this chapter:

$$\boxed{w^\star = \frac{1}{\gamma}\Sigma^{-1}\mu}$$

Three properties of this equation matter before anything else.

The weight is a ratio. It is expected return *divided by* risk, in a matrix sense. The division can make the weights very large. Every pathology in §2 comes from the inverted Σ.

$1/\gamma$ is only a scalar. It sets leverage and nothing else. The *direction* of the portfolio is $\Sigma^{-1}\mu$ regardless of risk aversion. The direction means the relative weights, which are the interesting part. This is Tobin's separation result. It means the sizing question (section 8.5 of Trend-Following in Financial Markets) and the composition question are genuinely separable. A portfolio can have the right direction and the wrong scale, or the reverse. These are different mistakes with different fixes.

Almost every method in the field is this formula with particular choices. The list includes equal weighting, minimum variance, risk parity, maximum diversification, Black–Litterman and hierarchical risk parity. The survey in §7 works through them. §8 shows that each is $\Sigma^{-1}\mu$ under a specific pair of assumptions about the value of μ and the regularisation of Σ. Several of them appear to have abandoned optimisation entirely, but none of them has.

1.4 The three claims, stated once

The rest of the chapter follows from three claims, and each section elaborates one of them. The chapter numbers them C1–C3. The slots S1–S5 of §8.1 use a separate numbering for a different purpose. The two numberings do not correspond.

> **C1 — the master form.** Every portfolio construction method is $w \propto \hat{\Sigma}^{-1}\hat{\mu}$ for some choice of $\hat{\mu}$ and some regularised $\hat{\Sigma}$. The choice of $\hat{\mu}$ is often implicit and often deliberately degenerate. The only true exception is hierarchical risk parity (§7.11), which reaches a similar result without ever forming the inverse.
>
> **C2 — the mechanism.** Inverting $\hat{\Sigma}$ places the largest bets in the directions with the smallest estimated variance. The estimate is most wrong in exactly those directions. The inputs imply the most confidence where it is least justified.
>
> **C3 — the diagnostic.** To first order, one number sets how badly this goes: the aspect ratio $q = N/T$. Neither N nor T sets it alone. Their ratio does. The ratio governs the *covariance* half of the problem. Calendar span, not the aspect ratio, limits the return view (§2.5). That is a separate and larger constraint.

The consequence, which §8 makes precise, is this:

> **Every repair in the field does the same thing. The repairs include shrinkage, factor models, position constraints, resampling, Bayesian priors, transaction-cost penalties and hierarchical clustering. Each adds structure that raises the smallest eigenvalues of $\hat{\Sigma}$, and so reduces the positions taken in those directions. The repairs differ in the source of the structure and in how they justify it. They do not differ in what they do.**

That last statement is why this chapter uses a taxonomy rather than a catalogue. Suppose a practitioner treats shrinkage, constraints and turnover penalties as three independent methods. The practitioner will then apply all three at full strength. The resulting portfolio is far more heavily regularised than intended.

1.5 What portfolio construction is not

People routinely use this name for several neighbouring problems. Naming them prevents much confusion.

Not this	Why it is different
Signal generation	μ arrives from outside. Construction takes it as given and is not responsible for whether it is any good. §2.5 shows that construction is responsible for how much damage a bad μ does
Position sizing	The scalar $1/\gamma$, or a volatility target. Independent of composition (§1.3), and a different problem
Diversification	Sometimes an outcome, and a poor objective. A portfolio can be maximally diversified and terrible
Risk management	A layer of constraints and monitoring *around* construction. It answers "what if the inputs are wrong", not "what is best"
Asset allocation	The same mathematics at $N \approx 10$, where estimation error is mild and the hard problems of §2 barely appear
Execution	The path from current weights to target weights. Genuinely separate, except that §10 shows costs feed back into the target itself

The first distinction causes the most trouble in practice. Practitioners nearly always blame the signal for a disappointing backtest. The construction step is at least as likely to be the cause, and it is far easier to fix.

1.6 A worked example

Take two assets. Both have 20% annual volatility, their correlation is ρ, and their expected excess returns are 6% and 5%. Set risk aversion so that the portfolio is fully invested. The question is what weights the master form gives.

Let $\Sigma = 0.04 \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$ and $\mu = (0.06, 0.05)'$. One line of algebra gives the answer. The inverse Σ^{-1} has a positive scalar factor $1/[0.04(1-\rho^2)]$, which normalisation removes anyway. Up to that factor,

$$\Sigma^{-1}\mu \;\propto\; \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix} \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} = \begin{pmatrix} \mu_1 - \rho\mu_2 \\ \mu_2 - \rho\mu_1 \end{pmatrix}.$$

The two entries sum to $(\mu_1 + \mu_2)(1-\rho)$. Dividing by that sum makes the weights add to one, and gives the fully invested optimum:

$$w_1 = \frac{\mu_1 - \rho\,\mu_2}{(\mu_1 + \mu_2)(1-\rho)}, \qquad w_2 = 1 - w_1.$$

The table evaluates these weights at five correlations.

ρ	w_1	w_2	Gross	Comment
0.00	0.55	0.45	1.0×	Mild tilt toward the better asset
0.50	0.64	0.36	1.0×	
0.90	1.36	−0.36	1.7×	Now short the second asset
0.95	2.27	−1.27	3.5×	
0.99	9.55	−8.55	18.1×	An 18× spread trade on a 1% return difference

This small table shows the central result of the chapter. As the two assets become more alike, the optimiser stops allocating between them and starts arbitraging between them. The leverage it applies grows without bound. The optimiser works as designed. Given *exact* inputs, these really are the optimal weights, and at $\rho = 0.99$ the spread really is nearly riskless.

The problem is that $\rho = 0.99$ is unknown. Analysts must estimate it. The large-sample standard error of a correlation estimated from T observations is roughly $(1-\rho^2)/\sqrt{T}$. Two years of daily data ($T = 500$) give a standard error of 0.0009.

A two-standard-error band around a sample correlation of 0.99 therefore runs from 0.988 to 0.992. The corresponding gross exposures run from 15.3× to 22.0×. In any other context, a band this narrow is negligible. Here the trade at the top end is 44% larger than the trade at the bottom.

The estimate in this example was good. It had four significant figures and a tight confidence interval. It had no misspecification, no outliers and no regime change. The instability comes from the $1/(1-\rho)$ in the denominator. That term is a property of the map from parameters to weights, not of the estimate. **Precision in the input does not imply stability in the output, and the optimiser reports no diagnostic that reveals which case applies.**

§2 scales this example from two assets to 500.

§1 Key takeaways

1. An optimiser cannot distinguish a forecast from an error. It amplifies both in proportion to the confidence its inputs imply. Better optimisation of bad inputs makes the result worse, not better.
2. The master form is $w^\star = \gamma^{-1}\Sigma^{-1}\mu$. Every method in the field chooses a return view and a way to regularise the inverse.
3. Composition and leverage separate cleanly. $1/\gamma$ sets the scale, and $\Sigma^{-1}\mu$ sets the direction. Treat them as two problems, because they fail in different ways.
4. The classical question, whether $w^\star$ is optimal, has been settled since 1952. The practical question is how a *decision rule* performs on a finite sample.
5. Inverting the covariance matrix makes the optimiser's largest positions spread trades between assets whose estimates say they are nearly identical. Those estimates are the least reliable part of the input.
6. Portfolio construction is not signal generation, sizing, diversification or risk management. Confusing it with signal generation is the most expensive of these errors, because it directs effort to the wrong fix.

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2. Why the naive answer fails

This section gives the mechanism behind everything that follows. §2.3 is the most important single part of the chapter.

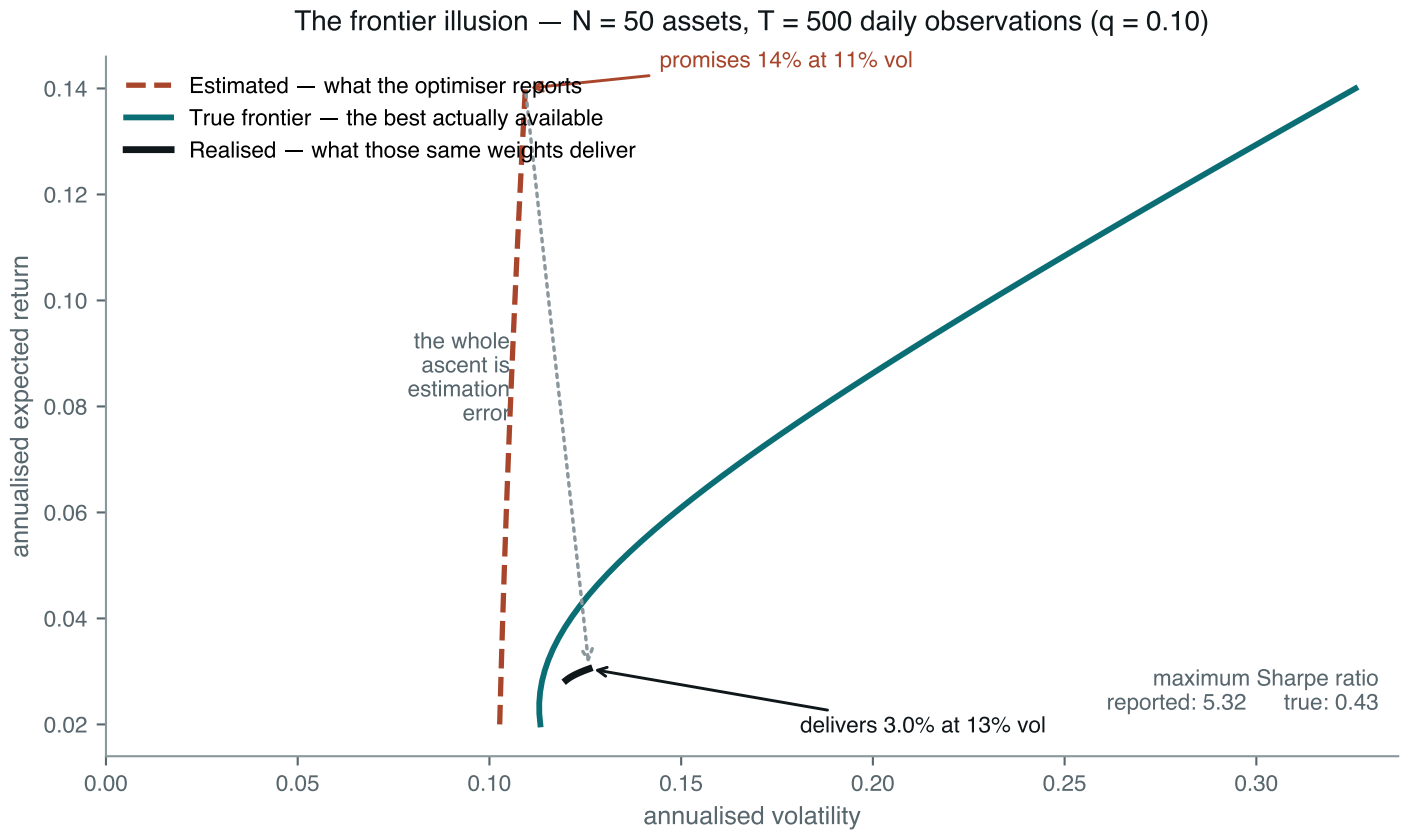
2.1 The demonstration

The experiment runs under conditions as favourable to the optimiser as possible. It uses 50 assets. A one-factor model, a market factor plus idiosyncratic noise, generates the returns. Betas lie between 0.6 and 1.4, and idiosyncratic volatilities lie between 15% and 35%. Expected returns are 5% per unit of beta plus a small alpha. The sample is two years of daily data, so $T = 500$ and $q = N/T = 0.10$. By industry standards that ratio is comfortable.

The simulation draws the data from exactly the model the optimiser assumes. Returns are IID, multivariate normal and stationary. The data have no fat tails, no regime changes, no missing data, no non-synchronous trading and no misspecification of any kind. Any failure is estimation error and nothing else.

The experiment computes the sample mean $\hat{\mu}$ and the sample covariance S, then traces out the mean-variance frontier. The figure plots three curves:

- the frontier the optimiser *reports*, using $\hat{\mu}$ and S.
- the frontier that is genuinely *available*, using μ and Σ.
- what the optimiser's chosen weights *actually deliver*, with the estimated weights evaluated at the true parameters.



Returns are drawn from exactly the model the optimiser assumes, so every part of the gap is estimation error — none of it is misspecification. Ask the estimated frontier for more return and it hands you more leverage on noise: the realised curve barely moves.

The table gives the numbers from a simulation run for this chapter.

Quantity	Value
True maximum Sharpe ratio	0.43
Sharpe ratio the optimiser reports	**5.32**
True expected returns, range across assets	1.9% to 7.8%
Sample mean returns, range across assets	**−38.9% to +44.9%**
Estimated frontier at its top end	14.0% return, 10.9% volatility
What those weights actually deliver	**3.0% return, 12.6% volatility**

Three observations follow, in increasing order of importance.

The reported Sharpe ratio is wrong by a factor of 12. The optimiser's error is far larger than slight optimism. Its report claims a portfolio 12 times better than the best portfolio in the universe it received. A research process that treats an in-sample optimised Sharpe ratio as evidence does not measure what it intends to measure.

The sample means are an order of magnitude more dispersed than the truth. The true spread of expected returns is about six percentage points. The sample spread is about 84. Essentially all of the cross-sectional variation in a two-year sample of mean returns is noise. This effect is not specific to small samples. The standard error of an annualised mean equals the annual volatility divided by the square root of the number of *years*. Annual expected returns are a few percent, while annual volatility is 20%. Over two years the standard error is therefore $0.20/\sqrt{2} \approx 14\%$, which is several times the quantity it estimates. [Fact] More data barely improves the ratio, because the standard error of a mean falls as $1/\sqrt{T}$ in *calendar time*, not in sample size. Sampling the same two years more finely does not help at all (Merton, 1980).

The realised frontier is flat. This finding should change behaviour. Asking the estimated frontier for 4% expected return delivers 2.8%. Asking for 14% delivers 3.0%. The dashed curve's entire ascent looks like the optimiser trading risk for return, and a committee would discuss it. That ascent adds only two-tenths of one percent of return. Otherwise it adds leverage on noise.

2.2 Error maximisation

Michaud (1989) named the phenomenon. He called mean-variance optimisers **estimation-error maximisers**. The argument is about selection, and it is short.

The optimiser overweights assets with a high estimated return, a low estimated variance and a low estimated correlation with the rest. In a large group of assets, the highest return estimates disproportionately belong to assets whose *errors* were positive. The lowest variance and correlation estimates disproportionately belong to assets whose errors were negative. The optimiser's largest positions therefore concentrate systematically in the assets whose inputs are most overstated. The effect grows with the number of assets the optimiser chooses among.

Writers often describe this failure loosely as "overfitting", which understates it. Overfitting usually means a model with too many parameters fitting the noise in a training set. Here the optimiser has no model and does no fitting. $\Sigma^{-1}\mu$ has no free parameters, and nobody chooses it to fit anything. **The optimiser takes a maximum, and a maximum over noisy inputs is biased upward by construction.** For the same reason, the best of 100 coin-flippers looks skilled. The optimiser selects a *direction in portfolio space*, not a parameter. It has effectively as many directions to choose from as there are assets.

2.3 The mechanism: what inversion actually does

The selection argument above is correct but qualitative. The precise version concerns eigenvalues. It deserves a slow treatment, because every method in §6 and §7 addresses it.

Write any covariance matrix in terms of its eigenvalues and eigenvectors:

$$\Sigma \;=\; \sum_{i=1}^{N} \lambda_i \, v_i v_i', \qquad \lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_N > 0.$$

This is a decomposition into **eigen-portfolios**. Each v_i is a combination of the assets, so it is a portfolio. Its weights have no sum constraint. Because eigenvectors are unit vectors, $v_i' \Sigma v_i \;=\; \lambda_i$ exactly. So λ_i *is* that portfolio's variance. The first eigen-portfolio v_1 is typically "everything, long". That is the market, and it has the largest variance. The last, v_N, is the lowest-variance combination available, an intricate long-short arrangement that hedges almost everything out.

Inversion replaces each eigenvalue with its reciprocal and leaves the eigenvectors unchanged:

$$\Sigma^{-1} \;=\; \sum_{i=1}^{N} \frac{1}{\lambda_i} \, v_i v_i'.$$

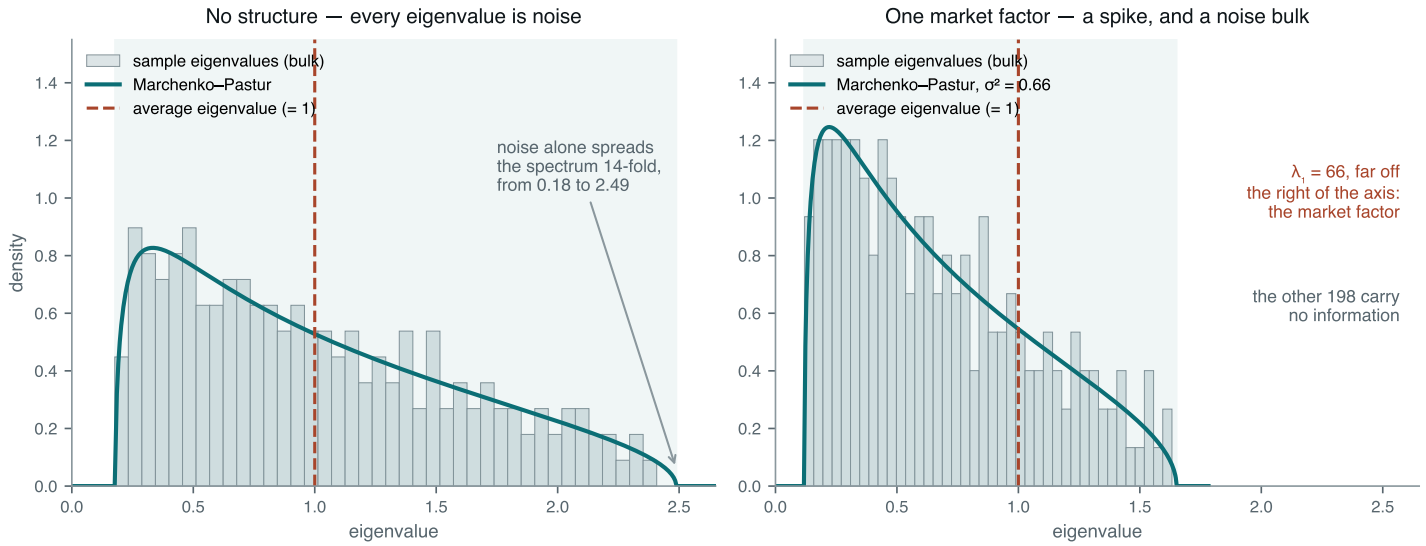
The optimal portfolio therefore decomposes as

$$w^{\star} \;\propto\; \Sigma^{-1}\mu \;=\; \sum_{i=1}^{N} \underbrace{\frac{v_i'\mu}{\lambda_i}}_{\text{how much of } v_i} \qquad v_i.$$

The optimiser holds each eigen-portfolio in proportion to its expected return divided by its variance. This is correct, and it is also the source of the problem. The optimiser divides the lowest-variance direction, v_N, by the smallest eigenvalue, so that direction receives the largest multiplier. *By construction, the optimiser's biggest bets are on the combinations of assets it estimates to be least risky.*

Now consider what the sample gets wrong. Estimating Σ from T observations does not give the eigenvalues independent, symmetric errors. The sample eigenvalues are **spread apart in a specific, predictable way**. The large ones are biased up, and the small ones are biased down. This is the Marchenko–Pastur result, stated in §5.5. The consequence here runs in four steps. The sample's smallest eigenvalue is too small. The optimiser therefore divides by a number that is too small. It therefore takes a position that is too large. That position lies in the direction where its estimate of risk is least reliable.

The figure shows the effect at $N = 200$ and $T = 600$, on data with *no structure at all*. The data are 200 independent series, so every true eigenvalue equals 1.



N = 200 assets, T = 600 observations, q = N/T = 0.33. Left: the truth is a single eigenvalue at 1, and the sample smears it across the shaded band. Right: once the market mode's share of the trace is taken out, what remains still matches the same law — which is exactly what makes it safe to discard.

The true spectrum is a single point at 1. The sample spreads it from 0.18 to 2.49, a 14-fold range. An optimiser given this matrix applies leverage of $1/0.18 = 5.6$ times to the lowest-variance direction and only $1/2.49 = 0.4$ times to the highest-variance direction. The result is a **14:1 spread of leverage across directions that are in truth identical.** All of that spread is noise, and the optimiser treats it as exact.

The summary is this:

> **The optimiser allocates in inverse proportion to estimated variance, and estimated variance is least accurate exactly where it is smallest. The largest positions therefore carry the largest errors. Regularisation is essential here. Without it, the optimiser builds the portfolio entirely from the least reliable part of the data.**

The right-hand panel shows how to repair the situation. Adding one real market factor raises its eigenvalue to 66, far outside the noise band. Every other eigenvalue stays inside a bulk. Once an analyst removes the factor's share of variance, the bulk keeps the Marchenko–Pastur shape. **Signal and noise separate cleanly in the spectrum.** This separation underlies eigenvalue clipping (§6.7), factor models (§6.8) and, in a sense, every method in §6. Theory says in advance that the bulk is noise. So an analyst can replace the bulk with something better behaved at no cost in information.

2.4 The diagnostic: everything is governed by $q = N/T$

The natural question after §2.3 is how much data is needed. That question leaves out the number of assets. The damage depends on $q = N/T$, not on T alone.

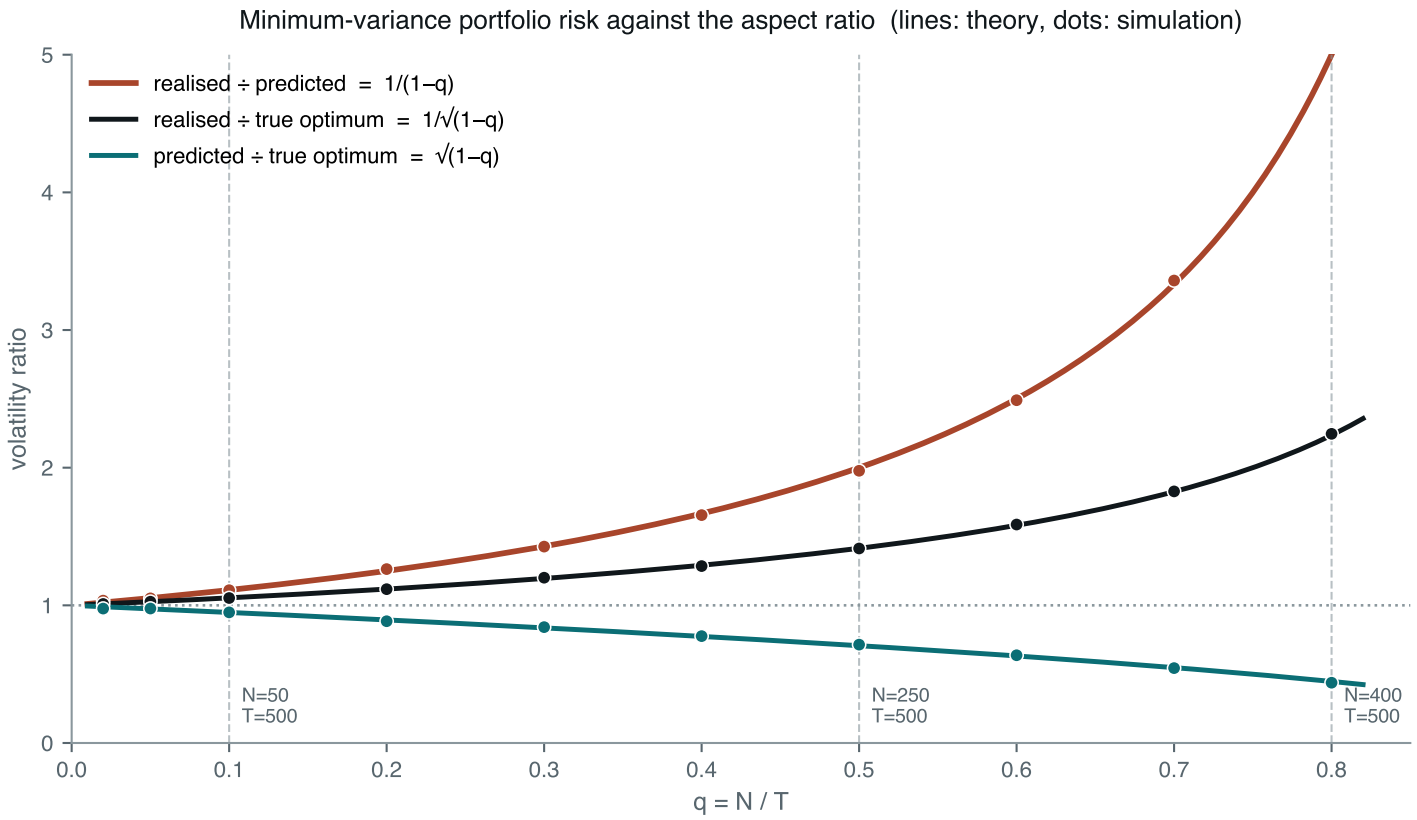
The global minimum-variance portfolio is the simplest case, because it uses no μ at all. Three of its quantities have clean large-sample limits. Write σ^2_{opt} for the variance of the true optimal portfolio. Then

$$\frac{\text{variance the optimiser predicts}}{\sigma^2_{\text{opt}}} \to 1 - q, \qquad \text{(it understates its own risk)}$$

$$\frac{\text{variance actually realised}}{\sigma^2_{\text{opt}}} \to \frac{1}{1-q}, \qquad \text{(it takes more risk than optimal)}$$

$$\frac{\text{realised}}{\text{predicted}} \to \frac{1}{(1-q)^2}. \qquad \text{(the two errors compound)}$$

In units of volatility rather than variance, the last line is simply $1/(1-q)$. Remember this number. §5.6 gives the derivation, and the figure confirms it by simulation.



The optimiser is simultaneously understating its own risk and taking more of it. Both errors come from the same place, and they compound.

The table gives the practical consequences.

N	T	q	Realised ÷ predicted volatility
50	500 (2 yr daily)	0.10	1.11×
100	500	0.20	1.25×
250	500	0.50	**2.00×**
400	500	0.80	**5.00×**
500	500	1.00	undefined — S is singular
500	2520 (10 yr daily)	0.20	1.25×

Two lessons follow. Both are counterintuitive, so the text states them flatly.

Doubling the universe is exactly as harmful as halving the history. Adding assets is diversification with a cost: it directly reduces the reliability of every position. A firm running 500 names on two years of data is in a qualitatively worse position than one running 50 names on the same data. No amount of care elsewhere compensates.

At $q \geq 1$ the sample covariance matrix is singular, and the optimiser is meaningless. With 500 names and two years of daily history, infinitely many "zero-risk" portfolios exist. These combinations had exactly zero variance in sample, purely by construction. The optimiser finds one and applies unlimited leverage to it. Code that fails silently in this case is worse than code that raises an error. [Practice] **Recommendation: check q before anything else.** The check is one line, and it can invalidate an entire research programme.

2.5 Which input hurts most

The inputs differ in how much damage their errors cause. The differences are large enough to drive design decisions.

Chopra & Ziemba (1993) measured the damage from each input. They perturbed each input and computed the *certainty-equivalent loss*: the amount of money an investor would give up by holding the perturbed portfolio instead of the true optimum. The result is among the most useful numbers in the field. [Fact] At a risk tolerance of 50, errors in **means** cost roughly **11 times** as much as errors in **variances**, and roughly **22 times** as much as errors in **covariances**. Risk tolerance is their scale

for the reciprocal of risk aversion. The ratio is approximately **22 : 2 : 1**, and the dominance of means grows further as risk tolerance rises.

The mechanism is the second observation of §2.1. Estimating a mean takes calendar time, and its standard error stays comparable to its own magnitude even over decades. A few months of daily data estimate a variance to within a few percent. Finer sampling improves variance estimates and does not improve mean estimates. [Fact] This asymmetry is structural. It is not a data problem, and a better dataset will not fix it.

The design conclusion is direct. It is the most consequential single decision in this chapter.

> **Without a genuine, tested return forecast, do not put one in.** Setting $\hat{\mu} \propto \mathbf{1}$ treats all assets as having the same expected return. It removes the dominant error term outright. This is the entire justification for the risk-based portfolios of §7.2–§7.6, and it explains why they perform so much better than their crudeness suggests.

The argument has limits. It does not say that expected returns are unforecastable. It does not forbid using a μ. It sets a high standard. A return forecast must beat zero, and it must also beat the substantial benefit of removing the largest source of estimation error from the problem. A weak but real signal can easily make the portfolio worse.

2.6 The 1/N challenge, and what it actually showed

The most cited empirical result in this literature comes from DeMiguel, Garlappi & Uppal (2009). Across seven datasets, they compared 14 portfolio rules against the naive rule of putting $1/N$ in each asset. The rules included sample mean-variance, Bayes–Stein, methods in the style of Black–Litterman, minimum variance and various constrained versions. [Fact] None of the 14 rules beat $1/N$ consistently on out-of-sample Sharpe ratio.

Their most quoted calculation gives the estimation window that sample-based mean-variance needs to beat $1/N$ reliably. The window is **about 3,000 months for 25 assets, and about 6,000 months for 50**: 250 and 500 years respectively. No such dataset exists, and none will.

Many sources report this result as "optimisation doesn't work". That reading is wrong.

Kritzman, Page & Turkington (2010) made the counter-argument directly. [Contested] The DeMiguel–Garlappi–Uppal comparison uses *sample means* as the return input. Replacing them with almost anything else lets optimised portfolios beat $1/N$ comfortably over long samples. The alternatives include a constant, a risk-premium estimate and a shrunk mean. They conclude that the finding is about the input and not about the method.

This chapter takes the view that Kritzman and co-authors have the better argument. §2.5 already contains the resolution:

> **DeMiguel, Garlappi and Uppal is a paper about $\hat{\mu}$. It is not a paper about optimisation.** It shows very convincingly that sample means are worse than useless as portfolio inputs. It does not show that the covariance matrix is useless. Minimum-variance portfolios use Σ and discard μ, and they are among the strongest performers in the paper's own tables and in most subsequent work.

[Contested] The remaining disagreement is narrower and genuine. It concerns whether risk-based optimisation beats $1/N$ *net of transaction costs and constraints* in a realistic implementation. There, turnover and the concentration of minimum-variance solutions remove much of the theoretical gain. The question is still open. Its answer differs across asset classes. §10 and §11 treat it.

§2 Key takeaways

1. In a simulation where the optimiser's model was exactly correct, it reported a Sharpe ratio of 5.3 against a true attainable 0.43. Its realised frontier was flat: asking for 14% return rather than 4% delivered an extra two-tenths of a percent.
2. The mechanism is inversion. The optimiser holds eigen-portfolios in proportion to expected return over variance, so the smallest eigenvalue gets the largest bet. The sample also gets the smallest eigenvalue most wrong.
3. Sample eigenvalues are spread apart in a known way. At $q = 1/3$, a truly flat spectrum appears spread 14-fold in the sample. Because the distortion is predictable, it is also correctable, which is the purpose of §6.
4. One number governs the damage: $q = N/T$. Realised volatility exceeds predicted volatility by $1/(1-q)$. At $q = 0.5$ the portfolio takes twice the risk it reports. At $q \geq 1$ the problem is not defined at all.
5. Doubling the number of assets does exactly as much harm as halving the history. Universe size is a statistical decision, not only a business one.

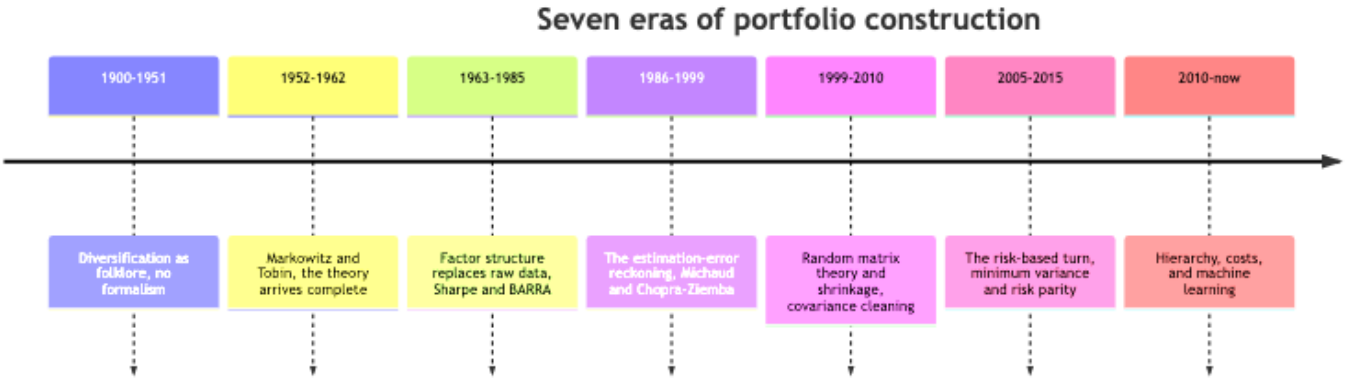
6. Errors in means cost roughly 11 times as much as errors in variances and 22 times as much as errors in covariances. Without a real return forecast, omitting μ entirely is the best available move. It is not a compromise.
7. The famous $1/N$ result is evidence against sample means, not against optimisation. It justifies dropping $\hat{\mu}$, not dropping $\hat{\Sigma}$.

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3. How the field evolved

The history has one thesis, and each change of era illustrates it:

> **The field's focus moved from "solve the optimisation" to "estimate the inputs", and then to "arrange not to need the inputs that analysts cannot estimate."**



Era I — Diversification as folklore (1900–1951). *Contribution:* the idea that spreading holdings reduces risk. It appeared in the practice of investment trusts and in Keynes's writings on portfolio policy. *What changed:* nothing formal. No method existed to say how much diversification was needed, or between which assets. *Limitations:* without a concept of covariance, the proverb "don't put all your eggs in one basket" cannot distinguish 20 correlated bets from five independent ones. *Lasting influence:* the intuition survives. §7.2 shows that the naive $1/N$ rule it implies is a far stronger competitor than the theory that displaced it.

Era II — The theory arrives complete (1952–1962). *Contribution:* Markowitz (1952) made risk a property of the *portfolio* rather than of its holdings, through the covariance matrix. He posed the problem as a quadratic programme. Tobin (1958) added the separation theorem: the composition of the risky portfolio does not depend on risk aversion. *What changed:* everything conceptual. "Risk" gained a definition that analysts could compute. *Limitations:* Markowitz needed $N(N+1)/2$ covariance estimates and computers that could invert the result. Neither existed at scale. More importantly, nobody yet understood that the estimates, not the arithmetic, were the binding constraint. *Lasting influence:* total. The master form of §1.3 is unchanged, and every later section builds on it.

Era III — Structure instead of data (1963–1985). *Contribution:* Sharpe (1963) proposed the single-index model. It models all co-movement through one market factor, and so replaces $N(N+1)/2$ covariances with $2N+1$ parameters. Barr Rosenberg's commercial risk models, at the firm that became BARRA, generalised this to fundamental models with many factors. *What changed:* analysts recognised that they should *model* a covariance matrix and not measure it. *Limitations:* factor structure is an assumption. The structure discards genuine residual correlation within industries. A statistical-arbitrage desk trades exactly that correlation. *Lasting influence:* very large. Factor covariance models (§6.8) remain the default in institutional equity risk management, and §8.2 shows that they are a form of shrinkage.

Era IV — The estimation-error reckoning (1986–1999). *Contribution:* a cluster of papers established that estimation error, not model error, was the binding constraint. Jorion (1986) applied Bayes–Stein shrinkage to the means. Michaud (1989) named the error-maximisation property. Best & Grauer (1991) computed analytically how sensitive optimal weights are to the means. Chopra & Ziemba (1993) measured the damage. Black & Litterman (1992) avoided it by starting from market-implied returns. *What changed:* the field stopped treating the optimiser as a solved problem. It started treating the optimiser as a source of error. *Limitations:* the diagnoses were more precise than the cures. Most cures were tuning parameters without theory. *Lasting influence:* decisive. Everything after this era is a response to it.

Era V — The physics import (1999–2010). *Contribution:* two independent literatures arrived at the same repair. Statistical physicists, in Laloux, Cizeau, Bouchaud & Potters (1999), showed that the bulk of the eigenvalue spectrum of a financial correlation matrix is indistinguishable from random-matrix noise. That gave a principled rule for what to discard. Statisticians, in Ledoit & Wolf (2003, 2004), derived in closed form the optimal intensity of shrinkage toward a structured target. Meanwhile Jagannathan & Ma (2003) proved that a no-short-sale constraint is *mathematically identical* to shrinking the covariance matrix. This explained why unsophisticated practice had been outperforming sophisticated theory. *What changed:* regularisation became principled rather than ad hoc, and the equivalences of §8.2 became visible. *Limitations:* almost all of this work addresses Σ, and §2.5 says that μ is the larger problem. *Lasting influence:* Ledoit–Wolf shrinkage is now the default in `scikit-learn` and in most risk systems. The ideas of this era are the methods practitioners use today.

Era VI — The risk-based turn (2005–2015). *Contribution:* practitioners acted on the conclusion of §2.5 and abandoned μ. Minimum-variance, maximum-diversification (Choueifaty & Coignard, 2008) and equal-risk-contribution (Maillard, Roncalli & Teiletche, 2010) portfolios became products. DeMiguel, Garlappi & Uppal (2009) supplied the negative result that justified this change. *What changed:* "we do not forecast returns" went from an admission to a selling point. *Limitations:* [Contested] much of the measured outperformance comes from exposure to the low-volatility and value factors and not from superior construction. The products are also capacity-constrained and crowded. *Lasting influence:* large and ongoing. Risk-based allocation is now the default starting point for multi-asset portfolios.

Era VII — Hierarchy, costs, and learning (2010–present). *Contribution:* three lines of work. López de Prado (2016) proposed hierarchical risk parity, which allocates with the covariance matrix without ever inverting it. Gârleanu & Pedersen (2013) solved the dynamic problem with transaction costs in closed form. They showed that the optimal portfolio is a weighted average of current holdings and a forward-looking target. Ledoit & Wolf (2017) extended shrinkage from a single intensity to a whole function of the spectrum. *What changed:* costs moved from an after-the-fact adjustment into the objective, and regularisation became nonparametric. *Limitations:* [Contested] researchers dispute the empirical advantage of HRP over well-regularised alternatives, and it appears sensitive to the test design. End-to-end allocation by machine learning remains mostly promising rather than demonstrated. *Lasting influence:* too early to judge for the machine-learning work. The cost-aware work is already standard.

§3 Key takeaways

1. Markowitz solved the optimisation problem in 1952, and it has never been the bottleneck. Every later era concerns the inputs.
2. Factor models (1963) were the field's first regularisation. Practitioners adopted them for computational reasons and kept them for statistical ones.
3. The 1986–1999 papers diagnosed estimation error precisely, but their cures were mostly tuning without theory.
4. Physics and statistics converged on the same repair around 2000. Analysts can identify the noisy part of the spectrum in advance and should replace it.
5. Jagannathan and Ma's 2003 proof that constraints *are* shrinkage is the central result of the whole history. It explained why crude practice beat refined theory, and it unified two literatures that appeared to disagree.
6. The risk-based turn applied the Chopra–Ziemba result. Dropping an input that analysts cannot estimate beats estimating it badly.

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4. Foundational references

This section groups the references by kind, because readers read each kind differently. Each entry says why the work matters. Where a free copy exists, the link points to it. The section marks entries available only behind a paywall.

4.1 The founding theory

- **Markowitz, H. (1952).** "Portfolio Selection." *Journal of Finance* 7(1), 77–91. — The original paper. It shows how carefully Markowitz already qualified his claims about the inputs. The field took 50 years to rediscover his caution.
- **Tobin, J. (1958).** "Liquidity Preference as Behavior Towards Risk." *Review of Economic Studies* 25(2), 65–86. — The separation theorem: leverage and composition are independent choices (§1.3).
- **Sharpe, W. F. (1963).** "A Simplified Model for Portfolio Analysis." *Management Science* 9(2), 277–293. — The single-index model, and the first argument that analysts should model a covariance matrix and not measure it.
- **Merton, R. C. (1980).** "On Estimating the Expected Return on the Market." *Journal of Financial Economics* 8(4), 323–361. — Explains why expected returns are hard to estimate and variances are not. The precision of the mean depends on calendar span, and the precision of the variance depends on sampling frequency. It is the most important single paper for understanding §2.5.
- **Stevens, G. V. G. (1998).** "On the Inverse of the Covariance Matrix in Portfolio Analysis." *Journal of Finance* 53(5), 1821–1827. — Few people read it, and §5.4 rests on it. Σ^{-1} decomposes into the coefficients and residual variance from regressing each asset on all the others. The link is the Federal Reserve working-paper version.

4.2 The estimation-error critique

- **Jorion, P. (1986).** "Bayes-Stein Estimation for Portfolio Analysis." *Journal of Financial and Quantitative Analysis* 21(3), 279–292. [[paywalled]] — Applies shrinkage to the means, a decade before shrinkage reached the covariance matrix.
- **Michaud, R. O. (1989).** "The Markowitz Optimization Enigma: Is 'Optimized' Optimal?" *Financial Analysts Journal* 45(1), 31–42. — Names error maximisation. It is short and non-technical, and it is the best single statement of why the failure in §2 happens.
- **Best, M. J. & Grauer, R. R. (1991).** "On the Sensitivity of Mean-Variance-Efficient Portfolios to Changes in Asset Means." *Review of Financial Studies* 4(2), 315–342. [[paywalled]] — The analytical companion to Michaud. It measures how far weights move per unit of change in μ.
- **Chopra, V. K. & Ziemba, W. T. (1993).** "The Effect of Errors in Means, Variances, and Covariances on Optimal Portfolio Choice." *Journal of Portfolio Management* 19(2), 6–11. [[paywalled]] — The 22 : 2 : 1 result of §2.5. It is the most important single number in this chapter.
- **Merton (1980)**, above, belongs here as much as in §4.1.

4.3 Estimating the covariance matrix

- **Laloux, L., Cizeau, P., Bouchaud, J.-P. & Potters, M. (1999).** "Noise Dressing of Financial Correlation Matrices." *Physical Review Letters* 83(7), 1467–1470. — Four pages. It showed that most of the spectrum of a financial correlation matrix is indistinguishable from noise. Also on arXiv.
- **Ledoit, O. & Wolf, M. (2003).** "Improved Estimation of the Covariance Matrix of Stock Returns with an Application to Portfolio Selection." *Journal of Empirical Finance* 10(5), 603–621. — Shrinkage toward a single-index target, with the optimal intensity derived rather than tuned.
- **Ledoit, O. & Wolf, M. (2004).** "Honey, I Shrunk the Sample Covariance Matrix." *Journal of Portfolio Management* 30(4), 110–119. — The version written for practitioners, and the most readable entry point to the whole subject. Start here.
- **Ledoit, O. & Wolf, M. (2004).** "A Well-Conditioned Estimator for Large-Dimensional Covariance Matrices." *Journal of Multivariate Analysis* 88(2), 365–411. — The theory behind the previous two.
- **Ledoit, O. & Wolf, M. (2017).** "Nonlinear Shrinkage of the Covariance Matrix for Portfolio Selection: Markowitz Meets Goldilocks." *Review of Financial Studies* 30(12), 4349–4388. — Shrinks each eigenvalue by its own amount rather than all of them by one intensity (§6.6).
- **Ledoit, O. & Wolf, M. (2022).** "The Power of (Non-)Linear Shrinking: A Review and Guide to Covariance Matrix Estimation." *Journal of Financial Econometrics* 20(1), 187–218. — A review of 15 years of the authors' own work, with practical guidance on which variant to use when. Use it as the efficient entry point if you read only one item from this subsection.
- **Bun, J., Bouchaud, J.-P. & Potters, M. (2017).** "Cleaning Large Correlation Matrices: Tools from Random Matrix Theory." *Physics Reports* 666, 1–109. — The definitive technical review of the random-matrix literature. It is long, and the first 30 pages are the useful ones.

- **Engle, R. F. (2002).** "Dynamic Conditional Correlation: A Simple Class of Multivariate GARCH Models." *Journal of Business and Economic Statistics* 20(3), 339–350. — Time-varying correlation, estimated in two tractable steps (§6.10).
- **Fan, J., Liao, Y. & Mincheva, M. (2013).** "Large Covariance Estimation by Thresholding Principal Orthogonal Complements." *Journal of the Royal Statistical Society, Series B* 75, 603–680. — POET: a factor model plus a *thresholded* residual matrix. It recovers the within-industry correlation that a pure factor model discards (§6.9).
- **Higham, N. J. (2002).** "Computing the Nearest Correlation Matrix — a Problem from Finance." *IMA Journal of Numerical Analysis* 22(3), 329–343. — The remedy for a matrix that is not positive semi-definite. Every implementation eventually needs it (§9.5).
- **Epps, T. W. (1979).** "Comovements in Stock Prices in the Very Short Run." *Journal of the American Statistical Association* 74(366), 291–298. [[paywalled]] — Measured correlation falls as the sampling interval shrinks. This is why §9.3 recommends estimating correlations weekly and volatilities daily.

4.4 Constraints, and why they work

- **Jagannathan, R. & Ma, T. (2003).** "Risk Reduction in Large Portfolios: Why Imposing the Wrong Constraints Helps." *Journal of Finance* 58(4), 1651–1684. — The central result of the field: a no-short constraint is *exactly* a shrinkage of the covariance matrix. Read next to Ledoit–Wolf, it unifies the two literatures. The link is the NBER working paper.
- **DeMiguel, V., Garlappi, L., Nogales, F. J. & Uppal, R. (2009).** "A Generalized Approach to Portfolio Optimization: Improving Performance by Constraining Portfolio Norms." *Management Science* 55(5), 798–812. [[paywalled]] — Generalises Jagannathan–Ma. Constraining any norm of w is equivalent to a corresponding regularisation. The identity between constraints and shrinkage therefore holds for a family of constraints and not only for one special case.

4.5 Risk-based portfolios

- **Choueifaty, Y. & Coignard, Y. (2008).** "Toward Maximum Diversification." *Journal of Portfolio Management* 35(1), 40–51. — The diversification ratio, and the portfolio that maximises it. [Contested] The authors' firm sells this strategy. The mathematics is sound. The performance claims come from a party with a financial interest.
- **Maillard, S., Roncalli, T. & Teiletche, J. (2010).** "The Properties of Equally Weighted Risk Contribution Portfolios." *Journal of Portfolio Management* 36(4), 60–70. — Risk parity done properly. It gives the existence and uniqueness results, and proves that ERC lies between $1/N$ and minimum variance.
- **Clarke, R., de Silva, H. & Thorley, S. (2011).** "Minimum-Variance Portfolio Composition." *Journal of Portfolio Management* 37(2), 31–45. — An analytic characterisation of *which* assets a minimum-variance portfolio ends up holding, and why so few. It is the best explanation for a reader surprised by a concentrated solution.
- **Asness, C. S., Frazzini, A. & Pedersen, L. H. (2012).** "Leverage Aversion and Risk Parity." *Financial Analysts Journal* 68(1), 47–59. — The economic argument for why risk parity should earn anything at all. [Contested] AQR runs risk-parity products.
- **López de Prado, M. (2016).** "Building Diversified Portfolios that Outperform Out of Sample." *Journal of Portfolio Management* 42(4), 59–69. — Hierarchical risk parity: allocation with the covariance matrix, without inverting it (§7.11).
- **Meucci, A. (2009).** "Managing Diversification." *Risk*, 74–79. — The effective number of bets, which is the right measure of whether a portfolio is actually diversified (§11.4).

4.6 Costs and dynamics

- **Gârleanu, N. & Pedersen, L. H. (2013).** "Dynamic Trading with Predictable Returns and Transaction Costs." *Journal of Finance* 68(6), 2309–2340. — The closed-form multi-period solution. Its result is the most useful single fact in §10: trade partway toward a forward-looking *aim* portfolio, not toward today's target.
- **Almgren, R. & Chriss, N. (2000).** "Optimal Execution of Portfolio Transactions." *Journal of Risk* 3, 5–39. — The treatment of execution that §10 relies on. It covers how a trader executes a given trade list, as distinct from the choice of the list.
- **Constantinides, G. M. (1986).** "Capital Market Equilibrium with Transaction Costs." *Journal of Political Economy* 94(4), 842–862. [[paywalled]] — The origin of the no-trade region, and of the result that its width scales as the cube root of the cost rather than linearly (§10.4).

4.7 The critiques

A bibliography that lists only a field's successes misleads the reader. These works belong next to everything above.

- **DeMiguel, V., Garlappi, L. & Uppal, R. (2009).** "Optimal Versus Naive Diversification: How Inefficient is the 1/N Portfolio Strategy?" *Review of Financial Studies* 22(5), 1915–1953. — Fourteen rules and seven datasets, and none beats equal weighting. Read it with the correction of §2.6 in hand.
- **Kritzman, M., Page, S. & Turkington, D. (2010).** "In Defense of Optimization: The Fallacy of 1/N." *Financial Analysts Journal* 66(2), 31–39. [[paywalled]] — The rebuttal: the problem is sample means, not optimisation. This chapter finds the rebuttal persuasive (§2.6).
- **Michaud (1989)** and **Chopra & Ziemba (1993)**, above, are critiques as much as contributions.
- **Lo, A. W. (2002).** "The Statistics of Sharpe Ratios." *Financial Analysts Journal* 58(4), 36–52. [[paywalled]] — The source of the standard error in §11.5, and of the effect of serial correlation on an annualised Sharpe ratio. Read it before believing the headline number of any backtest.

4.8 Books

- **Meucci, A. (2005).** *Risk and Asset Allocation.* Springer. — The most complete single treatment of estimation and allocation together. It is demanding and worth the effort.
- **Roncalli, T. (2013).** *Introduction to Risk Parity and Budgeting.* Chapman and Hall. — The definitive treatment of risk budgeting, with the algorithms written out.
- **Grinold, R. C. & Kahn, R. N. (1999).** *Active Portfolio Management*, 2nd ed. McGraw-Hill. — The practitioner's framework: the information ratio, breadth, the fundamental law, and the transfer coefficient, which measures how much of a signal survives construction (§11.4).
- **Michaud, R. O. & Michaud, R. O. (2008).** *Efficient Asset Management*, 2nd ed. Oxford. — Resampled efficiency (§7.9), by its inventors. [Contested] The authors patented the method and sell it. The diagnosis in the first chapters is excellent regardless.

4.9 If you only read six things

Read them in this order:

1. **Michaud (1989)** — why the problem exists, in 15 readable pages.
2. **Chopra & Ziemba (1993)** — which input to worry about, quantified.
3. **Ledoit & Wolf (2004), "Honey, I Shrunk…"** — the standard repair, explained for practitioners.
4. **Jagannathan & Ma (2003)** — the result that unifies constraints and shrinkage, and retroactively justifies most of what desks were already doing.
5. **DeMiguel, Garlappi & Uppal (2009)** *then* **Kritzman, Page & Turkington (2010)** — the field's central controversy, both sides, in that order.
6. **Ledoit & Wolf (2022)** — what to actually use, from the people who built it.

The list contains nothing about optimisation algorithms. That omission reflects the thesis of §3.

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5. The mathematics

This section gives the formal basis for §2. Most textbooks omit §5.4. Read §5.4 even if you skip the rest of the section.

5.1 The two canonical problems

Almost every allocation rule in §7 is one of two optimisations, or a constrained version of one.

Problem A — maximum Sharpe (tangency). Measure μ as excess return over cash. Then

$$\max_{w} \frac{w'\mu}{\sqrt{w'\Sigma w}} \qquad \Longrightarrow \qquad w_{\text{tan}} \;\propto\; \Sigma^{-1}\mu.$$

The objective does not fix the scale, because scaling the weights does not change the Sharpe ratio. This is the separation result of §1.3, seen as a property of the objective and not of the investor.

Problem B — global minimum variance (GMV). This problem uses no return forecast at all:

$$\min_{w} \; w'\Sigma w \quad \text{s.t.} \quad \mathbf{1}'w = 1 \qquad \Longrightarrow \qquad w_{\text{gmv}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}.$$

The solution has the form $\Sigma^{-1}\mu$ with $\mu = \mathbf{1}$, normalised to sum to one. **The minimum-variance portfolio is the mean-variance portfolio for an investor who assumes every asset has the same expected return.** This is the same formula, not an approximation or an analogy. It is the first entry in the equivalence table of §8.2. It is the reason the "risk-based" methods of §7 do not form a separate family.

5.2 The frontier, and two-fund separation

Adding a target return to Problem B and solving with Lagrange multipliers gives the full efficient frontier. Define the scalars

$$a = \mathbf{1}'\Sigma^{-1}\mathbf{1}, \qquad b = \mathbf{1}'\Sigma^{-1}\mu, \qquad c = \mu'\Sigma^{-1}\mu, \qquad d = ac - b^2.$$

The minimum-variance portfolio with expected return m is

$$w(m) = \Sigma^{-1}\left[\frac{c - bm}{d}\,\mathbf{1} + \frac{am - b}{d}\,\mu\right],$$

and its variance is $\sigma^2(m) = (am^2 - 2bm + c)/d$. Two consequences matter.

The frontier is a hyperbola in (σ, m) space. Its left vertex is the GMV portfolio, with $m_{\text{gmv}} = b/a$ and $\sigma^2_{\text{gmv}} = 1/a$.

Two-fund separation. For every m, $w(m)$ combines the same two fixed vectors, $\Sigma^{-1}\mathbf{1}$ and $\Sigma^{-1}\mu$, with coefficients that are affine in m. Normalising the two vectors to sum to one turns them into the GMV portfolio $\Sigma^{-1}\mathbf{1}/a$ and the tangency portfolio $\Sigma^{-1}\mu/b$. The coefficients on these two portfolios then sum to one for every m. Every efficient portfolio is therefore a weighted average of two fixed portfolios. This explains why the realised frontier of §2.1 could be flat. The optimiser moved along a line between two portfolios. One end, $\Sigma^{-1}\widehat{\mu}$, was almost pure noise. Moving further along that line adds leverage on noise and nothing else.

The quantity $c = \mu'\Sigma^{-1}\mu$ is the square of the maximum available Sharpe ratio. Estimation error inflates it more than it inflates anything else, and $\sqrt{\widehat{c}}$ was the 5.32 of §2.1.

5.3 What the covariance matrix contributes

Separate the two jobs that Σ does before inverting anything. Analysts estimate them with very different precision. Write $\Sigma = DCD$, where D is the diagonal matrix of volatilities and C is the correlation matrix.

Component	Parameters	Estimation difficulty
Volatilities D	N	Easy. Realised volatility from daily data is accurate within weeks. It is the most forecastable quantity in finance
Correlations C	$N(N-1)/2$	Hard. Quadratically many parameters, each individually noisy, and the errors interact through the inverse

[Fact] Essentially all of the difficulty lies in C, and so does essentially all of the N^2 growth. This suggests, correctly, that the most valuable regularisation acts on the correlation matrix and leaves the volatilities alone. §9.5 turns that into a concrete implementation rule.

5.4 What Σ^{-1} actually is: the regression identity

The following result makes everything else intuitive. It is standard in multivariate statistics. Stevens (1998) brought it into portfolio theory, where it remains underused.

For each asset i, regress its return on **all the other assets**:

$$r_i \;=\; a_i + \sum_{j \neq i} \beta_{ij}\, r_j \;+\; \varepsilon_i .$$

Let $s_i^2 \;=\; \mathrm{Var}(\varepsilon_i)$ be the residual variance and R_i^2 the coefficient of determination, so that $s_i^2 \;=\; \sigma_i^2\,(1 - R_i^2)$. The regression finds *which portfolio of the other $N-1$ assets best replicates asset i, and how well it replicates it.*

The entries of the inverse covariance matrix are then exactly

$$(\Sigma^{-1})_{ii} = \frac{1}{s_i^2}, \qquad (\Sigma^{-1})_{ij} = -\,\frac{\beta_{ij}}{s_i^2} \quad (j \neq i).$$

The off-diagonal expression looks asymmetric, because it carries s_i^2 and not s_j^2. It is symmetric. The identity $\beta_{ij}/s_i^2 \;=\; \beta_{ji}/s_j^2$ holds exactly, and it restates the symmetry of Σ^{-1}.

Substituting into $w \propto \Sigma^{-1}\mu$ and collecting terms gives

$$w_i \;\propto\; \frac{1}{s_i^2}\Big(\mu_i - \sum_{j \neq i} \beta_{ij}\mu_j\Big) \;=\; \frac{\alpha_i}{s_i^2} \;=\; \boxed{\frac{\alpha_i}{\sigma_i^2\,(1 - R_i^2)}}$$

Here $\alpha_i \equiv \mu_i - \sum_{j \neq i} \beta_{ij}\mu_j$ is the **residual alpha**. It is asset i's expected return *net of what its replicating portfolio would have earned instead*. Taking expectations through the regression shows that α_i equals the intercept a_i above. It is alpha in the ordinary sense of the word, measured against a benchmark made of the rest of the same universe.

This one line reorganises the whole subject. It supports four readings.

1. The optimiser responds to expected returns net of replication, not to raw expected returns. Suppose an asset's forecast return is exactly what its replicating portfolio already delivers. Then $\alpha_i = 0$, and the asset correctly receives *zero* weight, however attractive its headline number looks. This also explains why adding a highly correlated asset to a universe can change the existing weights drastically. The addition changes every other asset's α_i and R_i^2 at once.

2. Every position is a hedged bet, sized by its own Sharpe ratio. The form α_i/s_i^2 is exactly the sizing that would apply to a single stand-alone asset with mean α_i and variance s_i^2. It is that asset's own Sharpe ratio α_i/s_i, divided once more by s_i to turn a ratio into a position size. Mean-variance optimisation therefore does nothing exotic. It hedges each asset against all the others, then sizes each hedged residual independently. This describes the optimiser much better than "it balances risk and return across the portfolio."

3. $1/(1-R_i^2)$ is the leverage amplifier, and it has no bound. As an asset becomes more replicable, $R_i^2 \to 1$ and the weight diverges. The two-asset example of §1.6 is exactly this case. Regressing one asset on the other gives $R^2 = \rho^2$, so

at $\rho = 0.99$ the amplifier is $1/(1-0.99^2) \approx 50$. It acts on a residual alpha of $\alpha_1 = 0.06 - 0.99 \times 0.05 = 0.0105$, not on the 6% headline return. That product, divided by $\sigma_1^2 = 0.04$ and normalised so that the weights sum to one, produced the 18× gross exposure.

4. The key consequence follows. An analyst estimates R_i^2 by regressing on $N-1$ regressors with T observations. Suppose asset i has *no genuine relation* to the others. Under that null hypothesis, the expected in-sample R^2 is not zero. It is

$$\mathbb{E}[\hat{R}_i^2] \;=\; \frac{N-1}{T-1} \;\approx\; q.$$

The sample therefore indicates that every asset is about q-replicable, when none of them is replicable at all. Put $\hat{R}_i^2 = q$ into the boxed formula in place of the true $R_i^2 = 0$. The denominator shrinks by a factor of $1-q$, so the weight grows by roughly $1/(1-q)$. **§5.5 obtains the same constant from the Marchenko–Pastur spectrum, and §5.6 derives it again from the Wishart inverse.**

The agreement needs an exact statement. The three results are not independent confirmations. The regression identity above is an algebraic fact about *any* positive-definite matrix. Applying it to S and applying the inverse-Wishart expectation to S are the same statement expressed two ways: one asset by asset, the other over the whole matrix. The spectral integral states the same fact a third time, in the eigenbasis. **It is one fact expressed in three ways.**

The agreement therefore adds interpretation and no corroboration. The spectral form shows *where* the bias occurs: in the small eigenvalues. The regression form shows *which assets* carry it: the replicable ones. Only the regression form gives a quantity to compute per asset, which is the diagnostic below.

The identity also shows the form of the fix. The textbook correction for an inflated R^2 is the *adjusted* R^2. It inflates $1-R^2$ by $(T-1)/(T-N)$, which is the same factor to leading order. Under the null, that brings the expected value of one minus the adjusted R^2 back to exactly 1. In this light, shrinkage (§6.5), factor models (§6.8) and position constraints (§9.6) are all ways of discounting the sample's own $\hat{R}_i^2$.

A diagnostic that needs no optimiser. [Practice] For each asset, regress its returns on the rest of the universe and record R_i^2. Sort the values in descending order. Look at any asset with $\hat{R}_i^2$ above roughly 0.9 or, more carefully, well above the q expected from noise alone. Such an asset has an unstable optimiser weight, and its position will change sharply between rebalances. The check costs a few lines of code. It finds the dangerous assets before they cause losses, and it is the cheapest diagnostic in this chapter.

5.5 Random matrix theory: what noise looks like

§2.3 asserted that sample eigenvalues spread in a predictable way. The Marchenko–Pastur law is the precise statement.

Take T independent observations of N uncorrelated series with unit variance. The true correlation matrix is then the identity, and every true eigenvalue is 1. Let $N, T \to \infty$ with $q = N/T$ fixed in $(0, 1)$. The sample eigenvalues do not converge to 1. Instead, their empirical distribution converges to a density $f(\lambda)$ supported on $[\lambda_-, \lambda_+]$, with

$$\lambda_\pm = \left(1 \pm \sqrt{q}\,\right)^2, \qquad f(\lambda) = \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{2\pi q\, \lambda}.$$

The law has three useful properties.

The law is a prediction, not a description. Theory gives the noise spectrum *before* the analyst examines the data. Anything inside $[\lambda_-, \lambda_+]$ is consistent with pure noise, and anything outside is not. This turns "how much of the correlation matrix is real?" from a judgement call into a test. Laloux and co-authors (1999) used exactly this property.

The spread is severe at realistic q. The implied condition number of a pure-noise matrix is $\kappa = \lambda_+/\lambda_- = \left((1+\sqrt{q})/(1-\sqrt{q})\right)^2$. The table evaluates it at five values of q.

q	λ_-	λ_+	κ (noise only)
0.10	0.47	1.73	3.7
$1/3$	0.18	2.49	13.9
0.50	0.086	2.91	34.0

q	λ_-	λ_+	κ (noise only)
0.80	0.011	3.59	322
0.90	0.0026	3.80	1442

Take $q = 0.9$, which is 500 names on 555 days of history, a little over two years. Noise alone then produces a condition number of 1,442 in a matrix whose true value is the identity. An optimiser inverting that matrix places its largest bets on directions whose estimated variance is smaller than the truth by a factor of nearly 400.

Real matrices separate cleanly. Financial correlation matrices have a few eigenvalues far outside the bulk: the market mode, then the sector modes. Once an analyst removes the outliers' share of the trace, the bulk matches the law. That is the right-hand panel of the figure in §2.3, and it justifies discarding the bulk (§6.7).

The law and the inverse. The optimiser uses $1/\lambda$, not λ (§2.3). Integrating the law against $1/\lambda$ gives a closed form:

$$\int_{\lambda_-}^{\lambda_+} \frac{f(\lambda)}{\lambda}\,\mathrm{d}\lambda \;=\; \frac{1}{1-q}.$$

The average eigenvalue of a pure-noise correlation matrix is exactly 1, and the average of its reciprocals is $1/(1-q)$. That is the whole bias, taken directly from the spectrum. §5.6 derives the same constant by a different route.

Two numbers are easy to confuse here, because they differ a lot. $1/(1-q)$ is an **average** over the spectrum. At $q = 1/3$ it is 1.5, a 50% inflation. The *worst single direction* is a different and much larger quantity, $1/\lambda_- = (1-\sqrt{q})^{-2}$, which is 5.6 at the same q. The 14:1 leverage spread of §2.3 concerns the extremes. The $1/(1-q)$ of §2.4 and §5.6 concerns the average. Both follow from the same spreading. A portfolio concentrated in the lowest-variance direction suffers the extreme, and a diversified one suffers something closer to the average. That is why the ratio of §5.6 and the measured 1.155 of §9.8 are mild compared with the alarming figure in §2.3.

5.6 Why the minimum-variance portfolio misstates its own risk

This subsection derives the $(1-q)$ results quoted in §2.4. Let S be the sample covariance from T observations, so that $TS \sim W_N(\Sigma, T)$. This is a Wishart matrix with T degrees of freedom. That count assumes a known mean. Subtracting an estimated mean costs one degree of freedom and changes nothing below. The key fact is the inverse-Wishart expectation, valid whenever $T > N + 1$:

$$\mathbb{E}\big[S^{-1}\big] \;=\; \frac{T}{T-N-1}\,\Sigma^{-1}.$$

The inverse is biased upward, by a factor of about $1/(1-q)$. The sample covariance itself is unbiased. Its inverse is biased, and the optimiser uses the inverse. Inversion is convex, so Jensen's inequality guarantees a bias. The aspect ratio sets the size of the bias.

The GMV portfolio built on S *reports* the variance $\hat{\sigma}^2_{\text{pred}} = 1/(\mathbf{1}'S^{-1}\mathbf{1})$. This is the $1/a$ of §5.2, computed from the sample. The same subsection gives $\sigma^2_{\text{opt}} = 1/(\mathbf{1}'\Sigma^{-1}\mathbf{1})$. Taking expectations in the denominator gives

$$\mathbb{E}\big[\mathbf{1}'S^{-1}\mathbf{1}\big] = \frac{T}{T-N-1}\;\mathbf{1}'\Sigma^{-1}\mathbf{1} \approx \frac{1}{1-q}\cdot\frac{1}{\sigma^2_{\text{opt}}}.$$

To leading order, therefore, $\hat{\sigma}^2_{\text{pred}} \approx (1-q)\,\sigma^2_{\text{opt}}$. The last step swaps $\mathbb{E}[1/X]$ for $1/\mathbb{E}[X]$. The swap is valid here because $\mathbf{1}'S^{-1}\mathbf{1}$ concentrates around its mean as N and T grow together. The optimiser understates its risk by the factor $1-q$.

Evaluated at the true Σ, the same weights have a realised variance above the optimum:

$$\mathbb{E}\big[\hat{w}'_{\text{gmv}}\,\Sigma\,\hat{w}_{\text{gmv}}\big] \;\approx\; \frac{\sigma^2_{\text{opt}}}{1-q}.$$

Both statements hold to leading order, in the limit of large N and T with q fixed. The simulation in the figure of §2.4 tracks them closely up to $q \approx 0.8$. Dividing one by the other gives the main result. **Realised variance exceeds predicted variance by a factor of $1/(1-q)^2$. In volatility terms, the factor is $1/(1-q)$.**

The two errors compound and do not offset, so the error is dangerous and not merely inaccurate. The optimiser takes more risk than is optimal *and* reports less risk than it takes. Both errors have the same cause. Suppose a risk report uses the same covariance matrix that built the portfolio. The report will confirm that the portfolio is safe, using exactly the numbers that made it unsafe. §11.2 turns this into a monitoring rule.

5.7 The condition number as a working diagnostic

Numerical analysis gives a bound that guides practice. For $w = \Sigma^{-1}\mu$, a relative perturbation of size ϵ in Σ changes w by a relative amount of up to

$$\frac{\|\Delta w\|}{\|w\|} \lesssim \kappa(\Sigma)\,\epsilon, \qquad \kappa(\Sigma) = \frac{\lambda_1}{\lambda_N}.$$

The condition number is the amplification factor from input error to output error. [Practice] The table gives the rules of thumb.

κ	Interpretation
< 100	Comfortable. Weights are stable under resampling
10^2–10^4	Workable, but regularise, and expect visible turnover
10^4–10^8	Weights are dominated by the smallest eigenvalues. Do not trust them
$> 10^8$	Approaching double-precision limits. The inverse is numerically meaningless

Computing κ costs one `eigvalsh` call and needs no forecast. It therefore belongs in production monitoring next to q. A jump in κ between rebalances warns that two assets in the universe have become near-duplicates. This happens routinely. A merger announcement, the convergence of two share classes, or a market that stops trading and leaves stale prices can each cause it (§9.3).

§5 Key takeaways

1. Minimum variance *is* mean-variance with $\mu = \mathbf{1}$. Risk-based portfolios form no separate family. Each combines a choice of μ with a choice of how much structure to impose on Σ.
2. Every efficient portfolio mixes two fixed portfolios. Moving along the frontier moves the portfolio toward $\Sigma^{-1}\hat{\mu}$, the noisiest quantity in the problem.
3. The regression identity gives the weights: $w_i \propto \alpha_i/[\sigma_i^2(1-R_i^2)]$. Here α_i is expected return net of replication, and R_i^2 measures how replicable the asset is.
4. Mean-variance optimisation does two things. It hedges each asset against all the others, then sizes each hedged residual by its own Sharpe ratio.
5. Overfitting alone inflates the in-sample R_i^2 to about q, which inflates the weights by $1/(1-q)$. Random-matrix theory gives the same constant. The two routes describe the same effect.
6. The sample covariance is unbiased. Its inverse is biased up by a factor of about $1/(1-q)$. The optimiser uses the inverse.
7. The two errors compound and never offset. They have the same cause: the portfolio takes more risk than optimal and reports less than it takes. Never validate a portfolio's risk with the matrix that built it.
8. Production monitoring needs two numbers, the aspect ratio q and the condition number κ. Each takes one line of code.

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6. Estimating the covariance matrix

6.1 The one axis that organises this section

The dozen named estimators in this section share one idea:

Every estimator sets how much structure to impose in exchange for less estimation error.

The sample covariance sits at one end. It has no structure, no bias and maximal variance. A single number applied to everything sits at the other end. It has total structure, large bias and no variance. Every useful estimator lies between the two. The choice is a bias-variance trade-off. Because of §2, the variance term is almost always the binding one.

Estimators differ in *where* they apply structure. Recall the decomposition $\hat{\Sigma} = V\Lambda V'$. The eigenvectors V give the risk directions, and the eigenvalues Λ give the size of each one. This gives three families.

Family	What it modifies	Members
Rotationally invariant	Eigen*values* only; keeps the sample's eigenvectors	Nonlinear shrinkage, RMT clipping, linear shrinkage to a scaled identity
Structure-imposing	Eigen*vectors* too; asserts the sources of risk	Factor models, constant correlation, hierarchical, and linear shrinkage toward any of them
Reweighting	Which observations count	EWMA, DCC, and any window choice

Practitioners routinely combine the families. A factor model estimated on EWMA-weighted data with shrunk residuals uses all three. §8.3 orders the choices by how much they matter.

Each estimator below has the same six fields: intuition, definition, assumptions, cost, failure modes, and when to prefer it. The failure-modes field has the most practical value, so it gets the most space.

6.2 Sample covariance

Intuition. Measure what happened, and impose nothing.

Definition. $S = \frac{1}{T-1}\sum_{t=1}^{T}(r_t - \bar{r})(r_t - \bar{r})'$.

Assumptions. Returns are IID with finite fourth moments, and $T > N$ for invertibility. The process is stationary over the window.

Cost. $O(N^2T)$ to form, and $O(N^3)$ to invert or factorise.

Failure modes. The matrix is singular whenever $T \leq N$, and near-singular well before that. The table in §5.5 gives the numbers. The eigenvalue spreading of §2.3 is at its maximum here, because nothing reduces it. The estimator is highly sensitive to outliers. A single 10-sigma day enters squared and can dominate a covariance entry. The estimator also misestimates without any error message when assets trade in different time zones (§9.3).

When preferred. Use it when q is below about 0.05, which means a genuinely small universe with a long history. It also serves as the baseline for every other estimator to beat. Never use it in production above $q \approx 0.1$ without regularisation. [Practice] Compute it in every case, because most other estimators are functions of it.

6.3 Exponentially weighted (EWMA / RiskMetrics)

Intuition. Recent data are more relevant, so weight them more.

Definition. With decay $\theta \in (0,1)$, $\hat{\Sigma}_t = (1-\theta)\, r_{t-1}r_{t-1}' + \theta\, \hat{\Sigma}_{t-1}$. RiskMetrics popularised $\theta = 0.94$ for daily data.

Assumptions. Covariance drifts smoothly, and the recent past is the best guide. Implicitly, a single decay rate suits every entry of the matrix.

Cost. $O(N^2)$ per update. This is the cheapest estimator here. For that reason it appears in every real-time risk system.

Failure modes. Practitioners repeatedly misjudge the sample size of this estimator. The effective sample size of an EWMA with decay θ is the number of equally weighted observations that would carry the same estimation variance. It is

$$T_{\text{eff}} = \frac{1+\theta}{1-\theta}.$$

At the standard $\theta = 0.94$, the effective sample size is **32 observations**. A RiskMetrics covariance matrix on a universe of 100 assets therefore has an effective aspect ratio of $q_{\text{eff}} = 100/32 = 3.1$. That aspect ratio makes the matrix effectively singular, however many years of history it receives. [Practice] The matrix still inverts, because the arithmetic runs on all T rows. But the result is numerically meaningless, and the optimiser uses it anyway. **Compute N/T_{eff}, not N/T.** A universe of 100 names needs $\theta \gtrsim 0.995$, or $T_{\text{eff}} \approx 400$, before the matrix is usable for optimisation at all.

When preferred. EWMA works well for volatilities. Responsiveness genuinely helps there, and N parameters are cheap to estimate. EWMA is dangerous for correlations at any realistic N. [Practice] A sensible hybrid is common and rarely written down. Apply EWMA with a short decay to the volatilities. Estimate the correlation matrix on a much longer window. Then recombine them as $\hat{\Sigma} = \hat{D}\hat{C}\hat{D}$.

6.4 Constant correlation

Intuition. Estimate one correlation number instead of $N(N-1)/2$ of them.

Definition. $\hat{C}_{ij} = \bar{\rho}$ for $i \neq j$, where $\bar{\rho}$ is the average sample pairwise correlation, and $\hat{C}_{ii} = 1$. Then $\hat{\Sigma} = \hat{D}\hat{C}\hat{D}$, with sample volatilities on the diagonal.

Assumptions. All pairs are equally related. The assumption is plainly false, and the estimator makes it deliberately.

Cost. Trivial. The inverse has a closed form through Sherman–Morrison and never needs a decomposition.

Failure modes. The estimator discards all sector and industry structure. For equities, that structure is a large part of the true covariance, and strategies can trade on it. The estimator destroys any strategy whose edge comes from relative value within a sector.

When preferred. Rarely on its own. It is the single most useful *shrinkage target* (§6.5), and it is a surprisingly strong benchmark. Ledoit and Wolf's own work returns repeatedly to two targets, constant correlation and single index. Both capture the dominant market mode with almost no parameters.

6.5 Linear shrinkage (Ledoit–Wolf)

Intuition. The sample matrix is unbiased but noisy. A structured target is biased but stable. Take a weighted average of the two, and derive the weight rather than guessing it.

Definition.

$$\hat{\Sigma}_{\text{LW}} = \delta\,\Phi + (1-\delta)\,S, \qquad \delta \in [0,1].$$

Here Φ is a target with few parameters. Common choices are the identity scaled to the average variance, the constant-correlation matrix and a single-index matrix. Ledoit & Wolf (2003, 2004) derive the δ that minimises the expected squared Frobenius distance to the true Σ. The formula is closed-form and uses only the data. Roughly, $\delta^\star$ grows with the sampling noise in S and shrinks with the target's bias. It therefore rises with q automatically.

Assumptions. The chosen target is a reasonable central tendency. The optimal δ minimises a quadratic loss on the matrix. In practice the loss that matters is portfolio variance, which is a different loss. The failure modes explain the consequence.

Cost. $O(N^2T)$, dominated by forming S. `scikit-learn` provides `LedoitWolf` and `OAS`. Without it, the estimator takes a handful of lines to write.

Failure modes. The optimal intensity is optimal for *matrix* estimation. It is not optimal for *portfolio* performance, because the two objectives differ. Portfolio variance weights the small-eigenvalue directions far more heavily than Frobenius loss does. [Contested] In practice the Frobenius-optimal δ is usually somewhat too small for portfolio use, and shrinking harder often does better out of sample. Shrinking toward the identity also distorts the volatility structure, because it pulls up an asset with genuinely low volatility. For that reason a target in correlation space is generally better (§9.5). Finally, a single δ applies one affine map to the whole spectrum. §5.5 says the distortion is not uniform. The largest eigenvalue needs almost no correction, and the smallest needs a great deal. §6.6 fixes exactly this.

When preferred. [Practice] **Recommendation: this is the default.** If the covariance matrix receives only one treatment, use this one. The method is well tested and cheap, and it captures most of the available improvement. The method derives its

intensity instead of guessing it, but the user can still set a higher value. Treat $\delta^\star$ as a floor, not a final answer, for the reason given under failure modes. §9.8 shows what raising it buys. Every later estimator in this section is a refinement with a worse ratio of effort to benefit.

6.6 Nonlinear shrinkage

Intuition. §6.5 applies one correction to all eigenvalues. But the bias depends on the position in the spectrum. So give each eigenvalue its own correction.

Definition. Keep the sample eigenvectors, and replace each sample eigenvalue λ_i with a corrected value $\tilde{\lambda}_i$, so that $\hat{\Sigma} = \sum_i \tilde{\lambda}_i \, v_i v_i'$. The method estimates the map $\lambda_i \mapsto \tilde{\lambda}_i$ nonparametrically from the spectrum itself. The map targets the oracle that minimises loss given the sample eigenvectors (Ledoit & Wolf, 2017). The shape is intuitive. Large eigenvalues barely move. Those in the middle of the spectrum shrink toward the mean. The smallest eigenvalues rise substantially. This reverses the downward bias that §2.3 identified as the cause of the damage.

Assumptions. The sample eigenvectors are worth keeping. Random-matrix theory supports this for the bulk, but not for the leading directions in small samples. N and T are large, with q fixed.

Cost. $O(N^3)$ for the eigendecomposition, plus a numerical inversion of the Marchenko–Pastur relation. Implementations exist in R (`nlshrink`) and Python. Writing one from scratch is a substantial project.

Failure modes. The spectral estimation needs a reasonably large N to be meaningful. Below roughly 50 assets the estimate rests on too few eigenvalues, and linear shrinkage does as well. The theory assumes IID observations, so the method is as sensitive to non-stationarity as every other estimator. [Contested] Its out-of-sample advantage over well-tuned linear shrinkage is real but modest in most published comparisons. The advantage can vanish once the backtest includes transaction costs.

When preferred. Use it for large universes ($N \gtrsim 100$) with q between roughly 0.2 and 1. Use it when the user optimises the portfolio and does not merely report its risk, and when a few basis points of variance reduction justify the implementation effort. Ledoit and Wolf's own 2022 review guides the choice of variant.

6.7 Random-matrix eigenvalue clipping

Intuition. §5.5 identifies exactly which eigenvalues are consistent with pure noise. Those eigenvalues carry no information. Treat them as equal and replace them all with their average.

Definition. Compute the eigenvalues of the sample correlation matrix. Fit the Marchenko–Pastur edge λ_+, allowing for the outlier eigenvalues that absorb part of the trace. The figure in §2.3 uses this iterative fit. Keep every $\lambda_i > \lambda_+$ unchanged. Replace all the others by their common mean, which preserves the trace. Rebuild the matrix from the same eigenvectors.

Assumptions. Noise within the bulk is IID across assets and time. The sample estimates the eigenvectors of the retained directions well.

Cost. One eigendecomposition, $O(N^3)$, plus a short fitting loop.

Failure modes. The hard threshold is arbitrary at the margin. The method treats an eigenvalue just inside the edge as pure noise and one just outside as pure signal. The two are nearly indistinguishable. Nonlinear shrinkage (§6.6) is the smooth version, and it generally dominates clipping. Fat tails inflate the empirical edge, so a naive fit clips too much during turbulent periods. [Practice] Clipping also destroys genuine structure in the small eigenvalues, such as tightly cointegrated pairs. That is fatal if the strategy trades that structure.

When preferred. Use it when transparency matters. Clipping reports *how many* real risk factors the data support. That is a useful diagnostic in its own right, and it is easy to explain to a risk committee. As an estimator, §6.6 has largely replaced it.

6.8 Explicit factor models

Intuition. Assets move together because they share exposures. Model the exposures, and the N^2 correlations follow from a handful of factors.

Definition. $\hat{\Sigma} = F\Omega F' + \Psi$. Here F is the $N \times K$ matrix of loadings, Ω is the $K \times K$ factor covariance and Ψ is diagonal. The loadings come from *observable* characteristics, such as sector, size, value, momentum, duration and country. That distinguishes this family from §6.9. The parameter count drops from $O(N^2)$ to $O(NK)$.

Assumptions. The factor set spans the systematic risk, and the residuals are genuinely uncorrelated. The second assumption is the one that fails.

Cost. $O(NK^2+K^3)$. The inverse has a closed form through the Sherman–Morrison–Woodbury identity and never requires an $N \times N$ decomposition. That is a large practical advantage when N runs into the thousands.

Failure modes. A diagonal Ψ is a strong claim. Two airlines have correlated residuals after every standard factor. For a pairs strategy that trades that residual, the risk model reports the position as riskless. [Fact] This is the classic way a factor risk model understates the risk of a statistical-arbitrage book. The loadings are themselves estimated, and they drift. The model can also express only K systematic risk directions. The remaining $N - K$ eigenvalues are therefore confined to the range of the idiosyncratic variances on the diagonal of Ψ. This is a very aggressive form of implicit shrinkage. It is usually helpful and occasionally badly wrong.

When preferred. Use it for large equity universes, and wherever the factor structure is economically well understood. Use it also wherever the analysis must *attribute* risk as well as measure it. Factor models explain why the portfolio is risky, and rotationally invariant estimators do not. [Practice] Factor models have been the dominant choice in institutional equity risk management for 40 years, and deservedly so.

6.9 Statistical factor models and POET

Intuition. Let the data find the factors instead of specifying them. Then repair the factor model's worst assumption.

Definition. Take the leading K principal components of S as the factors. Choose K by the Marchenko–Pastur edge (§6.7) or by an information criterion. **POET** (Fan, Liao & Mincheva, 2013) adds a refinement. After removing the K principal components, it *thresholds* the residual covariance instead of forcing it to be diagonal. It keeps residual correlations that are large enough to be real and sets the rest to zero. This recovers the within-industry structure that §6.8 discards, and it still regularises.

Assumptions. The sample estimates the leading eigenvectors well. This holds when the corresponding eigenvalues lie far outside the bulk, and fails for marginal ones. Residual correlation is sparse.

Cost. $O(N^2T + N^3)$. The thresholding step is $O(N^2)$.

Failure modes. Statistical factors have no direct interpretation, and they rotate over time. Risk attribution therefore becomes unstable, and comparisons between periods are hard. A bad choice of K is costly in both directions. Too small a K puts real risk into the "idiosyncratic" residual. Too large a K brings back the noise that the method exists to remove. The thresholding level in POET is a genuine tuning parameter.

When preferred. Use it when no credible set of characteristics exists, as for futures, crypto or a cross-asset book. Use it also when the explicit factors may miss something. POET specifically suits a strategy that trades residual relationships, because the diagonal-Ψ assumption of §6.8 would harm that strategy.

6.10 Dynamic conditional correlation

Intuition. Volatilities and correlations both move over time, but they move differently. Model them separately.

Definition. Engle (2002) works in two steps. First, fit a univariate GARCH to each asset to get $\hat{D}_t$. Second, fit an autoregressive process with few parameters to the correlation of the standardised residuals. This gives $\hat{C}_t$. Then combine them as $\hat{\Sigma}_t = \hat{D}_t \hat{C}_t \hat{D}_t$. Two parameters govern the entire correlation dynamics, whatever the value of N.

Assumptions. Every pair's correlation follows the *same* two-parameter dynamics. This "correlation targeting" restriction makes the model tractable.

Cost. N univariate GARCH fits plus a small joint optimisation. The model is feasible up to a few hundred assets. Beyond that, the composite likelihood becomes hard to work with.

Failure modes. The two-parameter restriction is severe, and the data reject it. The correlation-targeting step is biased in high dimensions. [Contested] Researchers dispute the practical advantage over a well-shrunk static estimator for *portfolio construction*, as opposed to risk forecasting. Correlation persistence is real, but the portfolio gain from tracking it is smaller than the turnover it generates. Combining DCC with nonlinear shrinkage (DCC-NLS) addresses the dimension problem. It is the modern form.

When preferred. Use it when correlation *dynamics* are themselves the object of interest. Examples are tail-risk work, stress testing, and anything where the spike in crisis correlation matters. Section 12.4 of Market Regimes and Hidden Markov

Models reaches the same conclusion by a different analysis. The case is weaker when the portfolio needs a matrix only for one inversion a month.

6.11 Hierarchical and clustering-based estimators

Intuition. Assets cluster by sector, asset class and geography. Estimate the cluster structure, and impose it.

Definition. Convert the correlation matrix to a distance, $d_{ij} = \sqrt{2(1-\rho_{ij})}$, and cluster hierarchically. Then use the resulting tree to regularise, in one of two ways. Either average the correlations within and between clusters, which produces a block-structured matrix. Or use the tree for allocation directly, without ever inverting anything (§7.11).

Assumptions. A tree is a reasonable model of the dependence structure. This is a strong assumption, because real assets belong to several groupings at once.

Cost. $O(N^2 \log N)$ for the clustering.

Failure modes. Hierarchical clustering is notoriously unstable. Small perturbations to the correlation matrix can reorganise the tree entirely, and the change propagates into the weights. The linkage choice (single, average or Ward) is a free parameter with real consequences and no principled selection rule. A tree cannot represent an asset that belongs to two clusters.

When preferred. Use it when the universe has a genuine, stable and known group structure that the estimate should reflect explicitly. It works better as an ingredient in §7.11 than as a covariance estimator in its own right.

6.12 Robust estimation

Intuition. Covariance is a squared quantity, so one bad print moves it a lot. Down-weight the extremes.

Definition. Three routes exist. Winsorise or truncate returns before computing S. Use a robust scatter estimator such as the Minimum Covariance Determinant. Or fit a multivariate t and use its scatter matrix, which down-weights observations by their Mahalanobis distance.

Assumptions. Extreme observations are contamination and not information. In finance this assumption is often *false*. The crisis days are exactly the ones the risk model needs to know about.

Cost. Winsorising is free. MCD is expensive and scales badly in N.

Failure modes. When the assumption above fails, a robustified risk model may ignore exactly the days that matter. Robust estimators also interact badly with shrinkage. Both pull toward the centre, and the combined effect is easy to overdo.

When preferred. Use it for *data errors*, where the outlier is genuinely spurious. Examples are bad prints, stale quotes and mis-scaled corporate actions. [Practice] Treat this as a data-cleaning step (§9.2–§9.3), not a modelling step. Clean the errors robustly, then estimate on the cleaned series without further robustification.

6.13 Comparison

The table compares the estimators on the attributes that separate them. The ratings are this chapter's assessment, not measurements.

Estimator	Structure imposed	Cost	Needs $T > N$	Main risk
Sample	None	Low	Yes	Unusable above $q \approx 0.1$
EWMA	Recency	Very low	Effectively	Misjudged T_{eff} (§6.3)
Constant correlation	Extreme	Trivial	No	Destroys sector structure
Linear shrinkage	Moderate, tunable	Low	No	One δ for all eigenvalues
Nonlinear shrinkage	Moderate, adaptive	Medium	No	Implementation effort
RMT clipping	Moderate	Medium	No	Hard threshold; kills real spreads
Explicit factor	Strong, economic	Low	No	Diagonal Ψ is false
Statistical factor / POET	Strong, data-driven	Medium	No	K selection; uninterpretable
DCC	Dynamic	High	No	Two-parameter restriction
Hierarchical	Strong, tree	Low	No	Tree instability
Robust	Outlier down-weighting	Medium–high	Yes	Ignores the days that matter

§6 Key takeaways

1. Every estimator here sets how much structure to impose in exchange for less estimation error. Because of §2, the variance term is almost always the binding one. Err toward more structure.
2. Estimators modify eigenvalues (rotationally invariant), eigenvectors (structure-imposing) or observation weights (reweighting). The family determines what an estimator can and cannot fix.
3. Linear shrinkage is the default. It is cheap, needs no tuning, is well tested and captures most of the available gain. Apply it before anything else in this section.
4. EWMA's effective sample size is $(1+\theta)/(1-\theta)$. This is only 32 observations at the standard $\theta = 0.94$. A RiskMetrics correlation matrix is unusable for optimisation above about 30 assets, and it gives no warning.
5. Split the problem. Volatilities are easy and benefit from responsiveness. Correlations are hard and benefit from long windows. Estimate them separately, then recombine.
6. A factor model's diagonal residual assumption removes exactly what a relative-value book trades. If the edge comes from residual correlation, a standard factor risk model will report the position as riskless.
7. Robust estimation is a data-cleaning tool, not a modelling tool. In markets, the outliers are usually the information.

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7. From covariance to weights

7.1 How to read this section

Each rule below has the same six fields. One field is unusual: **what the rule implicitly assumes about μ and Σ.** Every rule here is $w \propto \hat{\Sigma}^{-1}\hat{\mu}$ for some choice, including the rules that explicitly disclaim optimisation. Naming that choice is the fastest way to see how a rule behaves when its assumptions fail. Filling in that field for all 11 rules gives the taxonomy of §8 directly.

A rule that "doesn't use expected returns" uses a *particular* vector of expected returns. It chooses that vector for statistical stability and not for plausibility. That is often a good trade. It is still a choice, and not an abstention. Practitioners who treat it as an abstention find their own portfolios surprising.

7.2 Equal weight ($1/N$)

Intuition. Estimate nothing.

Definition. $w_i = 1/N$.

Implied beliefs. All assets have identical expected returns, identical variances and identical pairwise correlations. Under those beliefs, $1/N$ is exactly the mean-variance optimum. The rule is therefore an extremely strong model, and not the absence of one.

Cost. None.

Failure modes. The rule ignores obvious differences in risk. An equal-weighted portfolio of a Treasury-bill fund and a leveraged biotech is unbalanced in risk terms. The rule concentrates risk in the most volatile assets. It requires rebalancing to maintain, and comparisons often omit that real cost. Its performance depends heavily on the universe. $1/N$ over a curated list of survivors behaves very differently from $1/N$ over all assets.

When preferred. Use it always as the benchmark that every other rule must beat. Use it as the answer when N is small, the assets are broadly comparable, and q is so large that nothing can be estimated. The result of DeMiguel and co-authors (2009) is a real result, whatever §2.6 says about its interpretation.

7.3 Inverse volatility, and inverse variance

Intuition. Give each asset the same risk budget, ignoring how the assets interact.

Definition. Inverse volatility sets $w_i \propto 1/\sigma_i$. Inverse variance sets $w_i \propto 1/\sigma_i^2$. These are different rules, and practitioners routinely confuse them.

Implied beliefs. Both rules discard correlation information. Beyond that, they differ. Inverse *variance* is the minimum-variance portfolio of uncorrelated assets ($\mu \propto \mathbf{1}$). Inverse *volatility* is the maximum-Sharpe portfolio of assets with **equal Sharpe ratios** ($\mu \propto \sigma$). This holds under a diagonal Σ, and equally under any equicorrelated one (§7.6). Inverse volatility is therefore the more aggressive of the two, because it assumes volatile assets have proportionally higher returns.

Cost. $O(N)$. The rules need only the diagonal, which is the part of Σ that the data estimate well (§5.3).

Failure modes. Ignoring correlation is a large approximation. Ten correlated European bank stocks and one gold miner receive 11 equal risk budgets, and 10 of them carry the same exposure. Volatility is also backward-looking. The rule therefore systematically underweights assets that were recently calm and are about to move.

When preferred. Use it very often, and more often than its crudeness suggests. [Practice] It is the sensible default when q is too large to trust any correlation estimate. It is also the right first implementation of almost any system. Get it working, then test whether correlations add anything.

7.4 Global minimum variance

Intuition. Make the portfolio's variance as small as possible, and do not forecast returns.

Definition. $w = \Sigma^{-1}\mathbf{1}/(\mathbf{1}'\Sigma^{-1}\mathbf{1})$, usually with a long-only or box constraint attached.

Implied beliefs. $\mu \propto \mathbf{1}$: all assets have the same expected return. This is a *strong* claim. It says that a biotech start-up and a utility have identical expected returns, which no one believes. Practitioners adopt it because §2.5 shows that a wrong but stable μ beats a noisily estimated one.

Cost. One linear solve, $O(N^3)$, or a small QP when constraints apply.

Failure modes. GMV is notoriously **concentrated**. Without constraints, it typically puts large weights on a handful of low-volatility assets and shorts others heavily. Clarke, de Silva & Thorley (2011) characterise exactly which assets receive positive weight. Broadly, these assets have a low beta to the dominant factor and low idiosyncratic volatility, and they are usually a small fraction of the universe. GMV is the rule most exposed to the $1/(1-q)$ risk understatement of §5.6, because the small-eigenvalue directions dominate it. Turnover is high without constraints.

When preferred. Use it when no return forecast exists but a usable correlation estimate does. [Fact] Long-only minimum-variance portfolios have outperformed cap-weighted benchmarks on a risk-adjusted basis over long samples in most equity markets. [Contested] Much of that outperformance comes from the low-volatility factor and not from skill in construction.

7.5 Maximum diversification

Intuition. Maximise the gap between the weighted-average volatility of the holdings and the volatility the portfolio actually realises. That gap *is* diversification.

Definition. Maximise the **diversification ratio** $\mathrm{DR}(w) = (w'\sigma)/\sqrt{w'\Sigma w}$, where σ is the vector of asset volatilities. The solution is $w \propto \Sigma^{-1}\sigma$ (Choueifaty & Coignard, 2008).

Implied beliefs. $\mu \propto \sigma$: every asset has the same Sharpe ratio. This is a genuinely defensible prior, and arguably more defensible than that of GMV. Equal Sharpe ratios are roughly what an efficient market with risk-averse investors would produce.

Cost. One solve, as for GMV.

Failure modes. The rule shares the concentration of GMV and its instability under Σ^{-1}. Adding near-duplicate assets can game the diversification ratio, because the measure rewards apparent breadth. Long-only constraints bind frequently and change the solution substantially.

When preferred. Use it when the equal-Sharpe prior is easier to accept than the equal-return prior. For a cross-asset universe it usually is. [Practice] Compute it alongside GMV. If the two disagree sharply, the disagreement shows which assets carry the portfolio's risk.

7.6 Equal risk contribution (risk parity)

Intuition. Every asset should contribute the same amount of risk. This means the same share of *portfolio* variance after accounting for correlations. Capital and stand-alone volatility need not be equal.

Definition. Asset i's marginal contribution to risk is $\partial\sigma_p/\partial w_i = (\Sigma w)_i/\sigma_p$, so its total contribution is $\mathrm{RC}_i = w_i(\Sigma w)_i/\sigma_p$. The ERC portfolio solves $\mathrm{RC}_i = \mathrm{RC}_j$ for all i, j, subject to $\mathbf{1}'w = 1$ and $w \geq 0$. In general no closed form exists, so a numerical solver finds the portfolio. Maillard, Roncalli & Teiletche (2010) prove existence and uniqueness for the long-only case when Σ is positive definite. For that reason ERC still needs $q < 1$, even though it never inverts anything.

Implied beliefs. Equal Sharpe ratios *and* equal pairwise correlations. Under exactly those two conditions, ERC is the tangency portfolio. Under equicorrelation alone, it reduces to inverse volatility (§7.3).

Cost. An iterative solve. Cyclical coordinate descent converges reliably. The cost exceeds that of a single linear solve, but not by much.

Failure modes. The volatility of ERC always lies between that of the minimum-variance portfolio and that of $1/N$. This guarantee is useful, and it also caps how much ERC can help. ERC never shorts and never concentrates, so it is *robust*. The robustness comes from ignoring most of the information in Σ. [Contested] The levered version scales risk parity up to equity-like volatility. Its results depend heavily on the bond leg and on borrowing costs, as is well known.

When preferred. Use it for multi-asset allocation, where the assets have genuinely different volatilities and the answer must be defensible, explainable and low in turnover. [Practice] ERC behaves best of the risk-based rules in production. Of these rules, it is also the easiest to explain to someone who will not read §5.

7.7 Mean-variance with an explicit μ

Intuition. A return forecast exists, so use it.

Definition. $w \propto \hat{\Sigma}^{-1}\hat{\mu}$, subject to constraints and scaled to a volatility target.

Implied beliefs. $\hat{\mu}$ is good enough to overcome the 11-fold penalty of §2.5.

Cost. One solve.

Failure modes. Every failure in §2 applies. Specifically, the portfolio concentrates on whichever assets have the most extreme $\hat{\mu}$, and those assets disproportionately carry the largest forecast errors. The damage scales with the *dispersion* of $\hat{\mu}$, not its accuracy. A signal that ranks correctly but is badly scaled will still wreck the portfolio.

When preferred. Use it when a real signal exists and out-of-sample tests have validated it. Even then, [Practice] the signal should almost always enter in a *moderated* form. Rank-transform it, winsorise it, scale it to a sensible dispersion and shrink it toward zero. A common industry practice converts a signal to cross-sectional z-scores, clips at ±2 or ±3, and multiplies by a modest assumed information coefficient. The practice is defensible, despite looking crude. The optimiser exploits any error of scale left in the input.

7.8 Black–Litterman

Intuition. Start from the returns implied by the market's own weights, and move away from them only where a view exists. This avoids forecasting returns from scratch.

Definition. Reverse-optimise the market portfolio to get equilibrium returns $\Pi = \gamma \, \Sigma \, w_{\text{mkt}}$. These are the μ for which the observed market portfolio is optimal. Then combine Π with explicit views through a Bayesian update. The views consist of a matrix P that picks out portfolios, a vector Q of expected returns on those portfolios, and a confidence matrix. The update produces a posterior μ_{BL}, which enters the ordinary master form.

Implied beliefs. The market portfolio is a sensible prior. This statement about market efficiency is the model's real content.

Cost. Trivial arithmetic. The only difficulty is specifying the views and confidences.

Failure modes. [Practice] The confidence parameters are the model, and no principled way exists to set them. In practice users tune them until the output looks acceptable. The method requires a defensible market portfolio. One exists for global equities, and one barely exists for a futures book. The formalism is often more elaborate than the underlying idea warrants.

When preferred. Use it when a natural equilibrium reference exists, together with a small number of well-articulated views on specific portfolios and not a full cross-sectional forecast. Its main lasting contribution is conceptual. **Shrink views toward a defensible prior.** This is the advice of §2.5 in formal form.

7.9 Resampled efficiency

Intuition. μ and Σ are unknown, and only one draw of them is available. So simulate many draws, optimise each one, and average the resulting weights.

Definition. Resample returns, by bootstrap or parametrically. For each sample, re-estimate $\hat{\mu}$ and $\hat{\Sigma}$ and re-optimise. Average the weight vectors across samples. Michaud (1989) proposed the method.

Implied beliefs. Averaging over the sampling distribution approximates the decision a Bayesian would make. This belief is heuristic rather than derived.

Cost. B optimisations for B resamples. This is the most expensive rule here, and it parallelises trivially.

Failure modes. [Contested] The method has no formal justification of optimality, and the averaged portfolio is not optimal for any coherent set of beliefs. Averaging weights that were individually extreme produces a less extreme portfolio. That is the whole benefit, and shrinkage achieves the same effect more cheaply and more transparently. The method also inherits any bias in the original estimates, because resampling from a bad estimate reproduces the bad estimate. A patent covers the method, which has limited its adoption.

When preferred. [Practice] Other methods have mostly replaced it. The idea remains valuable as a *diagnostic* and not as an estimator. Resample the inputs, re-optimise, and look at the spread of the resulting weights. Suppose asset 12's weight ranges from −40% to +60% across resamples. That range matters, and a point estimate does not reveal it.

7.10 Norm-constrained and regularised optimisation

Intuition. Solve the ordinary problem, but forbid extreme answers.

Definition. Minimise $w'\hat{\Sigma}w$ subject to $\|w\|_1 \leq c$ or $\|w\|_2 \leq c$, or add the norm as a penalty. The ℓ_1 constraint limits gross exposure and induces sparsity. The ℓ_2 constraint spreads weight out. DeMiguel, Garlappi, Nogales & Uppal (2009) show that these constraints nest $1/N$, minimum variance and shrinkage as special cases of one family.

Implied beliefs. The prior is that weights are small. This is the same prior as ridge or lasso regularisation in regression. Through §8.2, it is also the same as shrinking $\hat{\Sigma}$.

Cost. A convex programme. The ℓ_2 version is a QP. The ℓ_1 version becomes a QP once the formulation splits w into positive and negative parts. That split makes the constraint linear. Both are fast at realistic N.

Failure modes. c is a tuning parameter, so choose it out of sample. Otherwise the overfitting simply moves to c. The ℓ_1 constraint produces sparse portfolios. Sparsity sounds appealing, but the solution then jumps discontinuously as the inputs change, which raises turnover.

When preferred. Use it when explicit, explainable control over gross exposure and position size is required. Institutions usually require it. [Practice] Regularisation most often takes this form in production. A leverage limit is easier to defend to a risk committee than a shrinkage intensity, even though §8.2 shows that the two are the same thing.

7.11 Hierarchical risk parity

Intuition. The problem is the matrix inverse, so invert nothing. Use the correlation matrix to build a tree, then split capital down the tree.

Definition. The method has three steps (López de Prado, 2016). First, convert correlations to distances and cluster hierarchically. Second, reorder the covariance matrix so that similar assets are adjacent, which is called quasi-diagonalisation. Third, bisect recursively, splitting capital between the two halves in inverse proportion to each half's variance.

Implied beliefs. A tree describes the dependence structure well. The method trusts $\hat{\Sigma}$ for its diagonal, for the tree ordering and for each cluster's aggregate variance, but never for its inverse.

Cost. $O(N^2 \log N)$, with no inversion. The method works when $T < N$, a real advantage that none of §7.4–§7.6 shares.

Failure modes. Tree instability (§6.11) propagates directly into the weights. The linkage method is a free parameter without a principled choice. [Contested] Studies have not uniformly replicated the claimed out-of-sample advantage over well-regularised alternatives. Several studies find HRP comparable to inverse-variance weighting once both receive the same shrinkage treatment. This is expected, because the inner step of HRP *is* inverse-variance weighting.

When preferred. Use it when $q \geq 1$ and a rule is needed anyway. HRP degrades gradually where every rule based on inversion fails outright. Use it also when the universe has a real hierarchical structure that the portfolio should respect.

7.12 Growth-optimal (Kelly)

Intuition. Maximise the long-run growth rate of wealth rather than a one-period utility.

Definition. Maximise $\mathbb{E}[\log(1 + w'r)]$. Expanding the logarithm to second order gives $w'\mu - \frac{1}{2}w'\Sigma w$. Its maximiser is exactly $w = \Sigma^{-1}\mu$, the master form with $\gamma = 1$. The approximation enters through the expansion and not through the optimisation (§8.2).

Implied beliefs. μ and Σ are known, and the objective is to maximise long-run growth.

Cost. One solve.

Failure modes. Full Kelly is far too aggressive under parameter uncertainty. Its derivation assumes known parameters, and its drawdowns are severe even when the parameters are known. [Practice] Practitioners use fractional Kelly, typically a quarter to a half. That is exactly the master form with γ between 2 and 4. Fractional Kelly is better understood as a response to estimation error than as a different theory.

When preferred. Use it as a way of *thinking* about leverage. It answers "how large can this be" and not "how large should this be". The composition question it answers is the same as that of §7.7.

7.13 Comparison

The table compares the rules on the attributes that separate them.

Rule	Uses correlations	Implied μ	Concentration risk	Works at $q > 1$
$1/N$	No	$\propto \mathbf{1}$, with $\Sigma \propto I$	None	Yes
Inverse variance	No	$\propto \mathbf{1}$	Low	Yes
Inverse volatility	No	$\propto \sigma$	Low	Yes
Minimum variance	Yes	$\propto \mathbf{1}$	**High**	No
Maximum diversification	Yes	$\propto \sigma$	**High**	No
Equal risk contribution	Yes	$\propto \sigma$, equicorrelated	Low	No
Mean-variance	Yes	Explicit forecast	**Very high**	No
Black–Litterman	Yes	Views shrunk to equilibrium	Medium	No
Resampled	Yes	Explicit, averaged	Medium	No
Norm-constrained	Yes	$\propto \mathbf{1}$ or explicit	Controlled by c	Yes
Hierarchical risk parity	Ordering and cluster variance	$\propto \mathbf{1}$ within clusters	Low	Yes
Growth-optimal (Kelly)	Yes	Explicit forecast	**Very high**	No

§7 Key takeaways

1. Every rule here is $w \propto \widehat{\Sigma}^{-1}\widehat{\mu}$. A rule that "doesn't forecast returns" has chosen a μ, usually $\mathbf{1}$ or σ, for stability rather than plausibility. That is often right, and it is still a choice.
2. $1/N$ implies identical means, variances and correlations. It makes the strongest assumptions of any rule in the section.
3. Inverse volatility and inverse variance are different rules with different implied beliefs. Inverse volatility implies equal Sharpe ratios. Inverse variance implies equal expected returns. Know which one is running.
4. Minimum variance and maximum diversification are the same formula, with $\mu \propto \mathbf{1}$ and $\mu \propto \sigma$ respectively. Both concentrate, both depend on the small-eigenvalue directions, and both need constraints in production.
5. The volatility of equal risk contribution always lies between that of minimum variance and that of $1/N$. This property makes it safe, and it also limits it.
6. The durable idea of Black–Litterman is its instruction and not its algebra. The instruction is to shrink views toward a defensible prior.
7. Resampling works better as a diagnostic than as an estimator. Resample, re-optimise and look at the spread of weights. A position whose weight ranges from −40% to +60% shows more than any point estimate.
8. Hierarchical risk parity degrades gradually at $q \geq 1$, where every rule based on inversion fails. This is its main advantage.

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8. Taxonomy and equivalences

8.1 The master form, with slots

One expression covers everything in §6 and §7. With its slots named, it is the most compact statement of the field:

$$w \;\propto\; \Big(\underbrace{\widehat{\Sigma}}_{\text{S1}} + \underbrace{\Delta}_{\text{S2}}\Big)^{-1} \underbrace{\widehat{\mu}}_{\text{S3}} \qquad \text{subject to} \quad w \in \underbrace{\mathcal{C}}_{\text{S4}}, \qquad \text{scaled to} \quad \underbrace{\sigma^{\star}}_{\text{S5}}.$$

The table describes the five slots.

Slot	What it is	Choices	Section
S1	Covariance estimator	Sample, EWMA, factor, DCC, hierarchical	§6
S2	Regulariser added to it	Shrinkage, clipping, ridge, none	§6.5–§6.7
S3	Return view	**1**, σ, an explicit forecast, a Bayesian posterior	§7
S4	Constraint set	Long-only, box, gross-exposure, turnover, cardinality	§7.10, §9.6
S5	Leverage	Risk aversion, volatility target, Kelly fraction	§1.3, §10

Δ is whatever the regulariser adds to the raw estimate. It is νI for a ridge, $\delta(\Phi - \widehat{\Sigma})$ for shrinkage toward a target Φ (§6.5), and zero for no regulariser.

A design is a point in this five-dimensional space. A method's name, such as "minimum variance", "risk parity" or "HRP", specifies some coordinates and leaves the others implicit. Two firms running "risk parity" can therefore hold different portfolios, and both can use the label correctly.

8.2 The equivalences

This half page is the most valuable in the chapter. Each row states that two things that look different are the same, and says whether the identity is exact or only approximate. **A practitioner who does not know these identities will apply the same correction three times under three names.**

Exact identities.

This	Is exactly	Because
Minimum variance	Mean-variance with $\mu \propto \mathbf{1}$	Same formula (§5.1)
Inverse variance	Minimum variance with diagonal Σ	$\Sigma^{-1}\mathbf{1}$ has entries $1/\sigma_i^2$
Inverse volatility	Max-Sharpe, diagonal Σ, $\mu \propto \sigma$	$\Sigma^{-1}\sigma$ has entries $1/\sigma_i$
Maximum diversification	Max-Sharpe with $\mu \propto \sigma$	$w \propto \Sigma^{-1}\sigma$
$1/N$	Mean-variance with $\mu \propto \mathbf{1}$ and equal variances and correlations	$C^{-1}\mathbf{1} \propto \mathbf{1}$. $\Sigma \propto I$ is the special case
Equal risk contribution	Inverse volatility, under equicorrelation	Maillard et al. (2010)
Equal risk contribution	The tangency portfolio, under equal Sharpe ratios *and* equicorrelation	Same
Ridge on Σ: $\Sigma + \nu I$	Linear shrinkage to a scaled identity	Rearrange the convex combination. The leftover positive scalar drops out of w
…which is also	An ℓ_2 penalty on w	Add $\nu\|w\|^2$ to the objective
…which is also	A Gaussian prior on w centred at zero	MAP with a zero-mean Gaussian prior of variance $\propto 1/\nu$
Long-only constraint on GMV	**GMV on a shrunk covariance matrix**	**Jagannathan & Ma (2003)**

This	Is exactly	Because
An ℓ_1 or ℓ_2 constraint on w	A corresponding shrinkage of $\hat{\Sigma}$	DeMiguel, Garlappi, Nogales & Uppal (2009), §7.10
Black–Litterman with no views	Reverse-optimised market weights	The posterior equals the prior
Any efficient portfolio	An affine combination of GMV and tangency	Two-fund separation (§5.2). The two weights sum to one but may be negative

The bold row is the most consequential result in the practical history of the field. Jagannathan & Ma (2003) solved minimum variance subject to $w \geq 0$. They showed that the Karush–Kuhn–Tucker conditions make the solution *identical* to the unconstrained solution on a modified matrix, $\tilde{\Sigma} = \Sigma - \zeta\mathbf{1}' - \mathbf{1}\zeta'$. Here $\zeta \geq 0$ holds the multipliers on the non-negativity constraints. The modification reduces the covariances of assets that the unconstrained solution would short. That is precisely shrinkage, reached by a completely different derivation.

Two practical consequences follow.

A no-short constraint is a statistical estimator, and often a good one. It does not simply cost performance, as a business restriction would. This explains an otherwise puzzling record. Constrained naive optimisation kept beating unconstrained sophisticated optimisation.

Regularisation compounds. Suppose a portfolio shrinks its covariance matrix, imposes a long-only constraint, caps position sizes and adds a turnover penalty. It has applied four shrinkages, and not one strategy plus three prudent safeguards. The combined effect is frequently a portfolio far closer to $1/N$ than anyone intended. Worse, each component looks conservative in isolation. [Practice] **Recommendation: measure the total.** Compare the final weights with the unregularised solution *and* with $1/N$, and see which is nearer.

Approximate identities.

This	Approximately equals	Caveat
Michaud resampling	Shrinkage toward a less concentrated portfolio	No formal correspondence; empirically similar
A turnover penalty	Shrinkage toward the *current* portfolio	Exact for a quadratic penalty (§10.3)
RMT eigenvalue clipping	Nonlinear shrinkage with a step function	Clipping is the hard-threshold special case
EWMA with decay θ	A rolling window of $(1+\theta)/(1-\theta)$ observations	Different weight profile, same effective sample size
Kelly	Mean-variance with $\gamma = 1$	Exact only in the small-return limit
Fractional Kelly at f	Mean-variance with $\gamma = 1/f$	Same caveat

8.3 Which slot actually matters

The taxonomy yields this ordering. It is this chapter's assessment and not a measured result, though the evidence of §2.5 strongly supports its first item.

The list ranks the five *slots*. Data quality is not a slot, because it comes before the whole formula. §9.1 ranks it above everything here, and the ranking is literal. A bug in synchronicity (§9.3) outweighs every choice below. The ordering that follows assumes clean data and asks which slot matters most.

1. S3, the return view, dominates the other four combined. Whether a forecast enters the formula, and how dispersed it is, changes outcomes more than every other slot together. The $22:2:1$ ratio of Chopra–Ziemba is the quantitative version. The binary decision *"is there a real μ or not"* is the most consequential line in the codebase.

2. S2 and S4 jointly, but only as a binary. Whether to regularise matters enormously. *Which* regulariser to pick matters much less. Linear shrinkage, eigenvalue clipping, a long-only constraint and a cap on gross exposure all buy most of the same improvement, because §8.2 shows that they are the same act. Choose the one that can be explained and monitored.

3. S5, leverage. It sets the volatility that the portfolio actually realises, which is what investors notice. It is orthogonal to composition and easy to change.

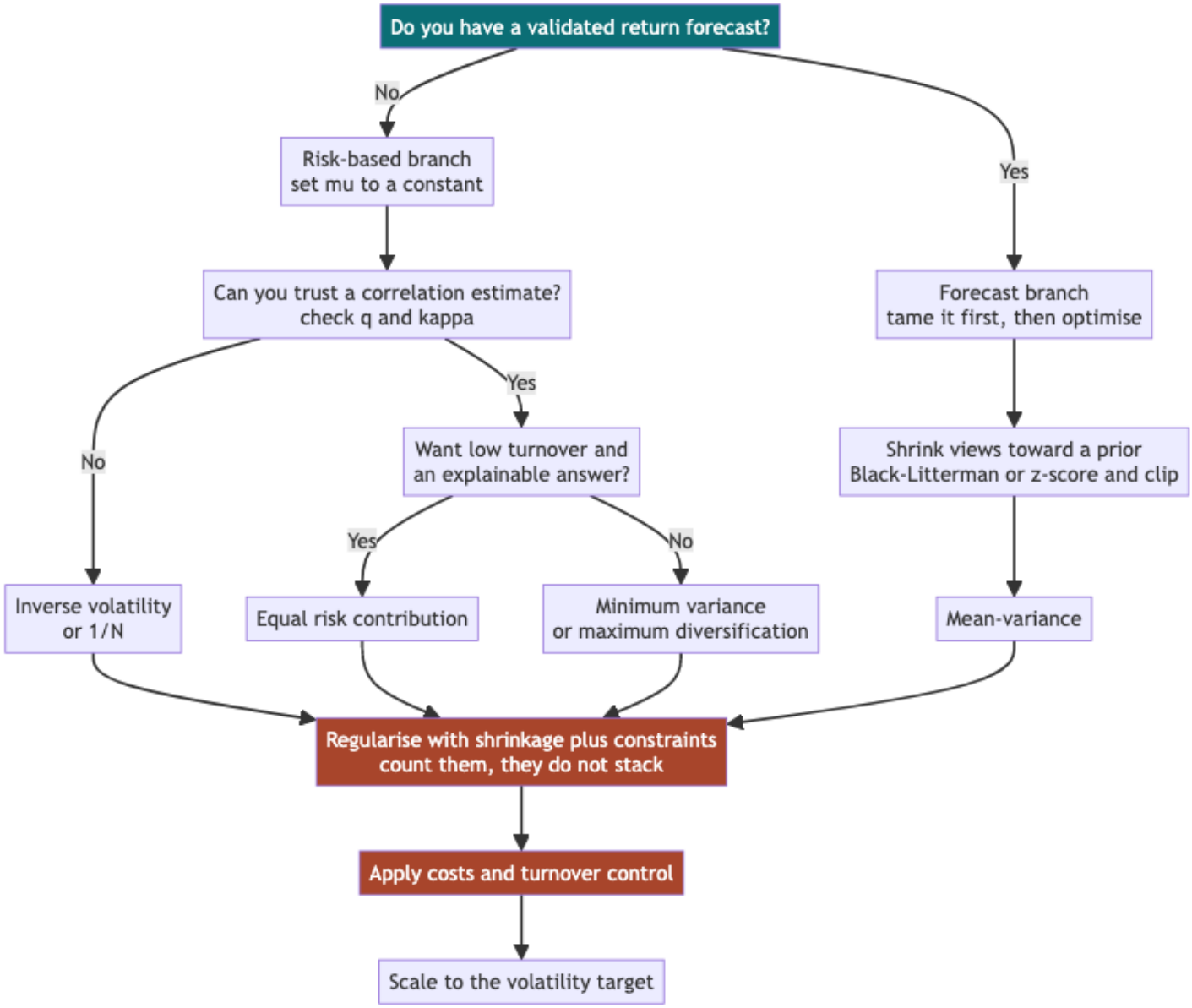
4. S1, the covariance estimator, matters least. *Conditional on some regularisation*, the choice between a shrunk sample matrix, a factor model and nonlinear shrinkage is real but second-order. It is typically worth a few percent of portfolio variance, not a multiple of it.

Point 4 is counterintuitive, and half of this chapter concerns covariance estimation.

> **The covariance estimator is the least important of the five slots. Its literature is the largest because the problem is mathematically interesting, and not because the estimator moves returns most. Suppose $\widehat{\mu}$ is a raw sample mean and gross exposure has no cap. Then choosing between nonlinear shrinkage and POET optimises the wrong slot by two orders of magnitude.**

8.4 The design space as a flowchart

The flowchart draws the design space as a sequence of decisions.



Every branch passes through the two red nodes, and practitioners most often skip those nodes. The *estimator* choice of §6 does not appear as a branch at all. It modifies the "regularise" node. This is the ordering of §8.3, in diagram form.

8.5 Same name, different thing

Several names in this field refer to two different things. The table separates them.

Name	Meaning A	Meaning B
Risk parity	Equal risk contribution: an allocation rule (§7.6)	A levered multi-asset product, typically bond-heavy
Hierarchical risk parity	López de Prado's tree algorithm (§7.11)	Often assumed to be ERC applied hierarchically. It is not, because it never equalises risk contributions
Minimum variance	The unconstrained $\Sigma^{-1}\mathbf{1}$ solution	The long-only constrained version, which is a different portfolio and a different estimator (§8.2)

Name	Meaning A	Meaning B
Shrinkage	Toward a structured *target* matrix (§6.5)	Of individual *eigenvalues* (§6.6). Different operations
Diversification	Number of holdings	The diversification ratio (§7.5), or the effective number of bets (§11.4). The three routinely disagree
Optimisation	The QP solve. Trivial, never the problem	The whole input-estimation pipeline. Hard, always the problem
Factor model	A *risk* model: explains covariance	A *return* model: predicts μ. Same word, opposite slots (S1 vs S3)

The last row causes more confusion than any other. A factor model used for risk and a factor model used for alpha share their mathematics and nothing else. One fills S1 and the other fills S3. §8.3 places those slots at opposite ends of the importance ordering.

§8 Key takeaways

1. The formula has five slots: covariance estimator, regulariser, return view, constraints and leverage. Every named method is a point in this space, and most names fix only two or three of the slots.
2. Minimum variance, maximum diversification, inverse volatility, ERC and $1/N$ are one formula. They use two choices of μ, $\mathbf{1}$ or σ, and five different amounts of structure imposed on Σ. They form no separate "risk-based" family.
3. The most important result is Jagannathan and Ma's identity. A long-only constraint *is* covariance shrinkage. Constraints are estimators, not concessions.
4. Regularisers compound and are not independent safeguards. Shrinkage plus long-only plus position caps plus a turnover penalty adds up to four shrinkages. Measure how close the result is to $1/N$. The portfolio may already be there.
5. The importance ordering is the return view, then whether to regularise at all, then leverage, and last the choice of covariance estimator.
6. The covariance literature is the largest because the problem is the most mathematically interesting, and not because covariance estimation moves returns most.
7. "Factor model" means a risk model in slot S1 and a return model in slot S3. These sit at opposite ends of the importance ordering and share only their algebra.

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9. Implementation: what actually matters

9.1 Effort ordering

Readers of this literature consistently misallocate effort, because the interesting problems and the important problems differ. The table ranks the decisions by what typically goes wrong, worst first.

Rank	Decision	Why it ranks here
1	Return data correctness — survivorship, corporate actions, synchronicity, stale prices	Corrupts everything downstream without warning, and produces *plausible* portfolios. An order of magnitude above the rest
2	Whether there is a μ at all	The 11× penalty of §2.5. A binary decision worth more than every refinement below
3	Budgeting $q = N/T$	Determines whether the problem is well-posed. Free to check, expensive to ignore
4	Applying *some* regularisation	Enormous as a binary; small as a choice among methods (§8.3)
5	Turnover control	Frequently dominates rank 4 in live P&L, and is invisible in a frictionless backtest
6	Which covariance estimator	Real, second-order, and the subject of most of the literature

The gap between rank 1 and rank 6 is larger than intuition suggests. [Practice] Switching covariance estimators does not recover what a synchronicity bug has cost.

9.2 The data layer

Define the universe as of each date, not as of today. The most common backtest error in this area builds a covariance matrix over the assets that exist now. Removing every dead ticker creates survivorship bias, and the bias makes covariance estimates look better than they are. The assets that disappeared are disproportionately the ones that moved violently.

Apply corporate actions before computing returns, not after. A missed 3-for-1 split produces a −67% return. That return enters the covariance squared and can reorder the eigenvalues on its own. [Practice] A cheap and effective guard is to flag any daily return beyond ±50% for a liquid name and inspect it, rather than winsorising it silently. A real −50% day is information, and a split artefact is a bug. Only inspection tells them apart.

Decide the return convention once. Log returns aggregate over time, and simple returns aggregate across assets. A portfolio is a cross-sectional aggregation, so portfolio weights multiply *simple* returns. Using log returns in a covariance matrix and then treating $w'r$ as the portfolio return is a small error at daily frequency and a real one at monthly frequency. The companion chapter Simple and Log Returns explains when the distinction matters.

Choose the estimation frequency deliberately. Higher frequency gives more observations. That lowers q, which §2.4 identifies as the key quantity. It also brings microstructure noise and the synchronicity problems of §9.3. [Practice] Daily data is the usual compromise. Weekly data is right when the universe spans time zones. Intraday data is right only with properly synchronised data and a reason to use it.

9.3 Non-synchronous and stale prices

This problem has its own subsection because it is the most damaging data problem that produces no error message and no obviously wrong number.

The mechanism. Consider daily close-to-close returns for a US stock and a Japanese stock. Tokyo closes roughly 14 hours before New York, so the two "daily" returns cover offset windows. News that arrives during New York hours on day t enters the US return for day t. Tokyo has already closed, so the same news enters the Japanese return for day $t + 1$. The contemporaneous correlation therefore **understates** the true relationship. Part of the co-movement appears in a lagged term that the estimator never computes.

The same thing happens without any time zones, whenever an asset trades infrequently. A bond or small-cap stock that did not trade near the close carries a stale price. Its measured return is then zero on days when its true return was not. These zeros depress measured volatility and pull measured correlation toward zero.

Why it is dangerous and not merely inaccurate. Both biases push in the same direction, *toward zero correlation and lower volatility*. In the notation of §5.4, they lower $\widehat{R}_i^2$ and $\widehat{\sigma}_i^2$, and both sit in the denominator of the weight. An asset that is illiquid or non-synchronous in measurement therefore looks like a genuine diversifier with low risk. The optimiser overweights exactly such assets.

> **Non-synchronous and stale pricing makes the optimiser systematically overweight the illiquid, the foreign and the infrequently traded. Investors can least afford to underestimate the risk of these assets, and they are the hardest to exit.** [Fact] This effect is a recognised contributor to the historical understatement of risk in portfolios that hold private, illiquid or appraisal-priced assets.

The frequency interaction. The bias worsens at higher frequency. Measured correlation between two assets falls monotonically as the sampling interval shrinks. Epps (1979) documented the effect, and the effect bears his name. Sampling the data more finely, which §9.2 recommended for lowering q, therefore makes this problem worse. The two goals genuinely trade off.

Remedies, in increasing order of effort:

1. **Lower the frequency for correlations.** Weekly returns largely absorb a one-day timing offset. Keep daily data for volatilities, where the problem is milder and the extra observations help. This is the split of §6.3 again, now for a different reason.
2. **Align to a common timestamp.** Where synchronous snapshots are available, such as 16:00 London for everything, use them instead of local closes. Futures make this easy. Cash equities make it much harder.
3. **Apply a lead–lag correction.** Estimate the covariance including one or more lagged cross-terms, and sum them. This follows the approach of the Scholes–Williams and Dimson corrections for beta. It is effective, and it costs parameters.
4. **Model staleness explicitly.** Unsmooth an appraisal-based or infrequently traded series before estimating. This is standard in private-asset work and rarely worth it elsewhere.

[Practice] A useful diagnostic counts the fraction of returns that are exactly zero for each asset. Above a few percent, a supposedly liquid instrument is stale, and its risk numbers are unreliable.

9.4 Budgeting the aspect ratio

Treat q as a budget to allocate rather than a number to discover.

$$q = \frac{N}{T_{\text{eff}}}, \qquad T_{\text{eff}} = \begin{cases} T & \text{rolling window} \\ \dfrac{1+\theta}{1-\theta} & \text{EWMA with decay } \theta \end{cases}$$

[Practice] The table gives the working targets recommended here.

q	Verdict
< 0.1	Comfortable. Sample covariance plus light shrinkage is fine
$0.1 - 0.3$	Normal operating range. Shrinkage is mandatory
$0.3 - 0.7$	Difficult. Use a factor model or nonlinear shrinkage, and constrain hard
> 0.7	Do not invert anything. Inverse volatility, HRP, or reduce N

Three levers move q, and they differ in cost.

- **Reduce N.** This is usually the most effective lever and the least popular. Group 500 stocks into 40 sector baskets, allocate across the baskets and equal-weight within them. That turns $q = 1.0$ into $q = 0.08$. The portfolio loses differentiation within sectors and gains a well-posed problem, which is often a good trade.
- **Increase T.** This is cheap, but a long window eventually breaks stationarity. [Practice] Beyond roughly five years of daily data, equity correlations average over structurally different regimes.
- **Increase the frequency.** This lowers q fastest, and §9.3 explains the cost.

9.5 Numerics

Small implementation choices here have large consequences, and most of them point the same way.

Never form the inverse. `np.linalg.inv(cov) @ mu` is slower and less accurate than `np.linalg.solve(cov, mu)`. For a symmetric positive-definite matrix, `scipy.linalg.cho_solve` on a cached Cholesky factor is better than both. At the condition numbers of §5.7, the difference matters for correctness.

Work in correlation space. Split $\hat{\Sigma} = \hat{D}\hat{C}\hat{D}$, regularise $\hat{C}$ and recombine. Shrinking Σ toward the identity pulls every *volatility* toward the average as a side effect. That is wrong, because the data estimate volatilities well (§5.3), and there is no reason to damage them. Shrinking C leaves them unchanged.

Choose the shrinkage target to match the data. [Practice] The identity is the wrong target for anything with a market factor, which is nearly everything. A constant-correlation or single-index target is nearly always better. §9.8 shows the size of the difference, which is large.

Check positive-definiteness, and fix it properly. Blends of estimators, pairwise deletion of missing data and lead–lag corrections can all produce a matrix with small negative eigenvalues. The right repair is the nearest-correlation-matrix projection of Higham (2002). The cheap approximation clips negative eigenvalues to a small floor and rescales to a unit diagonal. It is adequate when the violation is tiny. Adding ϵI until Cholesky succeeds is a crude fix that works. Log how large ϵ had to be, because a large value means an earlier step in the pipeline is faulty.

Keep units consistent, and assert it. Mixing annualised and per-period quantities is the most common numerical bug in this code. It produces results that are wrong by a factor of $\sqrt{252}$. It is easy to miss, because the portfolio still looks sensible. [Practice] Annualise once, at a boundary. Before using the covariance matrix, assert that its diagonal implies volatilities in a plausible range.

Log q and κ on every rebalance. Each takes one line (§5.7). When they drift, the weights are becoming unstable. Detect the instability before it appears in the P&L.

9.6 Constraints in practice

§8.2 established that constraints are regularisation. They are still not interchangeable, because they regularise in different directions and have different side effects, as the table shows.

Constraint	Form	What it does	Watch for
Long-only	$w \geq 0$	Strongest single regulariser available. Equivalent to a specific shrinkage	Discards genuine short alpha. Often binds on most assets
Box	$\lvert w_i \rvert \leq c$	Direct control of concentration	If it binds on many names, the cap chooses the weights and the optimiser does not
Gross exposure	$\sum_i \lvert w_i \rvert \leq L$	Controls leverage and, through it, costs	The most defensible constraint to a risk committee
Turnover	$\lVert w - w_{\text{prev}} \rVert_1 \leq \tau$	Shrinks toward the current portfolio (§10.3)	Makes the problem path-dependent. Backtest it as such
Cardinality	at most k non-zero	Limits operational complexity	Non-convex. Avoid unless it is a genuine requirement

[Practice] Two rules apply without exception. First, **count how many constraints bind at the solution.** If most positions sit at their cap, the optimiser has no effect, and the portfolio is a constrained naive portfolio. That may be acceptable, but the team should know it. Second, **always solve the unconstrained problem too**, purely as a diagnostic. The distance between the constrained and unconstrained solutions measures how much the prior shapes the result. §8.2 says to track exactly this quantity.

9.7 A reference implementation

The listing gives the whole pipeline, in the order it must execute. The code runs, and the numbers in §9.8 come from it.

```
import numpy as np
from scipy.linalg import cho_factor, cho_solve


def split_correlation(cov):
    """Separate a covariance matrix into correlations and volatilities."""
    vol = np.sqrt(np.diag(cov))
    return cov / np.outer(vol, vol), vol


def shrink_to_constant_correlation(corr, delta):
    """Pull every correlation toward the universe average. Volatilities untouched."""
    n = len(corr)
    off = ~np.eye(n, dtype=bool)
    target = np.full((n, n), corr[off].mean())
    np.fill_diagonal(target, 1.0)
    out = (1 - delta) * corr + delta * target
    np.fill_diagonal(out, 1.0)
    return out


def clip_to_psd(corr, floor=1e-8):
    """Clip negative eigenvalues, restore a unit diagonal. §9.5's cheap repair."""
    vals, vecs = np.linalg.eigh((corr + corr.T) / 2)
    if vals.min() >= floor:
        return corr
    out = vecs @ np.diag(np.clip(vals, floor, None)) @ vecs.T
    d = np.sqrt(np.diag(out))
    return out / np.outer(d, d)


def build_portfolio(returns, view=None, delta=0.3, max_weight=0.05,
                    target_vol=0.10, periods_per_year=252):
    t, n = returns.shape

    # 1. estimate, in correlation space                          # §9.5
    cov = np.cov(returns, rowvar=False) * periods_per_year
    corr, vol = split_correlation(cov)

    # 2. regularise the correlations only                        # §6.5
    corr = clip_to_psd(shrink_to_constant_correlation(corr, delta))
    cov_reg = corr * np.outer(vol, vol)

    # 3. solve. mu = 1 is minimum variance                       # §5.1
    view = np.ones(n) if view is None else view
    w = cho_solve(cho_factor(cov_reg), view)

    # 4. constrain, then scale to the risk target                # §9.6, §1.3
    w = np.clip(w / np.abs(w).sum(), -max_weight, max_weight)
    w = w * target_vol / np.sqrt(w @ cov_reg @ w)

    lam = np.linalg.eigvalsh(corr)
    return {"weights": w, "q": n / t, "gross": np.abs(w).sum(),
            "condition_number": lam[-1] / lam[0],
            "predicted_vol": np.sqrt(w @ cov_reg @ w)}
```

Five features of this listing matter most, and developers most often implement them wrongly.

The code splits the matrix into correlations and volatilities before regularising, and it regularises only the correlations. This is the rule of §9.5. It separates a shrinkage that helps from one that quietly damages the volatility estimates.

The shrinkage target is the constant-correlation matrix, not the identity. §9.8 shows the consequence of getting this wrong, and the consequence is large.

The solve uses `cho_solve` on a factorisation, not an explicit inverse. At the condition numbers of §5.7, this is a matter of correctness, not performance.

Constraining and scaling are separate steps, in that order. Clipping changes the portfolio's volatility, so the code must apply the risk target *after* the clip. Otherwise the portfolio misses the target. Reversing these two lines is a common bug, and it produces a portfolio that runs persistently below target. This order also determines what `max_weight` caps. It caps the gross-normalised *shape*, before the code applies leverage. A position at the cap ends up at `max_weight` times gross exposure. The 5% cap above, on a book running a gross exposure of 3.6, is therefore an 18% position. A hard cap on the weights actually held is a different constraint, and the code must impose it after the scaling.

q and κ come out with the weights. Log these diagnostics on every rebalance. Do not compute them after something has already gone wrong. The code measures κ on the regularised correlation matrix, which is the matrix that the shrinkage acts on.

The listing omits turnover control and costs, because they belong to the next section. This function returns a *target*, and §10 explains why a portfolio should usually not trade all the way to it.

9.8 What the shrinkage choice actually buys

The experiment runs the code above on simulated one-factor data and sweeps the shrinkage intensity. The data have $N = 120$ and $T = 750$ daily observations, so $q = 0.16$. The identity rows replace the constant-correlation target with I and change nothing else. The experiment computes realised volatility against the true covariance matrix. The simulation knows this matrix and the estimator does not. The numbers come from a simulation run for this chapter. Step 4 of §9.7 scales every portfolio to the target, so the predicted-volatility column is 10.0% by construction. Only the *realised* column and the ratio are informative.

Target	δ	$\kappa(C)$	Gross	Predicted vol	Realised vol	Realised ÷ predicted
—	0.00	178	4.35	10.0%	11.6%	1.155
Constant corr.	0.10	146	3.99	10.0%	10.7%	1.067
Constant corr.	0.30	108	3.60	10.0%	10.0%	**0.999**
Constant corr.	0.50	85	3.38	10.0%	9.9%	0.986
Constant corr.	0.80	64	3.20	10.0%	10.7%	1.070
Identity	0.30	62	3.38	10.0%	10.6%	1.064
Identity	0.80	**10**	2.18	10.0%	**17.8%**	**1.775**

The table supports four readings. The fourth matters most.

The main effect of shrinkage is an accurate risk forecast, not lower risk. At $\delta = 0.3$, the ratio of realised to predicted volatility is 0.999. The unshrunk portfolio took 16% more risk than it reported. This is consistent with the $1/(1-q) = 1.19$ of §5.6. Shrinkage removed essentially all of that gap.

Gross exposure falls monotonically, from 4.35 to 3.20. Costs scale with turnover. Turnover scales with gross exposure. So shrinkage has a large effect on P&L that the volatility column does not show.

Over-shrinking has a mild cost. At $\delta = 0.8$ the target's bias starts to affect the result, but the ratio only rises back to 1.07. The loss function is distinctly asymmetric. Too little shrinkage is much worse than too much. This is a good reason to err toward higher shrinkage.

The identity target gives a misleading condition number. At $\delta = 0.8$, the identity target produces the best-conditioned matrix in the table by a wide margin. Its condition number is $\kappa = 10$, against 64 for the comparable constant-correlation row. It also produces the *worst* risk forecast, understating realised volatility by 44%. This is worse than no shrinkage at all. A practitioner who monitors the condition number as a proxy for matrix quality would pick this row.

> [Practice] **The condition number shows whether a matrix is numerically invertible. It does not show whether the answer is right.** Shrinking toward a badly chosen target improves κ and destroys the structure that made the estimate useful. Monitoring κ does not reveal the damage. The only check that catches it compares realised with predicted risk, measured out of sample (§11.2).

§9 Key takeaways

1. Spend effort in this order: data correctness, whether a μ exists, q, whether to regularise at all, turnover, and last the choice of estimator. The literature emphasises roughly the reverse order.
2. Non-synchronous and stale prices bias correlations and volatilities *down*. This makes illiquid and foreign assets look like diversifiers. The optimiser then overweights the positions that are hardest to exit.
3. Sampling more finely lowers q and worsens the Epps attenuation. Resolve the conflict by splitting the estimates. Use daily data for volatilities and weekly data for correlations.
4. Regularise in correlation space. Shrinking Σ toward I damages volatility estimates that were not in doubt.
5. Never form an explicit inverse. Factor once and solve. Apply constraints before scaling to the volatility target, not after.
6. The main benefit of shrinkage is an accurate risk forecast, not lower risk. In the worked example it moved the ratio of realised to predicted risk from 1.16 to 1.00. It also cut gross exposure by 17%.
7. Err toward over-shrinking. The loss is markedly asymmetric. Too little shrinkage is far worse than too much.
8. A good condition number does not show that a matrix is good. The best-conditioned row of the table in §9.8 has the worst risk forecast.

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10. Costs, turnover, and rebalancing

10.1 Why turnover is the overlooked variable

Every section so far has treated portfolio construction as a one-shot problem. Given inputs, it produces weights. In production, portfolio construction is a *sequence* of such problems. The cost of moving links them.

Repetition changes the effect of the instability of §2. In a one-shot setting, an optimiser that produces very different weights from slightly different inputs is unreliable. In a repeated setting, the same instability *is* turnover. Turnover is a direct, compounding cost that the trader cannot recover. §2 and §5 characterised the sensitivity as a statistical problem. In the P&L it appears as trading costs.

> **Weight instability and transaction costs are one problem, measured in two units.** Anything that stabilises weights reduces costs. Examples are shrinkage, constraints, longer windows and fewer assets. Anything that reduces costs does so by stabilising weights.

For this reason the gross-exposure column of §9.8 mattered as much as its volatility column. For the same reason, a frictionless backtest systematically overstates the value of better estimation. The methods that look best without costs are typically the ones that trade most.

10.2 Where costs come from

Costs have three components, and each scales differently. Getting the scaling right matters more than getting the levels right. The levels are specific to each instrument, and the scaling determines what the optimiser does.

Component	Scales with	Typical magnitude
Spread and fees	Trade size, linearly	1–5 bp for liquid futures and large caps, and far more elsewhere †
Temporary impact	Roughly the square root of participation rate	The dominant term for institutional size
Permanent impact	Trade size, linearly	Hard to measure. Often folded into the above

† [Practice] Those magnitudes are indicative figures, not measurements. A desk's own fills are the only reliable source.

[Fact] The square-root *form* of impact is one of the more robust empirical regularities in market microstructure. Cost per share rises roughly with the square root of the fraction of daily volume traded, across venues, asset classes and decades. [Contested] Researchers debate the exponent itself, with estimates spanning roughly 0.4 to 0.6. They agree on the concavity. They dispute the precise power.

The useful simplification for portfolio construction is a **quadratic** total cost, $\frac{1}{2}(w-w_{\text{prev}})'\Lambda_c(w-w_{\text{prev}})$. Here Λ_c is diagonal, and its entries are inversely related to liquidity. The quadratic is not the true cost function. Square-root impact means total cost grows as size$^{3/2}$, not size^2. But the quadratic form makes the problem solvable in closed form. §10.3 shows that the resulting policy is qualitatively right.

10.3 Costs are the third regulariser

Adding a quadratic turnover penalty to the master form changes the objective to

$$\max_{w} \; w'\mu \; - \; \frac{\gamma}{2}\, w'\Sigma w \; - \; \frac{\nu}{2}\, \|w - w_{\text{prev}}\|^2.$$

This is the quadratic cost of §10.2 with $\Lambda = \nu I$, so ν scales the cost of moving. Differentiating with respect to w and setting the result to zero gives

$$\mu - \gamma\Sigma w - \nu\,(w - w_{\text{prev}}) = 0 \qquad \Longrightarrow \qquad (\gamma\Sigma + \nu I)\, w = \mu + \nu\, w_{\text{prev}},$$

so

$$\boxed{w = \left(\gamma\Sigma + \nu I\right)^{-1}\left(\mu + \nu\, w_{\text{prev}}\right)}$$

The cost term does two things. On the left, it adds νI to $\gamma\Sigma$. This is **exactly ridge regularisation**, with coefficient ν/γ on Σ. By the rearrangement in §8.2, it is the linear shrinkage of §6.5 toward a scaled identity, up to a positive scalar that changes only leverage. On the right, it adds $\nu\, w_{\text{prev}}$ to the return view. So one penalty enters both slot S2 and slot S3 of §8.1.

The phrase "shrinks the view toward the current portfolio" is important, so the right-hand half needs an exact statement. Substitute $\mu = \gamma\Sigma w^\star$, where $w^\star = \frac{1}{\gamma}\Sigma^{-1}\mu$ is the frictionless optimum of §1.3:

$$w \;=\; \left(\gamma\Sigma + \nu I\right)^{-1}\left(\gamma\Sigma\, w^\star + \nu\, w_{\text{prev}}\right).$$

Distributing the inverse gives two matrix coefficients that sum to the identity. So w is a matrix-weighted average of the frictionless optimum and the current position. Raising ν moves the portfolio from $w^\star$ toward w_{prev}. This is what §8.2 means by shrinkage toward the current portfolio. For a quadratic penalty it is an exact identity.

This completes the argument that §8.2 began:

> **The three main regularisers are covariance shrinkage, position constraints and transaction costs. They are the same operation, justified in three different ways.** Statistics justifies shrinkage, institutions justify constraints and economics justifies costs. All three add structure that reduces the positions taken in the small-eigenvalue directions. Their effects compound.

The practical consequence is the same warning as before, now with a third contributor. Suppose a firm shrinks its covariance matrix, imposes a long-only constraint, caps positions *and* runs a turnover penalty. It has applied four regularisations. Its weights will end up close to $1/N$, often without anyone having decided that. [Practice] The cheapest way to detect this is to compute the distance between the live weights and $1/N$ and track it. If the distance is small and falling, the extra sophistication adds nothing.

10.4 The no-trade region

The quadratic-cost result above implies moving partway toward the target. With *proportional* costs, such as a spread paid on every share regardless of size, the optimal policy has a different form. It defines a region around the current portfolio. Inside the region, the optimal action is to do nothing. Outside it, the optimal action is to trade only to the region's edge, never to the target.

[Fact] The classical asymptotic result comes from the line of work begun by Constantinides (1986). In it, the half-width of the no-trade band scales as the **cube root** of the proportional cost, not linearly with it. The non-linearity helps in practice. Quadrupling the cost estimate widens the band by only about 60%. So the policy is far less sensitive to mis-estimated costs than it appears. Approximately right costs are enough, because the band changes slowly with the cost estimate.

Two implementation forms are standard.

Buffering. Compute the target $w^\star$. Trade only if $\|w^\star - w_{\text{prev}}\|$ exceeds a threshold. Then trade to the edge of the band rather than to $w^\star$. The method is simple and robust, and it combines with any other method.

Partial adjustment. Move a fixed fraction of the way, $w = w_{\text{prev}} + \eta\,(w^\star - w_{\text{prev}})$ with $\eta \in (0, 1]$. This is the solution of §10.3 with its matrix coefficient replaced by a scalar. It is easier to reason about.

[Practice] Both forms work. The mistake that matters is to use neither. That mistake computes a fresh optimum each period and trades all the way to it. This is the default behaviour of every code sample in this literature, including the one in §9.7.

10.5 Aim at where the portfolio is going

The subtlest result in this area changes what the optimiser optimises, not only how fast the portfolio moves toward it.

Gârleanu & Pedersen (2013) solve the multi-period problem with predictable returns and quadratic costs. The optimal policy trades partway toward an **aim portfolio**. The aim portfolio is not today's Markowitz portfolio. It is a weighted average of the current optimal portfolio and all the expected future ones. The weighting systematically **discounts fast-decaying signals**.

Suppose a signal will have decayed by the time the trader has fully built the position. The trader then pays the full cost of establishing the position and captures only part of the return. A fast signal therefore needs less weight than its stand-alone Sharpe ratio implies, and a slow one needs more.

[Practice] **Signal persistence, not only signal strength, belongs in the portfolio construction step.** Two signals with identical information coefficients but different half-lives should receive different weights, and the difference can be large. Weighting each signal by its Sharpe ratio alone and applying costs afterwards gets this wrong, and no later cost adjustment repairs it.

This connects directly to the treatment of trading rate and buffering in section 8.7 of Trend-Following in Financial Markets. It is the strongest argument in this chapter for treating costs as part of the objective rather than as a later deduction.

10.6 Rebalancing policy

The table compares three rebalancing policies.

Policy	Mechanism	Verdict
Calendar	Rebalance monthly, quarterly	Simple, auditable and arbitrary. Trades on the calendar date, not when the portfolio needs it
Threshold	Rebalance when drift exceeds a band	§10.4's theory, and generally better
Cost-aware optimisation	Costs inside the objective	Best, and the only one that handles §10.5

[Practice] Two details matter more than the choice among these. First, **rebalance the risk model on a different schedule from the portfolio.** Re-estimating $\hat{\Sigma}$ daily while trading monthly is fine and often preferable. It keeps the risk view current without generating turnover. Second, **never rebalance everything on the same day.** Staggering trades across the universe reduces market impact and avoids a predictable monthly trading pattern that others can trade against.

§10 Key takeaways

1. Weight instability and transaction costs are the same problem in different units. A frictionless backtest overstates the value of better estimation, because the main benefit of better estimation is stability.
2. A quadratic turnover penalty is *exactly* ridge regularisation of Σ plus shrinkage of the view toward the current portfolio. Costs are provably a regulariser.
3. Shrinkage, constraints and costs are one operation with three justifications. Applying all three at full strength produces $1/N$, often unintentionally.
4. With proportional costs, do not trade to the target. Trade to the edge of a no-trade band whose width scales as the *cube root* of cost. That makes the policy robust to mis-estimated costs.
5. The aim portfolio is not today's optimum. Discount fast-decaying signals at the construction stage, because a position that the trader does not hold long enough to earn the signal's return still costs the full entry.
6. Signal persistence belongs in portfolio construction. Weighting signals by Sharpe ratio and applying costs afterwards gives a different and worse result.

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11. Evaluation and pitfalls

11.1 The staged tests

Test in the order of the table. Each stage is cheaper than the next, and each can fail the method outright.

Stage	Question	Cost
0	Is $q < 1$? Is $\hat{\Sigma}$ positive definite? Is κ sane?	Free
1	Does realised risk match predicted risk out of sample?	Cheap, and the highest-value test here
2	Are the weights stable — turnover, resampling spread, concentration?	Cheap
3	Does it beat $1/N$ and inverse volatility, at matched volatility?	Moderate
4	Does it survive realistic costs and constraints?	Expensive
5	Is the improvement statistically distinguishable from zero?	Requires care (§11.5)

[Practice] Most published and internal comparisons start at stage 3 and never run stages 1 and 2. This order is wrong. Stage 1 catches the failure mode that destroys portfolios. It does so without any return forecast, any backtest or any assumption about the future.

11.2 The single most important test

§5.6 established that a portfolio built on $\hat{\Sigma}$ and evaluated on $\hat{\Sigma}$ reports risk that is too low by construction. §9.8 showed a shrinkage target that made this worse while improving every other diagnostic. One test catches both problems:

Compute the ratio of realised to predicted portfolio volatility, out of sample, and track it over time.

At each rebalance, record the volatility that the risk model predicted for the portfolio actually held. Later, compute the volatility that the portfolio actually realised. The ratio should be close to 1. The table below explains how to read its value.

Ratio	Reading
≈ 1.0	The risk model is accurate. This is the goal
$1.1 - 1.3$	Normal for a lightly regularised model. Compare against $1/(1-q)$. If the ratio matches it, the regularisation has no effect
> 1.5	Something is badly wrong. The causes include q too high, a mis-chosen target (§9.8) and stale data (§9.3)
< 0.9	Over-shrunk, or the realised period was unusually calm. Less dangerous, still worth explaining

[Practice] This is one number. It needs no return forecast, and it uses data that the firm already stores. **Recommendation: put it on the first page of every risk report, ahead of the portfolio volatility itself.** A model that predicts 10% and delivers 10% is more valuable than one that predicts 8% and delivers 13%.

11.3 Errors specific to portfolio construction

Evaluating on the matrix that built the portfolio. This is the error described in §11.2. Use a genuinely out-of-sample window, not a different estimator on the same window.

Comparing at unmatched volatility. A method that produces 14% volatility usually beats one that produces 9% on raw return. That comparison is meaningless, because leverage is a free parameter (§1.3). **Always scale every candidate to the same target volatility before comparing.** This single discipline invalidates a surprising fraction of published comparisons.

Ignoring the universe's own selection. Testing on today's index members introduces survivorship into the covariance matrix. It flatters minimum-variance methods in particular, because the assets that vanished were the volatile ones.

Reporting one (N, T) pair. §2.4 shows that q governs everything, so a result at $q = 0.1$ says almost nothing about behaviour at $q = 0.6$. **Report a curve over q, not a point.** A method that wins at one aspect ratio and loses at another is common, and the sweep is cheap.

Free constraints. A long-only constraint improves out-of-sample results, and §8.2 explains why. Suppose a comparison constrains the sophisticated method and leaves the naive one unconstrained. It then measures the constraint, not the method. Constrain both identically.

Frictionless turnover. See §10.1. The methods that look best without costs are disproportionately the ones that trade most.

Tuning on the test set. Shrinkage intensities, thresholds, cluster linkages and band widths are all hyperparameters. Choosing them by out-of-sample performance and then reporting that performance is ordinary overfitting, applied to a matrix.

11.4 What to measure

Beyond return and volatility, five diagnostics actually discriminate between methods.

Realised ÷ predicted risk. See §11.2. It ranks first of the five.

Turnover, as an annualised two-way fraction of capital. This number converts directly into cost.

Concentration. Report the largest weight, the gross exposure and the Herfindahl index $\sum_i \tilde{w}_i^2$. The reciprocal of the Herfindahl index is the *effective number of positions*. It is a far more accurate count of positions than the number of non-zero weights. The normalisation is required, and it is easy to get wrong. The normalised weight is $\tilde{w}_i = w_i / \sum_j |w_j|$, each weight divided by **gross** exposure. On a long-only book that is the same as dividing by the sum of the weights. On a levered or long–short book the two differ. The reciprocal without the division measures leverage rather than breadth. Equal weights give exactly N. A 500-name portfolio with an effective count of 12 is a 12-name portfolio.

Effective number of bets. This is Meucci's refinement. Decompose the portfolio into uncorrelated risk sources, compute each source's share of total variance, and take the exponential of the entropy of that distribution. The Herfindahl index counts *positions*. This measure counts independent *risks*. The gap between the two is the accurate measure of diversification. [Practice] Consider a portfolio of 40 correlated bank stocks. Its effective number of positions is near 40, and its effective number of bets is near two. Only the second number predicts what will happen in a crisis.

Transfer coefficient. This measures how much of the signal survived construction. It is the *risk-adjusted* correlation between the portfolio held and the unconstrained optimum $w^\star$:

$$\mathrm{TC} = \frac{w' \Sigma w^\star}{\sqrt{w' \Sigma w}\,\sqrt{w^{\star\prime} \Sigma w^\star}}.$$

The Σ in the formula is needed because a plain correlation of the weight vectors is a different and more flattering number whenever volatilities differ widely. This is why the risk-adjusted form is standard. [Practice] A low value can be acceptable, because §8.2 shows that constraints are estimators. But a low value should come from a decision, not appear as a surprise.

11.5 Statistical significance, measured correctly

The arithmetic shows how little a short backtest can establish. For returns that are roughly IID, the standard error of an estimated Sharpe ratio over n years is approximately (Lo, 2002)

$$\mathrm{se}(\widehat{\mathrm{SR}}) \;\approx\; \sqrt{\frac{1 + \mathrm{SR}^2/2}{n}}.$$

At a true Sharpe ratio of 0.5 over 10 years, the standard error is 0.34. A 95% interval runs from about -0.16 to 1.16. This interval comfortably contains zero. The comparison that practitioners actually make is worse still. Take two methods, each measured over 10 years, one at 0.5 and one at 1.5. They differ by 1.0. The standard error of that difference is $\sqrt{0.34^2 + 0.46^2} \approx 0.57$. The difference is under two standard errors, so it is not distinguishable at any conventional level. That step treats the two estimates as independent. This is the conservative reading, and it is correct when the methods ran on *different* data. The closing paragraph below explains how much sharper the test becomes when they share a sample.

[Fact] **A 10-year backtest cannot distinguish a Sharpe ratio of 0.5 from one of 1.5.** Any claim that method A beat method B on 10 years of data, presented as a comparison of levels, is a claim about noise.

One method resolves this: **compare the difference, not the levels.** Two portfolio construction methods run on the same universe share almost all of their market exposure. The *difference* of their return series therefore has far lower volatility than either series alone. The paired comparison can be significant when neither level is. It also tests the actual question, "does B beat A", not "is B good".

[Practice] Three further disciplines are worth the effort. Use a **block bootstrap** on the return series rather than an IID one, because both volatility and correlation cluster. **Deflate for the number of configurations tried.** After evaluating 12 shrinkage intensities, the best of the 12 is biased upward. The deflated Sharpe ratio adjustment measures the size of that bias. **Test on several universes.** Portfolio construction methods are much more sensitive to the universe than signal research is. A method that wins on US large caps, loses on futures and wins on emerging markets shows a pattern that a single number cannot capture.

§11 Key takeaways

1. Test in this order: sanity checks, the accuracy of the risk model, weight stability, then performance. Most comparisons skip straight to performance and miss the failure that matters.
2. Realised ÷ predicted volatility, out of sample, is the most valuable single number in this chapter. It needs no forecast and no backtest.
3. Compare methods only at matched volatility. Leverage is a free parameter, so unmatched comparisons measure nothing.
4. Report a curve over q, not a point. A result at one aspect ratio does not transfer to another.
5. Give every candidate the same constraints. Constraints are estimators, so an asymmetric comparison measures the constraint.
6. The effective number of *bets* and the effective number of *positions* are different numbers. The gap between them measures the real diversification.
7. A 10-year backtest cannot separate a Sharpe ratio of 0.5 from 1.5. Compare the paired difference between methods, where the shared market exposure cancels.

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12. Synthesis

12.1 The framework in one page

The whole chapter reduces to three claims and one consequence.

The master form. Every portfolio construction method is

$$w \;\propto\; \left(\hat{\Sigma} + \Delta\right)^{-1} \hat{\mu} \quad \text{s.t.} \quad w \in \mathcal{C}, \quad \text{scaled to } \sigma^{\star}.$$

A method's name fixes some of the five slots and leaves others implicit. Minimum variance sets $\hat{\mu} = \mathbf{1}$. Risk parity sets $\hat{\mu} \propto \sigma$ and assumes equicorrelation. $1/N$ degenerates both $\hat{\Sigma}$ and $\hat{\mu}$ at once. Each of them fits the formula.

The mechanism. Inverting a covariance matrix allocates in inverse proportion to estimated variance. The eigendecomposition makes this exact. It gives $w \propto \sum_i (v_i'\mu/\lambda_i)\, v_i$, so the lowest-variance direction gets the largest position. This direction is also the one the sample estimates worst. Its eigenvalue is biased downward and its R_i^2 is biased upward. Both errors enter the same denominator. So the optimiser places its largest positions where the estimate is least reliable.

The diagnostic. On the covariance side, one number governs the damage: $q = N/T$. Realised volatility exceeds predicted volatility by $1/(1-q)$. Three derivations give this one result. The Marchenko–Pastur spectrum (§5.5), the inverse-Wishart expectation (§5.6) and, asset by asset, the regression identity (§5.4) each produce it. At $q = 0.5$ the portfolio takes twice the risk it reports. At $q \geq 1$ the problem is not defined. The variable q governs only the *covariance* problem. Calendar span alone governs the μ problem of §2.5. Sampling the same years more finely does not help it.

The consequence. Every repair in the field adds structure that raises the smallest eigenvalues and reduces the positions taken in those directions. Covariance shrinkage does this statistically, position constraints institutionally, transaction costs economically and factor models structurally. For the first three, the identity is exact and proved. §8.2 proves it for constraints and §10.3 for costs. For factor models and hierarchical methods, §6.8 and §6.11 give an argument, not a theorem. Either way, the effects compound instead of adding up to four independent safeguards. A practitioner who applies all of them at full strength has built $1/N$ without deciding to.

The chapter condenses into one sentence: **the return view and the aspect ratio are the binding constraints, in that order. The optimiser is not the problem, and the covariance estimator is not the solution.**

12.2 What to actually run

The table gives a configuration for each situation, with all five slots specified. These are the recommendations of this chapter, not measured optima.

Situation	Recommended configuration
No return forecast, $q < 0.3$	Minimum variance or ERC. Sample covariance, shrunk to constant correlation at $\delta \approx 0.3$–0.5, in correlation space. Long-only or box-constrained. Scaled to a volatility target
No return forecast, $0.3 < q < 0.7$	Equal risk contribution, shrunk harder ($\delta \approx 0.5$) and box-constrained. A factor covariance model where credible characteristics exist
No return forecast, $q > 0.7$	Inverse volatility, or HRP if the universe has real group structure. Do not invert anything. Reduce N by grouping where possible
Validated forecast, $q < 0.3$	Mean-variance. Rank-transform and clip the signal, scale it to a modest dispersion, shrink toward zero. Same covariance treatment as above. Gross-exposure cap
Validated forecast, $q > 0.3$	Factor covariance model or nonlinear shrinkage. Both stay invertible by construction where the sample matrix does not. Constrain hard. Consider allocating across signal-sorted baskets rather than individual names
Large universe, thousands of names	Explicit factor model (§6.8), for the Woodbury inverse and the risk attribution as much as the statistics

Situation	Recommended configuration
Strategy trades residual relationships	POET or a factor model with a thresholded residual. A standard factor model will report the position as riskless (§6.8)
Universe spans time zones	Weekly returns for correlations, daily for volatilities. Check the zero-return fraction per asset first (§9.3)

Some steps apply on every branch. Split into correlations and volatilities before regularising. Solve with a Cholesky factorisation rather than an inverse. Constrain before scaling to the risk target. Buffer the trades. Log q, κ and realised-over-predicted risk on every rebalance.

12.3 A staged build, with gates

Each stage has a gate. Do not proceed past a failed gate. Later stages will not fix the failure and will make it harder to see.

Stage 1 — Data. Build a point-in-time universe, apply corporate actions, fix one return convention and document the estimation frequency. *Gate:* the fraction of exactly-zero returns per asset is below a deliberately chosen threshold, and you have inspected every return beyond ±50% rather than clipping it.

Stage 2 — The naive baseline. Implement $1/N$ and inverse volatility end to end, with real rebalancing and real costs. *Gate:* both run in production shape, and you have recorded their Sharpe ratio, turnover and realised volatility. Everything after this stage must beat these two at matched volatility, or it is not worth its complexity.

Stage 3 — Risk model. Use the sample covariance, split into correlation and volatility, and shrunk toward constant correlation. *Gate:* the out-of-sample ratio of realised to predicted volatility is within about 10% of 1.0 (§11.2). If it is not, the fault lies in stage 1 or in q, not in the estimator. Go back rather than reaching for a better estimator.

Stage 4 — Allocation. Use minimum variance or ERC, constrained and scaled to target. *Gate:* the allocation beats stage 2 at matched volatility, on more than one universe, with the same constraints applied to both.

Stage 5 — Costs. Add turnover buffering or a cost term in the objective. *Gate:* the net-of-cost improvement over stage 2 is still positive. [Practice] This gate fails more often than any other, and it is a legitimate place for a project to stop.

Stage 6 — Return views. Add views only now, and only with a signal validated independently of the portfolio. *Gate:* the paired difference against stage 4 is significant (§11.5), and the result survives a sweep over q.

Stages 1 through 3 are infrastructure, and they seem uninteresting. They determine the outcome. [Practice] The characteristic failure of this project is to spend a month on stage 6 while a fault in stage 1 goes unnoticed. Stage 6 then produces plausible, presentable and wrong portfolios the whole time.

12.4 Advice for someone starting today

1. **Fix the data before touching the mathematics.** Build point-in-time universes, apply corporate actions, synchronise prices and count the zero returns per asset. This outranks everything below it. It is the least interesting work in the project, and it usually decides the outcome (§9.2, §9.3).
2. **Check $q = N/T$ next.** It takes one line, and it shows whether the problem is well-posed. A large amount of published and internal work runs at values of q where the problem is not well-posed.
3. **Decide explicitly whether a return forecast exists.** This is the line in the codebase with the largest effect on the result. If the answer is no, set $\widehat{\mu}$ to a constant. That deletes the dominant error term.
4. **Never skip $1/N$ and inverse volatility as baselines.** They are hard to beat and cheap to run. A method that cannot beat them at matched volatility does not justify its complexity.
5. **Regularise in correlation space, and shrink harder than feels comfortable.** Volatilities are the part that the data can estimate, so do not damage them while fixing the part that it cannot. The loss is asymmetric. §9.8 showed that too little shrinkage costs much more than too much.
6. **Count the regularisers.** Shrinkage, long-only, position caps and a turnover penalty are four applications of one idea. Applied together, they may produce $1/N$ through a more complicated route.
7. **Measure realised against predicted risk, out of sample, permanently.** It is one number, it requires no forecast, and it is the only diagnostic that catches the failure in §9.8.
8. **Compare at matched volatility, on several universes, over a sweep of q.** A comparison that violates these conditions measures something other than what it claims.
9. **Never form an explicit matrix inverse**, and never trade all the way to the target.

10. **Distrust a beautifully conditioned matrix.** A good κ means the arithmetic will complete. It does not mean the answer is right. Practitioners routinely confuse the two.

12.5 What is known, what is not, and where to bet

Known. [Fact] Estimation error dominates model error in this problem, and errors in means dominate errors in covariances by an order of magnitude. Sample eigenvalues spread in a predictable way governed by q, and the resulting distortion is correctable. Constraints and shrinkage are mathematically the same operation. This holds exactly for long-only minimum variance and for norm constraints. It does not hold for constraints in general. A cardinality constraint is not a shrinkage of anything. Minimum-variance portfolios have delivered better risk-adjusted returns than cap-weighting over long samples in most equity markets. [Contested] The record of risk parity is real but far more disputed, and it rests on a leverage assumption that minimum variance does not need. Non-synchronous and stale pricing biases correlations downward. As a result, optimisers systematically overweight illiquid assets.

Not known. [Contested] Four questions remain open.

- Whether the out-of-sample advantage of sophisticated covariance estimators over simple shrinkage survives realistic costs and constraints. The published evidence is mixed and depends heavily on universe, period and constraint set.
- Whether hierarchical methods genuinely beat well-regularised conventional ones, or merely match them while being easier to compute at $q \geq 1$.
- How much of the minimum-variance premium is skill in construction, and how much is exposure to the low-volatility factor.
- Whether end-to-end allocation by machine learning will add anything beyond what shrinkage already provides. The results so far are promising in sample and weak out of sample.

Where to bet. [Hypothesis] The view taken here makes three bets. They are views, not findings.

First, **the covariance-estimation literature has reached diminishing returns**, while the return-view and cost sides remain comparatively neglected. The former is mathematically tractable. The latter two are not. The field concentrates on the problems with the easiest mathematics.

Second, **most live portfolios are more heavily regularised than their owners believe**, for the compounding reason of §10.3. In a meaningful fraction of cases, the claim "our risk model is conservative" reflects four shrinkages applied together without anyone noticing.

Third, **for most practitioners the largest available improvement lies in §9.2 and §9.3 of this chapter, not in §6**. These sections cover point-in-time universes, correct corporate actions and synchronised prices. The claim is unglamorous, and this chapter holds it more firmly than any other.

In summary, portfolio construction is a problem where the mathematics is easy, the statistics are hard and the data work is harder. The field allocates its attention in almost exactly the reverse order.

§12 Key takeaways

1. Every method is $(\hat{\Sigma}+\Delta)^{-1}\hat{\mu}$ under constraints and a volatility target. Inversion places the largest positions where the estimate is worst. The ratio $q = N/T$ sets how large the resulting loss is.
2. Decide on the return view first, budget q second, and regularise once, in correlation space, while counting every regulariser.
3. Build in gated stages, starting from clean data and the $1/N$ baseline. Treat realised-over-predicted risk as the test that every later stage must pass.

Appendix A. Concepts and prerequisites

This appendix covers everything the main text relies on without stopping to explain. The intended reader is mathematically comfortable but does not work in this field. Such a reader meets "the tangency portfolio" or "complementary slackness" somewhere in §7. This reader wants the idea rather than a bare definition and prefers not to leave the chapter to get it.

The entries follow **dependency** order, not alphabetical order. Later entries use earlier ones. The five parts are also in dependency order, so the appendix can be read from first to last. Each entry gives the idea in words first, then the formal definition, then why it appears here, then where to go deeper. The notation follows the main text's notation block exactly. Where a standard formula from another field would conflict with it, the entry flags the conflict instead of inventing new symbols.

None of this is needed to follow the *argument* of the chapter. Much of it is needed to implement it.

Index. The table gives the section where each concept is first used.

Concept	First used	Concept	First used
Quadratic forms and ∇_w	§1.3	Euler's theorem	§7.6
Positive definiteness	§2.3	Coordinate descent	§7.6
Eigenvalues and the spectrum	§2.3	Realised volatility	§5.3
The trace	§5.5	GARCH	§6.10
Rotational invariance	§6.1	Principal component analysis	§6.9
The condition number	§5.5	Correlation as a distance	§6.11
The Frobenius norm	§6.5	Hierarchical clustering	§6.11
Cholesky, and solve vs invert	§9.5	Basis points and turnover	§6.6
Sherman–Morrison–Woodbury	§6.4	Excess returns, Sharpe ratio	§2.1
IID and stationarity	§2.1	Frontier and tangency	§5.1
The bias–variance trade-off	§6.1	CAPM and reverse optimisation	§7.8
Effective sample size (Kish)	§6.3	Cap-weighted benchmarks	§7.4
Wishart and inverse-Wishart	§5.6	The low-volatility anomaly	§7.4
James–Stein shrinkage	§3	The information coefficient	§7.7
Prior, posterior, MAP	§7.8	Transfer coefficient, the law	§11.4
R^2 and adjusted R^2	§5.4	Statistical arbitrage	§3
Mahalanobis distance	§6.12	Cointegration	§6.7
The bootstrap	§7.9	Market impact, participation	§10.2
Multiple testing	§11.5	Winsorising	§6.12
Shannon entropy	§11.4	Herfindahl, effective counts	§11.4
Convexity and the QP	§3	Effective number of bets	§11.4
Lagrange, KKT, slackness	§5.2	Standard error of a Sharpe	§11.5
ℓ_1, ℓ_2, ridge, lasso	§7.10	The deflated Sharpe ratio	§11.5

Part I — Linear algebra and numerics. The whole chapter manipulates the covariance matrix. Almost every claim in §2 and §5 concerns what happens when the optimiser inverts a matrix. These nine entries define the terms needed to state this precisely.

A.1 Quadratic forms, and differentiating with respect to a vector

The idea. Portfolio variance is a sum over every *pair* of holdings, with each covariance counted twice. It is not the sum of the assets' variances. The compact way to write that double sum is $w'\Sigma w$. An expression of that shape, a vector on each side of a matrix, is called a quadratic form. The quadratic form is the only non-linear object in this chapter. This makes the subject tractable, because differentiating a quadratic form gives a *linear* equation, and linear equations can be solved.

Formally. For symmetric $A \in \mathbb{R}^{N\times N}$, the quadratic form is the scalar $w'Aw = \sum_i \sum_j w_i A_{ij} w_j$. Two gradient rules are needed throughout:

$$\nabla_w (w'\mu) = \mu, \qquad \nabla_w (w'Aw) = (A + A')\, w = 2Aw \text{ for symmetric } A.$$

Differentiating a scalar by a vector produces a vector of partial derivatives, one per coordinate. Setting that vector to zero gives the first-order condition for a stationary point. For a convex objective (A.21), the stationary point is the optimum.

Why it appears here. §1.3 obtains the master form $w^\star = \gamma^{-1}\Sigma^{-1}\mu$ by applying exactly these two rules to $w'\mu - \frac{\gamma}{2}w'\Sigma w$ and setting the result to zero. §10.3 repeats the derivation with a turnover penalty attached. The marginal risk contribution $(\Sigma w)_i/\sigma_p$ of §7.6 is this derivative taken through a square root.

Deeper. Petersen & Pedersen, *The Matrix Cookbook* (2012), sec. 2, the standard lookup table for identities of this kind.

A.2 Positive definiteness and semi-definiteness

The idea. A covariance matrix cannot be arbitrary. Every portfolio formed from it has non-negative variance, because variance is an average of squares. This requirement *is* positive semi-definiteness. Strict positive definiteness adds that no portfolio has *exactly* zero variance. This is the condition for the matrix to be invertible, and therefore for the master form to be defined.

Formally. A symmetric A is **positive semi-definite** (PSD) if $x'Ax \geq 0$ for every $x \in \mathbb{R}^N$. It is **positive definite** (PD) if $x'Ax > 0$ for every $x \neq 0$. Equivalently, all eigenvalues are ≥ 0 (PSD) or > 0 (PD). A true covariance matrix is always PSD. The sample covariance S from T observations has rank at most $\min(N, T-1)$. It is therefore PD only when $T > N$, and otherwise merely PSD, which means singular.

Why it appears here. This formalises the statement in §2.4 that "at $q \geq 1$ the sample covariance matrix is singular, and the optimiser is meaningless". With $N \geq T$, non-zero vectors w exist with $w'Sw = 0$ exactly. The optimiser finds one and scales it without limit. The positive-definiteness check of §9.5 and the repair of Higham (2002) restore the property after blending or patching has destroyed it.

Deeper. Horn & Johnson, *Matrix Analysis*, 2nd ed. (CUP, 2012), chapter 7.

A.3 Eigenvalues, eigenvectors, and the spectrum

The idea. A symmetric matrix acts in a geometrically simple way, although its entries do not show it. It has a set of mutually perpendicular directions in which it acts purely by stretching, with no rotation. Those directions are the eigenvectors, and the stretch factors are the eigenvalues. Together they form the *spectrum*. For a covariance matrix these objects are concrete. Each eigenvector is a portfolio, and its eigenvalue is that portfolio's variance. So §2.3 can discuss "eigen-portfolios" without extra theory.

Formally. For symmetric Σ, the **spectral theorem** guarantees an orthonormal basis of eigenvectors $v_1, \ldots, v_N$, so that $v_i'v_j = 0$ for $i \neq j$ and $v_i'v_i = 1$. The corresponding real eigenvalues $\lambda_1 \geq \ldots \geq \lambda_N$ satisfy $\Sigma v_i = \lambda_i v_i$. Collect them into $V = [v_1 \cdots v_N]$ and $\Lambda = \text{diag}(\lambda_i)$. Then

$$\Sigma = V\Lambda V' = \sum_{i=1}^{N} \lambda_i\, v_i v_i', \qquad \Sigma^{-1} = V\Lambda^{-1}V' = \sum_{i=1}^{N} \frac{1}{\lambda_i}\, v_i v_i'.$$

Because v_i has unit length, $v_i'\Sigma v_i = \lambda_i$. So the eigenvalue *is* the variance of the portfolio v_i. Inversion leaves every direction unchanged and replaces each λ_i by $1/\lambda_i$. This reverses the order of importance completely.

Why it appears here. This is the mechanism of the whole chapter. §2.3 decomposes $w^\star \propto \sum_i (v_i'\mu/\lambda_i)v_i$ and shows that the smallest eigenvalue receives the largest bet. §5.5 describes how the sample spreads the λ_i apart. §6.6 and §6.7 repair the spectrum directly. The key fact is that the eigenvalues of Σ are the variances of N uncorrelated portfolios that together span every possible holding.

Deeper. Strang, *Introduction to Linear Algebra*, 5th ed. (2016), chapter 6, for the geometry. Golub & Van Loan, *Matrix Computations*, 4th ed. (2013), chapter 8, for how the decomposition is actually computed.

A.4 The trace, and trace preservation

The idea. The trace is the sum of a matrix's diagonal entries. For a covariance matrix, it is the sum of the individual asset variances. The same number is also the sum of the eigenvalues. Total variance can therefore be counted two ways, asset

by asset or eigen-portfolio by eigen-portfolio, and the two counts agree. So the trace is a fixed total of variance. An estimator that redistributes variance across directions should neither create nor remove any. "Preserving the trace" is the constraint that enforces this.

Formally. $\text{tr}(A) = \sum_i A_{ii}$, and for symmetric A, $\text{tr}(A) = \sum_i \lambda_i$. The trace is invariant under similarity and cyclic under products: $\text{tr}(AB) = \text{tr}(BA)$. For a correlation matrix every diagonal entry is 1. So $\text{tr}(C) = N$, and the eigenvalues of any correlation matrix average to exactly 1.

Why it appears here. The eigenvalue clipping of §6.7 replaces the whole noise bulk by its common mean, *"preserving the trace"*. This stops the repair from changing the portfolio's overall level of risk. §6.7 notes that the fit must allow for the outlier eigenvalues that absorb part of the trace. This is the same constraint seen from the other side. If the market mode carries an eigenvalue of 66, only $N - 66$ of the total remains for the bulk. Fit the Marchenko–Pastur edge to what remains.

Deeper. Horn & Johnson, *Matrix Analysis*, 2nd ed. (CUP, 2012), sec. 1.2.

A.5 Rotational invariance

The idea. Some estimators assume *which* combinations of assets are risky. Others assume only *how much* risk lies in each direction the sample already found. The second kind is called rotationally invariant. The name is literal: if an arbitrary orthogonal mixing relabels the assets, the estimator's output is mixed the same way and keeps the same form. Such an estimator can only adjust eigenvalues. It has no information with which to move an eigenvector.

Formally. An estimator $\hat{\Sigma}(\cdot)$ applied to returns is **rotationally invariant** if $\hat{\Sigma}(UR) = U\hat{\Sigma}(R)U'$ for every orthogonal U (with $U'U = I$) and every return matrix R. Any such estimator has the form $\hat{\Sigma} = \sum_i \tilde{\lambda}_i \, v_i v_i'$. Here the v_i are the *sample* eigenvectors, and only the map $\lambda_i \mapsto \tilde{\lambda}_i$ is chosen.

Why it appears here. The three-family table of §6.1 uses this as its top-level division. Nonlinear shrinkage (§6.6), eigenvalue clipping (§6.7) and shrinkage toward a scaled identity keep the sample's eigenvectors, so they are rotationally invariant. Factor models and constant correlation assert where the risk is, so they are not. The distinction determines what an estimator can fix. A rotationally invariant estimator cannot repair a badly estimated leading eigenvector, because it never changes one.

Deeper. Bun, Bouchaud & Potters (2017), *Physics Reports* 666, sec. 5, which develops the class under exactly this name.

A.6 The condition number

The idea. Inverting a matrix amplifies input errors by some factor. If an input to the linear system is wrong by one part in a thousand, the answer can be wrong by that factor times one part in a thousand. The condition number is that factor. It is a property of the matrix alone, and one can compute it before any data error exists. So it can warn of an error before it occurs, instead of diagnosing it afterwards.

Formally. For a symmetric positive definite Σ with eigenvalues $\lambda_1 \geq \ldots \geq \lambda_N > 0$, the spectral condition number is

$$\kappa(\Sigma) = \frac{\lambda_1}{\lambda_N} \geq 1.$$

For w solving $\Sigma w = \mu$, a relative perturbation of size ϵ in Σ or μ changes w by at most about $\kappa(\Sigma)\,\epsilon$ in relative terms. A singular matrix has $\kappa = \infty$. As a rule of thumb, a solve loses roughly $\log_{10}\kappa$ decimal digits of accuracy, so at $\kappa \approx 10^{16}$ double precision retains none.

Why it appears here. §5.7 makes κ one of the two numbers that belong in production monitoring. §5.5 shows that pure noise at $q = 0.9$ produces $\kappa = 1442$ in a matrix whose true value is the identity. §9.8 gives the most important result. The best-conditioned row of its table has the *worst* risk forecast. So κ certifies that the arithmetic will complete. It does not show whether the answer is right.

Deeper. Trefethen & Bau, *Numerical Linear Algebra* (SIAM, 1997), lectures 12 and 18.

A.7 The Frobenius norm

The idea. Measuring how close one matrix is to another requires a distance. The simplest distance treats both matrices as long lists of numbers and takes the ordinary Euclidean distance between the lists. This is the Frobenius norm. Its advantage is that it is differentiable and leads to closed-form answers. Its drawback is that it weights every entry equally, whereas portfolio risk does not weight errors equally.

Formally. For $A \in \mathbb{R}^{N\times N}$,

$$\|A\|_F^2 = \sum_i \sum_j A_{ij}^2 = \mathrm{tr}(A'A) = \sum_i \lambda_i(A'A).$$

For symmetric A, it is therefore the sum of squared eigenvalues. It is the norm induced by the inner product $\langle A, B\rangle = \mathrm{tr}(A'B)$, which makes minimising it a least-squares problem.

Why it appears here. The shrinkage intensity δ of §6.5 minimises $\mathbb{E}\|\widehat{\Sigma}_{\mathsf{LW}} - \Sigma\|_F^2$. The most important caveat in the same section applies to that choice. Portfolio variance depends on Σ^{-1}, which weights small-eigenvalue directions enormously, while the Frobenius loss weights them like any other entry. That mismatch is why the Frobenius-optimal δ is usually somewhat too small for portfolio use. It is also why §9.8 finds the loss function asymmetric in favour of over-shrinking.

Deeper. Ledoit & Wolf (2004), "Honey, I Shrunk the Sample Covariance Matrix," which sets up the loss explicitly.

A.8 Cholesky factorisation, and why to solve instead of inverting

The idea. The inverse Σ^{-1} itself is almost never needed. The calculation needs the vector w with $\Sigma w = \mu$. Computing the inverse and then multiplying is the slow and inaccurate way to get it. The fast and accurate way factorises Σ once, into a triangular factor and its transpose. Solving then takes two sweeps of back-substitution. A positive definite matrix has exactly one such factorisation. So the check of A.2 and this algorithm are the same check.

Formally. A symmetric positive definite Σ has a unique lower-triangular factor G with positive diagonal such that

$$\Sigma = GG'.$$

The factorisation costs about $N^3/3$ operations, half the cost of a general LU factorisation. Solving $\Sigma w = \mu$ is then $Gy = \mu$ followed by $G'w = y$, each $O(N^2)$. Forming Σ^{-1} explicitly costs roughly three times as much. It also adds a second round of rounding error in the multiplication that follows. Cholesky *fails* exactly when the matrix is not positive definite, because no positive diagonal exists. So it also tests definiteness at no extra cost.

Why it appears here. §9.5 states the rule: never form the inverse. §9.7 implements it with `cho_factor` and `cho_solve`, and its commentary calls this "a matter of correctness, not performance". At the condition numbers of §5.7, an explicit inverse loses digits that the computation cannot afford to lose.

Deeper. Golub & Van Loan, *Matrix Computations*, 4th ed. (Johns Hopkins, 2013), sec. 4.2.

A.9 The Sherman–Morrison–Woodbury identity

The idea. Suppose a matrix is already inverted and then receives a low-rank correction. Examples are a market factor, a handful of industry factors or a single common correlation. The inversion need not start over. Woodbury's identity expresses the corrected inverse as the old inverse plus a correction term. This term requires inverting only a matrix as large as the correction's rank. For a factor model this reduces an $N \times N$ problem to a $K \times K$ one. With N in the thousands, this makes the problem feasible.

Formally. For invertible A ($N \times N$) and Ω ($K \times K$) and any $F \in \mathbb{R}^{N\times K}$,

$$\left(A + F\Omega F'\right)^{-1} = A^{-1} - A^{-1}F\left(\Omega^{-1} + F'A^{-1}F\right)^{-1}F'A^{-1}.$$

The **Sherman–Morrison** special case is $K = 1$. Apply the identity to the factor model $\widehat{\Sigma} = F\Omega F' + \Psi$ with Ψ diagonal. Then $A^{-1} = \Psi^{-1}$ is trivial, and the only genuine inversion is of a $K \times K$ matrix.

Why it appears here. §6.8 cites exactly this as the practical advantage of explicit factor models: "the inverse has a closed form … and never requires an $N \times N$ decomposition". §12.2 recommends a factor model for very large universes "for the Woodbury inverse … as much as the statistics". The constant-correlation matrix of §6.4 is the rank-one case. $C = (1-\bar\rho)I + \bar\rho\,\mathbf{1}\mathbf{1}'$ inverts in closed form with no decomposition at all.

Deeper. Golub & Van Loan, *Matrix Computations*, 4th ed. (2013), sec. 2.1.4; Hager, "Updating the Inverse of a Matrix," *SIAM Review* 31(2) (1989), 221–239.

Part II — Probability and estimation. Part I described what the matrix *is*. This part describes what goes wrong when it must be estimated from T observations, which is the entire subject of §2 and §5.

A.10 IID, and stationarity

The idea. Every estimator here averages over history. An average estimates a quantity only if the quantity stayed constant during the sample. Two assumptions are involved, and practitioners routinely conflate them. *Independence* means that today's return carries no information about tomorrow's. *Stationarity* means that the statistical properties do not change over time. Stationarity can hold without independence, and volatility clustering is exactly that case. Stationarity is the assumption actually needed.

Formally. Observations $r_1, \dots, r_T$ are **IID** if they are mutually independent and share one distribution. A process is **strictly stationary** if the joint law of $(r_t, \dots, r_{t+k})$ does not change when t shifts. **Covariance (weak) stationarity** is the version used in practice. It requires only that $\mathbb{E}[r_t]$, $\mathrm{Var}(r_t)$ and $\mathrm{Cov}(r_t, r_{t-k})$ exist and do not depend on t. IID implies stationarity, and the converse fails badly for returns.

Why it appears here. The demonstration of §2.1 runs on data that are IID, multivariate normal and stationary, with no fat tails and no regime changes. That design ensures the damage it measures can only be estimation error. Without this design, the argument would not hold. §6.2 lists IID with finite fourth moments as the assumption of the sample covariance. §9.4 warns that "beyond roughly five years of daily data, equity correlations average over structurally different regimes". That warning means that stationarity, not sample size, limits T.

Deeper. Hamilton, *Time Series Analysis* (Princeton, 1994), chapter 3. The companion chapter Market Regimes and Hidden Markov Models covers what to do when stationarity fails.

A.11 The bias–variance trade-off

The idea. An estimator can be wrong in two ways. It can be biased, so that its average is at the wrong value. Or it can be erratic, so that it varies widely around the right value from sample to sample. Total error is the sum of both, and the two trade off against each other. Imposing structure moves the estimate toward a wrong but steady answer. Imposing none leaves an answer that is right on average but varies widely. Zero bias is usually not the best choice.

Formally. For an estimator $\hat{\vartheta}$ of a scalar ϑ, the mean squared error decomposes as

$$\mathrm{MSE}(\hat{\vartheta}) = \underbrace{\left(\mathbb{E}[\hat{\vartheta}] - \vartheta\right)^2}_{\text{bias}^2} + \underbrace{\mathrm{Var}(\hat{\vartheta})}_{\text{variance}},$$

and the cross term vanishes identically. The decomposition extends entrywise to matrices under the Frobenius loss of A.7.

Why it appears here. §6.1 names this trade-off as the single question that organises all of §6. The question is how much structure to impose in exchange for less estimation error. The sample covariance is at one extreme, with zero bias and maximum variance. A single number applied to everything is at the other extreme. The chapter repeatedly instructs the reader to shrink harder than feels comfortable (§9.8, §12.4). That instruction rests on the claim that, in this problem, the variance term is almost always the binding one.

Deeper. Hastie, Tibshirani & Friedman, *The Elements of Statistical Learning*, 2nd ed. (Springer, 2009), sec. 7.3.

A.12 Effective sample size, in the Kish sense

The idea. When observations are weighted unequally, the number of rows in the data is no longer the number of observations actually available. A weighting scheme that puts most of its weight on a handful of days is statistically equivalent to a handful of days. The effective sample size is the count of *equally weighted* observations that would give an estimator the same sampling variance. It can be far smaller than T, and nothing in the code shows this.

Formally. For non-negative weights $u_1, \dots, u_T$, Kish's effective sample size is

$$T_{\text{eff}} = \frac{\left(\sum_t u_t\right)^2}{\sum_t u_t^2}.$$

It equals T when all weights are equal, and it falls otherwise. For the EWMA of §6.3, $u_t \propto \theta^t$ for $t = 0, 1, 2, ...$, so $\sum u_t = 1/(1-\theta)$ and $\sum u_t^2 = 1/(1-\theta^2)$. That gives

$$T_{\text{eff}} = \frac{(1-\theta^2)}{(1-\theta)^2} = \frac{1+\theta}{1-\theta}.$$

Why it appears here. The last line derives $T_{\text{eff}} = (1+\theta)/(1-\theta)$ in §6.3 and explains the pitfall that section describes. At the RiskMetrics value $\theta = 0.94$ the effective sample is 32 observations. A 100-asset EWMA correlation matrix therefore has $q_{\text{eff}} = 3.1$, and it is meaningless however many years of data it receives. §9.4 makes N/T_{eff}, not N/T, the quantity to budget.

Deeper. Kish, *Survey Sampling* (Wiley, 1965), sec. 8.2, where the formula originates in the context of survey designs with unequal probabilities.

A.13 The Wishart and inverse-Wishart distributions

The idea. A single sample variance has a chi-squared distribution. The matrix generalisation, the sampling distribution of a whole sample covariance matrix, is the Wishart. It describes how S varies across repeated samples. Its most useful property for this chapter concerns S^{-1}, not S. The inverse of an unbiased estimate is a biased estimate of the inverse, and the Wishart gives the exact size of the bias.

Formally. Suppose $r_1, \dots, r_T$ are IID $\mathcal{N}(0, \Sigma)$ in $\mathbb{R}^N$. Then $A = \sum_t r_t r_t' \sim W_N(\Sigma, T)$, the Wishart distribution with scale Σ and T degrees of freedom. It has $\mathbb{E}[A] = T\Sigma$, so $S = A/T$ is unbiased. The distribution of A^{-1} is the **inverse-Wishart**. Provided $T > N + 1$,

$$\mathbb{E}\big[A^{-1}\big] = \frac{\Sigma^{-1}}{T-N-1} \qquad \Longrightarrow \qquad \mathbb{E}\big[S^{-1}\big] = \frac{T}{T-N-1}\,\Sigma^{-1} \approx \frac{1}{1-q}\,\Sigma^{-1}.$$

Estimating the mean costs one degree of freedom and changes nothing to leading order. The inverse-Wishart also serves as the conjugate prior for Σ in Bayesian work (A.15). That is a separate use of the same family.

Why it appears here. This is §5.6 in one line. The sample covariance is unbiased. Its inverse is biased by a factor of about $1/(1-q)$. The optimiser uses the inverse. §12.1 lists three derivations of this result. The other two are the Marchenko–Pastur integral of §5.5 and the inflated $\hat{R}_i^2$ of §5.4. The inverse-Wishart expectation states the result for the whole matrix at once. The condition $T > N + 1$ is the invertibility requirement of A.2, reappearing as the condition for the expectation to exist.

Deeper. Muirhead, *Aspects of Multivariate Statistical Theory* (Wiley, 1982), chapter 3, the standard reference for both distributions and for the derivation of the expectation above.

A.14 James–Stein shrinkage

The idea. In 1961 James and Stein proved a counterintuitive result. Suppose three or more means are estimated at once, and the score is total squared error. Then the obvious estimator, which uses each sample mean for its own quantity, is *inadmissible*. Pulling all of the estimates toward a common point always beats it, whatever the true means are. The individual estimates get worse, and the collection gets better. Every shrinkage estimator in this chapter, including Ledoit–Wolf, derives from that result.

Formally. Observe $\hat{\mu} \sim \mathcal{N}(\mu, \sigma^2 I_N)$ with $N \geq 3$. The James–Stein estimator that shrinks toward the origin is

$$\hat{\mu}^{\text{JS}} = \left(1 - \frac{(N-2)\,\sigma^2}{\|\hat{\mu}\|^2}\right)\hat{\mu}.$$

It has strictly smaller total mean squared error than $\hat{\mu}$ for every μ. Shrinking toward the grand mean instead of the origin replaces $N-2$ by $N-3$. The shrinkage depends on the data. It pulls noisy, dispersed estimates strongly and tight ones barely at all.

Why it appears here. The linear shrinkage of §6.5 is this idea applied to a matrix, with the intensity derived rather than guessed. Era IV of §3 records that Jorion (1986) applied it to the means a decade before it reached the covariance. It also

contradicts the instinct that unbiasedness is a virtue. The argument of §2 is that the unbiased estimator is the poor choice here.

Deeper. Efron & Morris, "Stein's Paradox in Statistics," *Scientific American* 236(5) (1977), 119–127. It is six pages long with no prerequisites, and it makes the result feel inevitable rather than paradoxical.

A.15 Prior, posterior, the Bayesian update, and MAP

The idea. A Bayesian treats the unknown parameter as itself uncertain. The analysis starts with a distribution describing beliefs before seeing data, called the prior. The data supply a likelihood. The two combine into a distribution describing beliefs afterwards, called the posterior. The posterior combines prior and data, weighted by their relative precision. So "shrinkage" and "a prior" are the same operation described in two vocabularies. When a single number is wanted rather than a distribution, the posterior's mode gives the MAP estimate.

Formally. With prior $p(\vartheta)$ and likelihood $p(\text{data} \mid \vartheta)$, Bayes' rule gives the posterior $p(\vartheta \mid \text{data}) \propto p(\text{data} \mid \vartheta)\, p(\vartheta)$. The **maximum a posteriori** estimate is $\widehat{\vartheta}_{\text{MAP}} = \arg\max_{\vartheta} p(\vartheta \mid \text{data})$. Equivalently, it maximises $\log p(\text{data} \mid \vartheta) + \log p(\vartheta)$. That is a penalised likelihood, with the prior supplying the penalty. In the Gaussian conjugate case, the posterior mean is a precision-weighted average of the prior mean and the sample estimate.

Why it appears here. Black–Litterman (§7.8) is precisely this. The equilibrium returns Π are the prior, the stated views are the likelihood and μ_{BL} is the posterior. The equivalence chain of §8.2 then completes the link. A ridge penalty $\nu\|w\|^2$ *is* a MAP estimate under a zero-mean Gaussian prior on w with variance proportional to $1/\nu$. That is why "add a penalty", "impose a prior" and "shrink the covariance" name one operation. §7.9 describes resampling as an attempt to approximate the same decision without the Bayesian formalism.

Deeper. Gelman, Carlin, Stern, Dunson, Vehtari & Rubin, *Bayesian Data Analysis*, 3rd ed. (CRC, 2013), chapters 1–2.

A.16 The coefficient of determination, and its adjusted form

The idea. R^2 is the fraction of a variable's variation that the regressors account for. Its main defect is that it can only rise when regressors are added, even regressors made of pure noise. With enough regressors, a regression explains any data perfectly and reveals nothing. The adjusted version penalises each regressor, and the size of that penalty is the central result of §5.4.

Formally. Take a regression with T observations and k regressors plus an intercept. Then $R^2 = 1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$, the fraction of variance explained. Under the null that no regressor has any relationship to the target, $\mathbb{E}[R^2] = k/(T-1)$, not zero. The adjusted version corrects for the degrees of freedom consumed:

$$1 - \bar{R}^2 = (1 - R^2)\,\frac{T-1}{T-k-1},$$

which restores $\mathbb{E}[1 - \bar{R}^2] = 1$ under that null.

Why it appears here. §5.4 defines R_i^2 as the coefficient of determination from regressing asset i on all $N-1$ others. So $k = N-1$, and the penalty is $(T-1)/(T-N)$. The overfitting inflation $\mathbb{E}[\widehat{R}_i^2] = (N-1)/(T-1) \approx q$ follows directly from the formula above. Asset i's weight carries $1/(1-R_i^2)$ in its denominator. That inflation is therefore exactly the $1/(1-q)$ weight inflation that the rest of §5 derives by two other routes.

Deeper. Greene, *Econometric Analysis*, 8th ed. (Pearson, 2018), sec. 3.5, which derives both the expectation and the adjustment.

A.17 Mahalanobis distance

The idea. Euclidean distance is a poor measure of how unusual an observation is. It treats a 3% move in a volatile asset and a 3% move in a low-volatility one as equally unusual. It also treats two correlated assets moving together as twice as unusual as one moving alone. Mahalanobis distance corrects both problems by measuring in units of the data's own covariance. It equals the Euclidean distance after whitening the data, so that every direction has unit variance and no direction is correlated with another.

Formally. For an observation r with mean μ and covariance Σ,

$$d_{\mathcal{M}}(r)^2 = (r-\mu)' \, \Sigma^{-1} \, (r-\mu).$$

This is a quadratic form (A.1) in the *inverse* covariance. Under multivariate normality, $d^2_{\mathcal{M}}$ is chi-squared with N degrees of freedom, which supplies a calibrated threshold for "unusual". The distance is invariant to any non-degenerate linear recombination of the assets.

Why it appears here. The robust estimators of §6.12 down-weight observations by this distance, and fitting a multivariate t does so automatically. The same quantity makes an outlier at the *portfolio* level detectable when no single asset looks extreme. It is also the same algebra as the maximum squared Sharpe ratio $c = \mu' \Sigma^{-1} \mu$ of §5.2. The optimiser's own objective is a Mahalanobis length.

Deeper. Mardia, Kent & Bibby, *Multivariate Analysis* (Academic Press, 1979), sec. 1.6 and chapter 3.

A.18 The bootstrap, and the block bootstrap

The idea. The goal is to find the sampling distribution of a statistic, meaning how much the statistic would vary across alternative histories. Only one history is available. The bootstrap resamples that history with replacement, recomputes the statistic on each resample and measures the spread. It fails when the data are serially dependent. Resampling one observation at a time destroys the serial structure the analysis relies on. The block bootstrap repairs this. It resamples *blocks* of consecutive observations instead of single points.

Formally. Given data $r_1, \dots, r_T$ and a statistic $\widehat{\vartheta}$, draw B resamples of size T with replacement. Compute $\widehat{\vartheta}^{(1)}, \dots, \widehat{\vartheta}^{(B)}$, and use their empirical distribution as an estimate of the sampling distribution. The **moving-block** bootstrap instead draws blocks of ℓ consecutive observations and concatenates T/ℓ of them. That preserves dependence up to lag ℓ. The **stationary** bootstrap randomises ℓ geometrically, so that the resampled series is itself stationary. The choice of ℓ trades bias against variance. Too short a block loses the dependence. Too long a block leaves too few distinct blocks.

Why it appears here. The resampled efficiency of §7.9 bootstraps returns, re-optimises on each draw and averages the weight vectors. §7.9's better recommendation keeps the same procedure as a *diagnostic*. It examines the spread of asset 12's weight across resamples rather than the average. §11.5 requires the block version for significance testing, "because both volatility and correlation cluster". An IID bootstrap of financial returns reports standard errors that are too small.

Deeper. Efron & Tibshirani, *An Introduction to the Bootstrap* (Chapman & Hall, 1993); Politis & Romano, "The Stationary Bootstrap," *JASA* 89(428) (1994), 1303–1313.

A.19 Multiple testing and selection bias

The idea. A 5% significance level means that one test in 20 comes back positive on pure noise. Running 20 configurations and reporting the best one produces a significant result from noise alone. This happens whether or not the researcher thinks of the work as running 20 tests. The maximum of many noisy quantities is biased upward. The bias grows with the number of things examined, including the ones tried, discarded and forgotten.

Formally. For M independent test statistics under a global null, the chance of at least one rejection at level α is $1-(1-\alpha)^M$. At $M = 20$ and $\alpha = 0.05$, that is 0.64. The classical corrections control either the family-wise error rate (**Bonferroni**: test each at α/M) or the false discovery rate (**Benjamini–Hochberg**). For a maximum, the quantity that matters is its expected value. For M independent standard normals, $\mathbb{E}\big[\max_m Z_m\big] \approx \sqrt{2 \ln M}$. The best of 100 tries therefore scores about 3.0 by luck alone.

Why it appears here. §11.3 calls this "ordinary overfitting, applied to a matrix". Shrinkage intensities, thresholds, linkage choices and band widths are all hyperparameters. Tuning them on the evaluation window and then reporting that window's performance is a failure of multiple testing, whatever it is called. The instruction in §11.5 to deflate is the quantitative form (A.46). The error-maximisation argument of §2.2 is the same phenomenon, acting on portfolio directions rather than on model configurations.

Deeper. Harvey, Liu & Zhu, "…and the Cross-Section of Expected Returns," *Review of Financial Studies* 29(1) (2016), 5–68. This is the finance-specific version. It argues that the field's usual hurdle of $t > 2$ should be roughly $t > 3$.

A.20 Shannon entropy

The idea. Entropy measures how spread out a distribution is. It measures how many outcomes genuinely share the probability, not distance from a centre. A distribution concentrated on one outcome has zero entropy, and a uniform distribution

over n outcomes has the maximum. Its exponential is therefore an *effective count*, the number of equally likely outcomes that would be this spread out. Entropy appears in a chapter on portfolios for this reason.

Formally. For a discrete distribution $p_1, \dots, p_n$ with $p_i \geq 0$ and $\sum_i p_i = 1$,

$$H(p) = -\sum_{i=1}^{n} p_i \ln p_i \ \in [0, \ln n],$$

with $0 \ln 0 = 0$ by convention. $H = 0$ when one p_i is 1, and $H = \ln n$ when all are equal. The **perplexity** $\exp(H)$ lies in $[1, n]$, and it equals n exactly when the distribution is uniform.

Why it appears here. The effective number of bets in §11.4 is $\exp(H)$ applied to the shares of portfolio variance carried by uncorrelated risk sources (A.44). The "effective count" reading makes it comparable to the reciprocal Herfindahl index of A.43. The two constructions are similar. Both take a vector of shares and return a count, and both return n when the shares are equal.

Deeper. Cover & Thomas, *Elements of Information Theory*, 2nd ed. (Wiley, 2006), chapter 2.

Part III — Optimisation. The chapter states that optimisation is not the hard part. This is true, but it does not justify being vague about optimisation. These five entries supply what is needed to read §5.2, §7.6, §7.10 and §8.2 without assuming any result.

A.21 Convexity, and the quadratic programme

The idea. In a convex problem, every local optimum is the global optimum. A solver that improves the objective step by step therefore cannot get stuck at a worse point. This property separates the optimisation problems that have effective solutions from those that do not. Mean-variance optimisation is among the solved problems. A *quadratic programme* is the convex form this chapter uses repeatedly. It has a quadratic objective with linear constraints. Reliable solvers for it have existed for decades.

Formally. A set is convex if it contains the segment between any two of its points. A function J is convex if $J(\alpha x + (1-\alpha)y) \leq \alpha J(x) + (1-\alpha)J(y)$ for $\alpha \in [0,1]$. A twice-differentiable J is convex exactly when its Hessian is positive semi-definite. So $w'\Sigma w$ is convex precisely because Σ is PSD (A.2). A **quadratic programme** is

$$\min_{w} \ \tfrac{1}{2}\, w'\Sigma w - w'\mu \quad \text{s.t.} \quad w \in \mathcal{C},$$

with $\mathcal{C}$ defined by linear equalities and inequalities. That is the constraint set of §8.1. The programme is convex whenever Σ is PSD, and it is solvable in polynomial time. **Jensen's inequality**, $\mathbb{E}[J(X)] \geq J(\mathbb{E}[X])$ for convex J, is the probabilistic companion. Matrix inversion is convex in the relevant sense. That is why §5.6 can assert a bias in $\mathbb{E}[S^{-1}]$ before computing it.

Why it appears here. §3 records that Markowitz posed the problem as a quadratic programme in 1952. The remark in §1.1 that the optimisation is "a first-year exercise" is a statement about convexity. Convexity also sets the limit of what is easy. The cardinality constraint of §9.6, at most k non-zero positions, is non-convex. That is why its row in the table says to avoid it.

Deeper. Boyd & Vandenberghe, *Convex Optimization* (CUP, 2004), chapters 2–4, free at web.stanford.edu/~boyd/cvxbook.

A.22 Lagrange multipliers, KKT conditions, and complementary slackness

The idea. At a constrained optimum, the constraints exactly cancel the direction in which the objective would improve. A Lagrange multiplier measures the size of that cancelling force. It is also a price. The multiplier equals the improvement in the objective when the constraint loosens by one unit. *Inequality* constraints add a feature that is the most consequential fact in §8. A constraint either binds, and then it has a positive price, or it is slack, and then its price is zero. It is never both. This is complementary slackness. It converts a constrained solution into the unconstrained solution of a modified problem.

Formally. For $\min_w J(w)$ subject to $g_j(w) \leq 0$ and $h_m(w) = 0$, form the Lagrangian $\mathcal{L}(w, \zeta, \xi) = J(w) + \sum_j \zeta_j g_j(w) + \sum_m \xi_m h_m(w)$. The **Karush–Kuhn–Tucker** conditions are necessary at any optimum under mild regularity, and sufficient when the problem is convex:

$\nabla_w \mathcal{L} = 0$	(stationarity)
$g_j(w) \leq 0, \quad h_m(w) = 0$	(primal feasibility)
$\zeta_j \geq 0$	(dual feasibility)
$\zeta_j\, g_j(w) = 0$ for every j	(complementary slackness)

The last line says that a product of two non-negative numbers is zero, so at least one of them is zero. Either the constraint is tight ($g_j = 0$), or its multiplier equals zero.

Why it appears here. §5.2 uses equality multipliers to trace the efficient frontier. Every scalar in that section (a, b, c, d) follows from the resulting linear system. More importantly, the main result of §8.2 is a KKT argument. Minimising variance subject to $w \geq 0$ gives multipliers $\zeta \geq 0$ on the non-negativity constraints. Complementary slackness then makes the constrained solution *identical* to the unconstrained solution on $\tilde{\Sigma} = \Sigma - \zeta\mathbf{1}' - \mathbf{1}\zeta'$. This reduces the covariances of the assets that the unconstrained solution would short, which is shrinkage. Jagannathan & Ma (2003) establish this result. So the chapter can say that constraints *are* estimators and not concessions.

Deeper. Boyd & Vandenberghe, *Convex Optimization* (CUP, 2004), chapter 5. Sec. 5.5 covers the KKT conditions. Sec. 5.6 covers the interpretation of the multipliers as sensitivities.

A.23 The ℓ_1 and ℓ_2 norms; ridge and lasso

The idea. The ℓ_1 and ℓ_2 norms are two ways to keep the answer small, and they behave differently. The ℓ_2 norm penalises the squares of the weights. It penalises one large position heavily and many small positions lightly, so it *spreads* weight across assets. The ℓ_1 norm penalises absolute values. It penalises one position of size $2c$ and two positions of size c equally. Its optimum tends to lie at corners where many weights are exactly zero, so it *selects* assets. Ridge and lasso are the regression names for the same two penalties. The geometry is identical.

Formally. $\|w\|_1 = \sum_i |w_i|$ and $\|w\|_2 = (\sum_i w_i^2)^{1/2}$. Adding $\nu\|w\|_2^2$ to a quadratic objective (**ridge**) leaves it a QP. By §8.2, this equals replacing Σ by $\Sigma + (\nu/\gamma)I$. Adding $\nu\|w\|_1$ (**lasso**) keeps the problem convex but makes it non-differentiable at zero. Write $w = w^+ - w^-$ with both parts non-negative. This makes the constraint linear and restores a QP. In portfolio terms, $\|w\|_1$ *is* gross exposure. So an ℓ_1 constraint and a leverage cap are the same constraint.

Why it appears here. §7.10 builds on this pair. DeMiguel and co-authors (2009) show that the resulting family nests $1/N$, minimum variance and shrinkage. §8.2 shows that four things are one operation: an ℓ_2 penalty on w, a ridge on Σ, a Gaussian prior centred at zero and linear shrinkage to a scaled identity. §10.3 shows that a quadratic turnover penalty is the same operation centred on w_{prev} rather than on zero.

Deeper. Hastie, Tibshirani & Friedman, *The Elements of Statistical Learning*, 2nd ed. (Springer, 2009), sec. 3.4. It includes a figure of the two constraint regions. The figure explains why one penalty sets coefficients to zero and the other does not.

A.24 Euler's theorem, and why risk contributions add up

The idea. The statement "Asset i contributes 8% of portfolio risk" follows from the mathematics. It is not an accounting convention chosen for convenience. Doubling every weight doubles portfolio volatility exactly. Any function with that property has a unique decomposition into per-asset contributions that sum to the whole. Risk parity is well posed only because that decomposition exists.

Formally. A function f is **homogeneous of degree one** if $f(\alpha w) = \alpha f(w)$ for $\alpha > 0$. **Euler's theorem** then gives $\sum_i w_i \, \partial f/\partial w_i = f(w)$. Portfolio volatility $\sigma_p(w) = \sqrt{w'\Sigma w}$ is homogeneous of degree one. Its partial derivatives are the marginal contributions of §7.6. So

$$\frac{\partial \sigma_p}{\partial w_i} = \frac{(\Sigma w)_i}{\sigma_p}, \qquad \text{RC}_i = w_i \frac{(\Sigma w)_i}{\sigma_p}, \qquad \sum_{i=1}^{N} \text{RC}_i = \frac{w'\Sigma w}{\sigma_p} = \sigma_p.$$

Variance does *not* work. It is homogeneous of degree two. Its Euler decomposition sums to $2\, w'\Sigma w$.

Why it appears here. The equal-risk-contribution portfolio of §7.6 sets $\text{RC}_i = \text{RC}_j$ for all pairs. That target is meaningful only because the RC_i sum exactly to σ_p. So §11.4 can report risk shares as percentages of a whole and not as loose indicators.

Deeper. Roncalli, *Introduction to Risk Parity and Budgeting* (Chapman & Hall, 2013), chapter 2. It writes out the decomposition and the algorithms that use it.

A.25 Cyclical coordinate descent

The idea. Some problems have no closed-form solution but become simple once all variables but one are fixed. Coordinate descent uses this. It cycles through the coordinates and optimises each in turn with the others held fixed. It repeats until no coordinate changes. It has no tuning parameters. For the problems it suits, it beats more sophisticated methods, because each step is a scalar solve and not a matrix operation.

Formally. To minimise $J(w)$ over $w \in \mathbb{R}^N$, sweep repeatedly over $i = 1, \ldots, N$, setting

$$w_i \leftarrow \arg\min_x \; J(w_1, \ldots, w_{i-1}, \, x, \, w_{i+1}, \ldots, w_N),$$

until a sweep changes w by less than a tolerance. Suppose J is convex and equals a smooth part plus a *separable* non-smooth part. A separable part is a sum of per-coordinate terms such as $\nu\|w\|_1$. Then the iteration converges to a global optimum. It fails when the non-smooth part couples coordinates. This is the standard cautionary case.

Why it appears here. §7.6 notes that solvers find the ERC portfolio numerically and that cyclical coordinate descent converges reliably. With all other weights fixed, the equal-risk condition for w_i reduces to a scalar quadratic with one positive root, so each update has a closed form. The same algorithm makes lasso (A.23) practical at scale.

Deeper. Roncalli, *Introduction to Risk Parity and Budgeting* (2013), sec. 2.2, for the risk-parity version. Friedman, Hastie & Tibshirani, "Regularization Paths for Generalized Linear Models via Coordinate Descent," *Journal of Statistical Software* 33(1) (2010), for the general method.

Part IV — Time series and clustering. This part covers two smaller topics. The first supports estimators that let Σ change over time (§6.3, §6.10). The second supports estimators that impose group structure on Σ (§6.11, §7.11).

A.26 Realised volatility

The idea. Volatility is not observed, but unlike expected return it can be measured almost directly. Divide a period into many small intervals, square each return and add the squares. The sum measures that period's variation, and it becomes more precise as the intervals get shorter. This asymmetry recurs throughout the chapter. The precision of a mean depends on how *long* the data run. The precision of a variance depends on how *finely* the data are sampled. A researcher can buy finer data but cannot buy a longer history.

Formally. Divide a period into n sub-intervals with returns $r_1, \dots, r_n$. The realised variance is

$$\text{RV} = \sum_{j=1}^{n} r_j^2.$$

For a continuous price process, it converges to the period's integrated variance as $n \to \infty$. In practice the convergence stops and reverses at high frequency. At high frequency, microstructure noise such as bid-ask bounce and discrete ticks dominates the signal and inflates RV. The standard remedies are to sample every five minutes rather than at every tick, or to use an estimator that is robust to the noise.

Why it appears here. The table of §5.3 depends on realised volatility. Volatilities are "the most forecastable quantity in finance", and correlations are not. So §9.5 regularises in correlation space and leaves D alone. For the same reason, §6.3 recommends EWMA for volatilities and long windows for correlations. The Epps effect of §9.3 is the same relationship between sampling frequency and precision in reverse, where finer sampling *hurts*.

Deeper. Andersen, Bollerslev, Diebold & Labys, "Modeling and Forecasting Realized Volatility," *Econometrica* 71(2) (2003), 579–625.

A.27 GARCH

The idea. Volatility clusters, so volatile days tend to follow volatile days. GARCH is the minimal model of this, and it consists of one recursion. Today's variance is a blend of a long-run level, yesterday's return shock and yesterday's variance. Two of those three terms give a smoothed average of past squared returns, which is EWMA. The third term makes GARCH different. It pulls the forecast back toward a long-run mean. Without it, the forecast drifts wherever the data go.

Formally. GARCH(1,1) models the conditional variance σ_t^2 of a zero-mean return ε_t as

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \qquad \omega > 0,\ \ \alpha_1, \beta_1 \geq 0.$$

The process is stationary when $\alpha_1 + \beta_1 < 1$. Its unconditional variance is then $\omega/(1-\alpha_1-\beta_1)$. Forecasts decay toward it at rate $\alpha_1 + \beta_1$. Setting $\omega = 0$ and $\alpha_1 + \beta_1 = 1$ recovers the EWMA of §6.3 with $\theta = \beta_1$. That is why RiskMetrics is sometimes called integrated GARCH. The coefficients α_1 and β_1 follow the universal GARCH notation. They differ from the residual alpha α_i or the regression betas β_{ij} of §5.4.

Why it appears here. The DCC model of §6.10 fits a univariate GARCH to each asset to obtain $\hat{D}_t$. Only then does it model correlation separately. This turns the split of §5.3 into an algorithm. Mean reversion also shows why EWMA is not enough. An EWMA forecast at any horizon equals today's estimate. A GARCH forecast reverts toward the long-run variance, so it falls back after a volatile period.

Deeper. Bollerslev, "Generalized Autoregressive Conditional Heteroskedasticity," *Journal of Econometrics* 31(3) (1986), 307–327; and Engle (2002) for the multivariate step.

A.28 Principal component analysis

The idea. Given many correlated series, find the single combination that captures the most variation. Then find the best combination orthogonal to it, and so on. The result is a new set of coordinates in which the data are uncorrelated and ordered by the variance they carry. For returns, the first coordinate is almost always "everything, long", which is the market. PCA *is* the eigendecomposition of the covariance matrix from A.3, with a statistical interpretation.

Formally. With $\Sigma = V\Lambda V'$ as in A.3, the ith **principal component** is the portfolio v_i. Its variance is λ_i. The components are mutually uncorrelated: $v_i'\Sigma v_j = 0$ for $i \neq j$. The share of total variance explained by the first K components is $\sum_{i \le K} \lambda_i \big/ \sum_i \lambda_i$, where the denominator is $\text{tr}(\Sigma)$ by A.4. Truncating at K gives the best rank-K approximation to Σ in the Frobenius norm.

Why it appears here. The statistical factor models of §6.9 take the leading K principal components as factors and choose K by the Marchenko–Pastur edge. This replaces the usual visual inspection of a scree plot, because §5.5 states in advance where noise ends. The effective number of bets in §11.4 (A.44) uses the same decomposition for a different purpose. It counts independent risks and does not reduce dimension. The known weakness is that principal components rotate over time and are hard to interpret. §6.9 lists this as the main failure mode of these models.

Deeper. Jolliffe, *Principal Component Analysis*, 2nd ed. (Springer, 2002), chapters 1–3.

A.29 The correlation-to-distance metric

The idea. Clustering algorithms need distances, and a correlation is not a distance, for three reasons. It runs the wrong way, since a high value means close. It is not zero for identical objects. It does not obey the triangle inequality. The standard conversion fixes all three. The converted quantity is exactly the Euclidean distance between the two assets' return series, after demeaning each series and scaling it to unit length. So the conversion is not an arbitrary rescaling.

Formally. For a correlation $\rho_{ij} \in [-1, 1]$, define

$$d_{ij} = \sqrt{2\,(1 - \rho_{ij})} \;\in [0, 2].$$

This is a proper metric. It has $d_{ii} = 0$, it is symmetric, and it satisfies the triangle inequality. To see this, let x and y be the two demeaned return series, normalised so that $\|x\| = \|y\| = 1$. Then $x'y = \rho_{ij}$ and $\|x - y\|^2 = 2 - 2\rho_{ij} = d_{ij}^2$. Perfectly correlated assets are at distance 0, uncorrelated ones at $\sqrt{2}$ and perfectly anti-correlated ones at 2.

Why it appears here. §6.11 and §7.11 both open with this conversion. It is also the first step of hierarchical risk parity. The conversion treats anti-correlated assets as maximally *far apart*. This is right for a taxonomy. It is arguably wrong for a portfolio, where a strong hedge is closely related to the asset it hedges. A common variant uses $\sqrt{2(1 - |\rho_{ij}|)}$ instead. The choice is a modelling decision and not a detail.

Deeper. Mantegna, "Hierarchical Structure in Financial Markets," *European Physical Journal B* 11 (1999), 193–197. This paper introduced the metric to finance.

A.30 Hierarchical clustering, and linkage

The idea. Given distances between objects, the algorithm repeatedly merges the two closest into a group. It then treats that group as an object and continues until everything forms one tree. The output is a nested hierarchy called a dendrogram, and not a single partition. Cutting the dendrogram at any height gives any desired number of clusters. One decision is genuinely arbitrary: how to measure the distance between two *groups* once they contain more than one member. That choice is called the linkage.

Formally. Agglomerative clustering starts with N singletons and merges the nearest pair at each of $N - 1$ steps. For clusters $\mathcal{A}$ and $\mathcal{B}$, the common linkages are

single: $\min_{i \in \mathcal{A},\, j \in \mathcal{B}} d_{ij}$ complete: $\max_{i \in \mathcal{A},\, j \in \mathcal{B}} d_{ij}$

average: $\dfrac{1}{|\mathcal{A}||\mathcal{B}|} \displaystyle\sum_{i \in \mathcal{A}} \sum_{j \in \mathcal{B}} d_{ij}$ Ward: merge that minimises the increase in within-cluster variance

Single linkage readily builds long, thin clusters, a behaviour called chaining. Complete linkage and Ward produce compact groups of roughly equal size. The cost is $O(N^2 \log N)$.

Why it appears here. §6.11 uses the tree to impose block structure on the correlation matrix. §7.11 uses it to order and bisect the universe without ever inverting anything. Both sections flag the same two weaknesses, and both are real. First, the linkage is "a free parameter with real consequences and no principled selection rule". Second, the tree is unstable. Small

perturbations to $\hat{C}$ can reorganise it entirely, and the change passes directly into the weights. A tree also cannot represent an asset that genuinely belongs to two groups, and most assets do.

Deeper. Hastie, Tibshirani & Friedman, *The Elements of Statistical Learning*, 2nd ed. (Springer, 2009), sec. 14.3.12; Murtagh & Contreras, "Algorithms for Hierarchical Clustering: An Overview," *WIREs Data Mining and Knowledge Discovery* 2(1) (2012), 86–97.

Part V — Finance. This part defines the financial terms. Several entries are simple. A basis point is just a unit. A reader who must guess these terms will misread §10 and §11 entirely. The last few entries carry more content than their names suggest.

A.31 Basis points, and two-way turnover

The idea. The cost and evaluation sections use two units of account. The basis point exists because percentages are too coarse for trading costs. A spread of "0.03%" is awkward to say and easy to mistype. "3 bp" is neither. Turnover needs a denominator, and the convention is capital. The convention also double-counts by design, because every rebalance both sells something and buys something. Both legs cost money.

Formally. One **basis point** (bp) is 10^{-4}, so 1% is 100 bp and 5 bp is 0.05%. **Two-way turnover** over a rebalance is

$$\text{TO} = \sum_{i=1}^{N} \left| w_{i,t} - w_{i,t-1} \right|.$$

It is the total absolute weight traded, buys plus sells, as a fraction of capital. Annualise it by multiplying by the number of rebalances per year. Selling one asset entirely and buying another entirely gives $\text{TO} = 2$, or 200%. The one-way convention would call the same trade 100%, so the qualifier is necessary. The cost per rebalance is then roughly TO$\times$ (cost in bp), with the caveats of A.41.

Why it appears here. §10.2 quotes spreads and fees as "1–5 bp for liquid futures and large caps". §11.4 measures turnover "as an annualised two-way fraction of capital". This number converts directly into cost. §6.6 justifies nonlinear shrinkage by "a few basis points of variance reduction". Using the wrong convention, one-way or two-way, puts a factor-of-two error into every downstream cost estimate.

Deeper. Any primer on institutional trading-cost analysis. The conventions are industry usage rather than theory.

A.32 Excess returns, and the Sharpe ratio

The idea. Return alone cannot rank strategies, because leverage can raise return. The Sharpe ratio corrects for this. It is return per unit of volatility, measured above the risk-free return. Each part of the ratio has a purpose. Subtracting the risk-free rate makes it measure the reward for taking risk and not for holding cash. Dividing by volatility makes it invariant to leverage. So two managers running the same strategy at different sizes get the same score.

Formally. With r_f the risk-free rate, the **excess return** is $r - r_f$, and the chapter's μ always contains excess returns. For a portfolio with mean excess return $w'\mu$ and volatility $\sigma_p = \sqrt{w'\Sigma w}$,

$$\text{SR} = \frac{w'\mu}{\sqrt{w'\Sigma w}}.$$

The ratio is scale-invariant, so $\text{SR}(\alpha w) = \text{SR}(w)$ for $\alpha > 0$. By convention, analysts annualise a per-period Sharpe ratio by multiplying it by $\sqrt{\text{periods per year}}$. This is valid only under serial independence.

Why it appears here. Problem A of §5.1 maximises it. Its scale invariance leaves the leverage of the tangency portfolio undetermined. This is the separation result of §1.3, seen as a property of the objective. The headline finding of §2.1 compares Sharpe ratios, 5.32 reported against 0.43 available. The rule in §11.3 to compare methods only at matched volatility applies the same invariance as a safeguard.

Deeper. Lo (2002), "The Statistics of Sharpe Ratios," including what serial correlation does to the $\sqrt{\text{periods}}$ annualisation.

A.33 The efficient frontier, and the tangency portfolio

The idea. Plot every available portfolio as a point, with volatility on one axis and expected return on the other. The points have a left boundary. For each level of return, the boundary gives the lowest risk that achieves it. This boundary is the efficient frontier. Now add cash, which lies on the vertical axis at zero risk. Mixing cash with any risky portfolio traces a straight line from that point. The *best* such line touches the frontier at one point. That point is the tangency portfolio, and the slope of the line is the highest available Sharpe ratio. This explains the name, which the main text never states.

Formally. Let $a = \mathbf{1}'\Sigma^{-1}\mathbf{1}$, $b = \mathbf{1}'\Sigma^{-1}\mu$, $c = \mu'\Sigma^{-1}\mu$ and $d = ac - b^2$. The frontier is the hyperbola $\sigma^2(m) = (am^2 - 2bm + c)/d$ in (σ, m) space, with its left vertex at the GMV portfolio $(1/\sqrt{a},\ b/a)$. The tangency portfolio is

$$w_{\text{tan}} = \frac{\Sigma^{-1}\mu}{\mathbf{1}'\Sigma^{-1}\mu} = \frac{\Sigma^{-1}\mu}{b}.$$

The **capital market line** runs from the origin through this portfolio. Its slope is $\sqrt{c} = \sqrt{\mu'\Sigma^{-1}\mu}$, the maximum attainable Sharpe ratio.

Why it appears here. §5.2 derives all of this. Its two-fund separation says that every efficient portfolio is an affine mix of the GMV and tangency portfolios. The practical importance comes from the flat realised frontier of §2.1. Moving along the frontier means moving toward the $\Sigma^{-1}\hat{\mu}$ end. §5.2 identifies that end as the noisiest quantity in the problem. The reported $\sqrt{\hat{c}}$ was the 5.32 of §2.1.

Deeper. Markowitz (1952) for the frontier; Tobin (1958) for the separation that introduces the tangency point.

A.34 CAPM, equilibrium returns, and reverse optimisation

The idea. Suppose everyone solves the same mean-variance problem with the same inputs. In equilibrium, the sum of everyone's holdings must be the market itself. The market portfolio therefore *is* the tangency portfolio, and expected returns must be the values that make this true. Reversing the logic gives a useful method. Instead of forecasting returns and deriving weights, take the observed market weights and derive the returns that would justify them. These are the equilibrium returns. They are a defensible neutral starting point, and they require no forecasting skill.

Formally. CAPM states $\mathbb{E}[r_i] - r_f = \beta_i\,(\mathbb{E}[r_{\text{mkt}}] - r_f)$, with $\beta_i = \text{Cov}(r_i, r_{\text{mkt}})/\,\text{Var}(r_{\text{mkt}})$. **Reverse optimisation** inverts the master form of §1.3. If w_{mkt} is optimal at risk aversion γ, then

$$\Pi = \gamma\,\Sigma\,w_{\text{mkt}},$$

the vector of implied equilibrium excess returns. This is $\mu = \gamma\Sigma w$ solved for μ rather than for w. It needs no estimate of the means, only Σ, the observed weights and a scalar.

Why it appears here. Black–Litterman (§7.8) starts from exactly this Π and updates it with views (A.15). §8.2 records that with no views the posterior reduces to reverse-optimised market weights. The same technique underlies the advice in §9.6 to "always solve the unconstrained problem too". Reverse-optimising any portfolio reveals what its holder must believe, which is often more informative than asking.

Deeper. Sharpe, "Capital Asset Prices," *Journal of Finance* 19(3) (1964), 425–442; and Black & Litterman (1992) for the reverse-optimisation step in use.

A.35 Cap-weighted benchmarks

The idea. The default way to build an index is to hold every constituent in proportion to its market value. Cap-weighting has two properties that no other weighting has. First, it is the only weighting that everyone can hold simultaneously, because it is what the market actually consists of. Second, it needs no rebalancing as prices move, because a stock that doubles also doubles its own weight. Zero turnover and universal capacity make it the benchmark. Both properties follow from arithmetic, not from any performance claim.

Formally. Asset i's weight is $w_i = \text{Cap}_i / \sum_j \text{Cap}_j$, where Cap_i is shares outstanding times price, usually adjusted for free float. Because Cap_i moves with price, the weights adjust automatically between index reconstitutions. Under CAPM (A.34) this portfolio is the tangency portfolio, so cap-weighting is mean-variance optimal exactly when CAPM holds.

Why it appears here. §7.4 reports that long-only minimum-variance portfolios "have outperformed cap-weighted benchmarks on a risk-adjusted basis over long samples in most equity markets", and §12.5 repeats it as a [Fact]. The comparison is informative because cap-weighting is mean-variance optimal exactly when CAPM holds. Beating cap-weighting on risk-adjusted terms is therefore evidence against CAPM, so the finding is of scientific as well as commercial interest. The comparison also omits a cost. Every alternative weighting, including $1/N$, requires rebalancing that cap-weighting does not, and §7.2 warns that comparisons routinely omit it.

Deeper. Sharpe, "Indexed Investing: A Prosaic Way to Beat the Average Investor" (1976, reprinted widely), for the arithmetic argument. Any index provider's methodology document covers the mechanics of free float and reconstitution.

A.36 The low-volatility anomaly

The idea. CAPM says that more risk earns more return. Empirically the relationship is far flatter than CAPM predicts, and over long samples it has often *reversed*. Portfolios of low-volatility, low-beta stocks have delivered similar or better returns than high-volatility ones, with far less risk. This is one of the most persistent anomalies in asset pricing. It matters here because any strategy that systematically prefers low-volatility assets captures this effect, whether or not the designer intends it.

Formally. Sort stocks on trailing volatility or on market beta. A portfolio long the lowest quintile and short the highest has produced positive risk-adjusted returns in most equity markets over samples of several decades. The leading explanations are leverage constraints and benchmarking. Investors who want more return and cannot borrow bid up high-beta stocks instead. A manager measured against a cap-weighted index treats low-beta stocks as risky.

Why it appears here. §7.4 and §12.5 both flag this as the standing [Contested] caveat on minimum-variance performance. Much of the measured outperformance may be exposure to this factor rather than skill in construction. Era VI of §3 makes the same point about the risk-based turn generally. This gives a practical test. If a test cannot separate a construction method's contribution from a low-volatility tilt, the test has not measured construction.

Deeper. Frazzini & Pedersen, "Betting Against Beta," *Journal of Financial Economics* 111(1) (2014), 1–25; Black, Jensen & Scholes (1972) for the original finding of a flat line.

A.37 The information coefficient

The idea. The information coefficient measures forecast quality as the cross-sectional correlation between the prediction and the outcome. It is not the share of correct signs, and it is not an R^2. The measure is useful because it has no scale. It grades the *ranking*, which is what a cross-sectional portfolio uses. It ignores calibration, which is a separate and easier problem. The realistic range is small. A valuable equity signal has an IC of perhaps 0.03 to 0.05. Treat an out-of-sample IC above 0.10 as wrong until proven otherwise.

Formally. For forecasts $\widehat{\mu}_i$ and realised returns r_i across N assets in one period,

$$\mathrm{IC} = \mathrm{Corr}\left(\widehat{\mu}_i,\ r_i\right).$$

Analysts usually compute it on ranks (Spearman) rather than levels, and average it over periods. It maps directly to a return forecast. Under a standard normalisation, $\mathbb{E}[r_i \mid \widehat{\mu}_i] \approx \mathrm{IC} \cdot \sigma_i \cdot z_i$, where z_i is the cross-sectional z-score of the signal. This is exactly the industry method described in §7.7.

Why it appears here. The tamed mean-variance method of §7.7 converts a signal to cross-sectional z-scores, clips at ±2 or ±3 and multiplies by "a modest assumed information coefficient". That multiplication sets the *dispersion* of $\widehat{\mu}$. §7.7 states that dispersion, and not accuracy, determines the damage. §10.5 then notes that two signals with identical ICs but different half-lives deserve different weights. In that case the IC alone is not sufficient.

Deeper. Grinold & Kahn, *Active Portfolio Management*, 2nd ed. (McGraw-Hill, 1999), chapter 6.

A.38 The transfer coefficient, and the fundamental law of active management

The idea. Grinold's law gives the performance a manager *should* achieve from two numbers: the quality of each forecast and the number of independent forecasts the manager makes. Performance is skill times the square root of breadth. The original version assumes that the manager can hold whatever the forecasts imply, and no manager can. The extended version therefore adds a third factor. This factor measures how much of the intended portfolio survives the constraint set. It is the transfer coefficient. It turns "our constraints cost us something" from a complaint into a number.

Formally. Take IC as in A.37, and let the breadth BR be the number of independent bets per year. The information ratio then satisfies $\mathrm{IR} \approx \mathrm{IC}\sqrt{\mathrm{BR}}$. The generalised form inserts the **transfer coefficient**:

$$\mathrm{IR} \approx \mathrm{TC} \cdot \mathrm{IC}\sqrt{\mathrm{BR}}, \qquad \mathrm{TC} = \frac{w' \Sigma\, w^{\star}}{\sqrt{w' \Sigma w}\,\sqrt{w^{\star\prime} \Sigma w^{\star}}}.$$

TC is the correlation between the portfolio actually held and the unconstrained optimum $w^{\star}$, measured in risk-adjusted terms. So Σ appears in the inner product, and the measure is not a plain correlation of the weight vectors. Typical values for long-only institutional portfolios are 0.3 to 0.6. The construction step therefore discards most of the signal.

Why it appears here. §11.4 lists TC as one of the five diagnostics that actually discriminate, and it gives the right reading. A low value is not necessarily bad, because §8.2 shows that constraints are estimators. A low value should still be a decision and not a surprise. §9.6 advises always solving the unconstrained problem too. TC is the number that quantifies the answer.

Deeper. Grinold & Kahn, *Active Portfolio Management*, 2nd ed. (1999), chapter 6; Clarke, de Silva & Thorley, "Portfolio Constraints and the Fundamental Law of Active Management," *Financial Analysts Journal* 58(5) (2002), 48–66, for the transfer coefficient itself.

A.39 Statistical arbitrage and relative-value books

The idea. A relative-value position bets that two assets that normally move together have diverged and will reconverge. The position is long one asset and short the other, so the shared exposure to the market and the sector cancels. The remainder is a small, uncorrelated spread with high turnover. Statistical arbitrage does this systematically across hundreds of pairs or residuals at once. It appears in a chapter on covariance because it operates entirely on the part of the matrix that most risk models discard.

Formally. Using the notation of §5.4, a relative-value book holds positions in residual space. The book hedges asset i against its replicating portfolio and sizes the position by α_i/s_i^2, where $s_i^2 = \sigma_i^2(1 - R_i^2)$ is the residual variance. R_i^2 is high by construction, because the aim is to pick assets that are well replicated. Such books therefore operate exactly where the $1/(1 - R_i^2)$ factor is largest.

Why it appears here. §6.8 states the failure. A factor model with diagonal Ψ asserts that residuals are uncorrelated. In §6.8's words, "Two airlines have correlated residuals after every standard factor. A pairs strategy that trades that residual will be told by its risk model that the position is riskless." §6.7 adds the same warning for eigenvalue clipping, which destroys tightly cointegrated structure. Era III of §3 makes the general point. A statistical-arbitrage desk trades exactly what factor structure discards. For such a business, the recommendation is POET (§6.9).

Deeper. Avellaneda & Lee, "Statistical Arbitrage in the US Equities Market," *Quantitative Finance* 10(7) (2010), 761–782.

A.40 Cointegration

The idea. Two prices can each rise or fall without limit while the *gap* between them stays bounded. This is cointegration, and it is a much stronger statement than correlation. Correlation measures the co-movement of returns over some window, and it can be high by accident. Cointegration is a long-run equilibrium relationship in levels, which implies that the spread must eventually revert. It is the formal basis of every pairs trade.

Formally. Consider series that are individually non-stationary with a unit root, which means integrated of order one. They are **cointegrated** if some linear combination of them is stationary. That is, there exists $u \neq 0$ with $u'p_t$ stationary, where p_t stacks the levels. The Engle–Granger procedure tests this by regressing one series on the other and testing the residual for a unit root. The Johansen procedure handles several series and several cointegrating relationships at once. Cointegration implies an error-correction representation, in which returns respond to the current size of the disequilibrium.

Why it appears here. §6.7 warns that clipping "destroys genuine structure in the small eigenvalues, such as tightly cointegrated pairs. That is fatal if the structure is what the strategy trades." This warning states the central tension in §6. The smallest eigenvalues contain the noise *and* the real spread relationships, and no rotationally invariant estimator (A.5) can tell them apart. For a book that holds cointegrated positions, the aggressive spectral repairs of §6.6 and §6.7 remove the structure the book trades, and not only the noise.

Deeper. Engle & Granger, "Co-integration and Error Correction," *Econometrica* 55(2) (1987), 251–276; Hamilton, *Time Series Analysis* (1994), chapters 19–20.

A.41 Market impact, and the participation rate

The idea. Trading moves the price against the trader. The shape of this effect is specific and robustly measured. Cost per share grows roughly with the *square root* of the fraction of the day's volume that the trader takes, and not linearly. Large trades are therefore cheaper per share than a linear model predicts, but total cost still grows faster than size. The governing variable is the fraction of available liquidity consumed, and not the amount of money traded.

Formally. Let x be the order size and ADV the average daily volume. The **participation rate** is POV $= x/\text{ADV}$, and the standard impact model is

$$\text{cost per share} \;\approx\; \text{const} \times \sigma \times \sqrt{\text{POV}},$$

with σ the asset's daily volatility. Total cost is that times x, so it grows as $x^{3/2}$. The growth is superlinear, so size is costly. Convention splits impact into a **temporary** part and a **permanent** part. The price recovers from the temporary part once trading stops. It does not recover from the permanent part.

Why it appears here. §10.2 states the square-root form as [Fact] and the exponent, between roughly 0.4 and 0.6, as [Contested]. For construction, it instead uses a *quadratic* cost $\frac{1}{2}(w - w_{\text{prev}})'\Lambda_c(w - w_{\text{prev}})$. This is not the true cost function, but it makes the closed form of §10.3 possible. The proportional costs of §10.4 are a third shape, which produces a no-trade band. Three shapes give three qualitatively different policies. So §10.2 states that the scaling matters more than the level. The Λ_c of §10.2 is a diagonal cost matrix. It is not the eigenvalue matrix of the notation block.

Deeper. Almgren & Chriss (2000), "Optimal Execution of Portfolio Transactions". Tóth et al., "Anomalous Price Impact and the Critical Nature of Liquidity," *Physical Review X* 1 (2011), 021006, covers the empirical status of the square-root law.

A.42 Winsorising

The idea. A single extreme observation can dominate a variance estimate, because variance squares each deviation. Winsorising limits this by moving outliers to a threshold and does not remove them. The observation still counts, but the cap limits its influence. Deleting a point changes the sample size and can bias the estimate. Capping it keeps every observation and only bounds its influence.

Formally. Choose a quantile level u, commonly 0.01. Winsorising replaces every observation below the u quantile with that quantile's value. It replaces every observation above the $1 - u$ quantile with that quantile's value. **Truncation**, or trimming, discards them instead. In cross-sectional signal work, analysts usually apply the same operation to z-scores with a fixed cap, clipping at ±2 or ±3, rather than at an empirical quantile.

Why it appears here. Winsorising has two opposite roles, and the chapter separates them carefully. §6.12 lists it as a robust *covariance* technique and then argues against it, because in markets the outliers are usually the information. The rule of §9.2 is to flag any return beyond ±50% and *inspect* it, because a winsoriser cannot distinguish a real crash from a missed split. In §7.7, clipping signal z-scores is the role where winsorising is unambiguously right, because there the extremes are estimation error by construction.

Deeper. Huber & Ronchetti, *Robust Statistics*, 2nd ed. (Wiley, 2009), chapters 1–2.

A.43 The Herfindahl index, and effective counts

The idea. The Herfindahl index comes from industrial organisation, where it measures how concentrated an industry is. To compute it, square every firm's market share and add the squares. The reciprocal is the useful quantity. If n things share the total equally, the sum of squared shares is $1/n$ and the reciprocal is n. The reciprocal therefore always gives "the number of equally sized things this is equivalent to", however unequal the actual distribution is. This is the same "effective count" construction as Kish's sample size (A.12) and the perplexity of A.20. Recognising them as one construction saves learning three.

Formally. For shares $p_i \geq 0$ that sum to one, $\text{HHI} = \sum_i p_i^2 \in [1/n, 1]$, and the **effective number** is $1/\text{HHI} \in [1, n]$. For a portfolio, the shares must be normalised by gross exposure, $p_i = |w_i| / \sum_j |w_j|$. On a levered or long–short book, the raw squared weights measure leverage and not breadth.

Why it appears here. §11.4 includes this among the five diagnostics worth reporting. It gives this example: "A 500-name portfolio with an effective count of 12 is a 12-name portfolio." The index pairs with the warning in §7.4 that unconstrained minimum variance is notoriously concentrated. It also complements the characterisation by Clarke, de Silva & Thorley (2011) of exactly which assets survive. But it counts *positions*. The next diagnostic in §11.4 exists because positions do not diversify a portfolio.

Deeper. Hirschman, "The Paternity of an Index," *American Economic Review* 54(5) (1964), 761. It is a one-page note on the index's own history.

A.44 The effective number of bets

The idea. Forty bank stocks are 40 positions and one bet. The effective number of bets counts *independent risks* and not holdings, so it captures what the Herfindahl index cannot see. The method rotates the portfolio into uncorrelated components and measures the share of total variance that each component carries. It then applies the effective-count construction to those shares. The components are uncorrelated by construction, so holding the same bet many times cannot inflate the result.

Formally. Decorrelate the portfolio's risk, either by PCA (A.28) on Σ or by Meucci's minimum-torsion rotation, which stays closer to the original assets. Write p_i for the fraction of portfolio variance carried by the ith uncorrelated component, so that $\sum_i p_i = 1$. Then

$$\text{ENB} = \exp\big(H(p)\big) = \exp\Big(-\sum_{i=1}^{N} p_i \ln p_i\Big) \in [1, N],$$

which is the perplexity of A.20. It equals N when every independent risk source carries an equal share, and 1 when one source carries everything.

Why it appears here. §11.4 calls the gap between this number and the effective number of positions "the accurate measure of diversification". Its example is the portfolio of 40 bank stocks, with an effective number of positions near 40 and an effective number of bets near two. It states, "Only the second number predicts what will happen in a crisis." The measure also checks the diversification ratio of §7.5. That section admits that adding near-duplicates can inflate the ratio. This tactic raises the DR and leaves the ENB flat.

Deeper. Meucci (2009), "Managing Diversification," which introduces the measure; Meucci, Santangelo & Deguest (2015) for the minimum-torsion rotation.

A.45 The standard error of a Sharpe ratio

The idea. A Sharpe ratio is an estimate, and like any estimate it has a confidence interval. The interval is very wide. A Sharpe ratio is essentially a t-statistic divided by the square root of the sample length. Its precision therefore depends only on the *calendar span*. Sampling the same three years more finely does not help. This is the asymmetry of A.26 again, seen from the side of performance measurement.

Formally. For IID returns, the Sharpe ratio estimated over n years has

$$\text{se}\big(\widehat{\text{SR}}\big) \approx \sqrt{\frac{1 + \text{SR}^2/2}{n}}.$$

Distinguishing a true Sharpe ratio of 0.5 from zero at 95% confidence therefore needs roughly 16 years. Distinguishing 0.5 from 1.5 needs longer still. Serial correlation inflates the standard error further, and Lo (2002) gives the corrected form. Positively autocorrelated returns are common in illiquid books. They make the naive annualisation overstate the Sharpe ratio as well as its precision.

Why it appears here. §11.5 works the arithmetic. At a true Sharpe ratio of 0.5 over 10 years, the standard error is 0.34. Two methods at 0.5 and 1.5 differ by 1.0 against a standard error of 0.57. That is under two standard errors, so it does not establish a difference. §11.5 avoids this problem by comparing the *paired difference* of the two return series rather than their levels. Two construction methods on the same universe share nearly all of their market exposure, so the difference has far lower volatility than either return series.

Deeper. Lo (2002), "The Statistics of Sharpe Ratios," *Financial Analysts Journal* 58(4), 36–52.

A.46 The deflated Sharpe ratio

The idea. When researchers try many configurations and report the best, the reported number is the maximum of a sample of noisy quantities, and maxima are biased upward. It does not estimate that configuration's quality. The deflated Sharpe ratio tests this directly. Given M trials, it computes how large the best result would have been *by luck alone* and checks whether the observed result exceeds that value. This converts "the best backtest scored 1.4" into a testable statement. The test often shows that 1.4 is unremarkable.

Formally. Combine the multiple-testing arithmetic of A.19 with the standard error of A.45. Suppose M independent trials each have a Sharpe ratio with standard error se. Under a null of no skill, the expected best of them is about se $\sqrt{2 \ln M}$. The benchmark to beat therefore rises with $\sqrt{\ln M}$ and does not stay at zero. The **deflated Sharpe ratio** is the probability that the observed maximum exceeds this null-adjusted threshold. The version of Bailey and López de Prado also corrects for skewness, excess kurtosis and the effective number of *independent* trials. That number is smaller than M when the configurations are correlated.

Why it appears here. §11.5 prescribes it directly. It states, “After evaluating 12 shrinkage intensities, the best of the 12 is biased upward, and the deflated Sharpe ratio adjustment quantifies by how much.” The trial count grows quickly here, because shrinkage intensity, target matrix, window length, rebalance frequency, constraint set and linkage method each add trials. §11.3 calls this the “ordinary sin of overfitting, applied to a matrix”.

Deeper. Bailey & López de Prado, “The Deflated Sharpe Ratio,” *Journal of Portfolio Management* 40(5) (2014), 94–107; Harvey & Liu, “Backtesting,” *Journal of Portfolio Management* 42(1) (2015), 13–28.

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